



HEALTHCARE ANALYTICS PLATFORM

TECHNICAL DOCUMENTATION

VERSION 1.0

Business Intelligence Portfolio Project



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TECHNOLOGIES

- PostgreSQL
- SQL
- Tableau Desktop



DASHBOARD MODULES

-  Comparative Analytics
-  Revenue Distribution
-  Patient Demographics
-  Clinical & Medical Conditions
-  Healthcare Operations



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Transforming Data into Insights.
Empowering Better Healthcare Decisions.



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Healthcare Analytics Platform

Technical Documentation

CHAPTER 1

Executive Summary

"Transforming healthcare data into actionable business intelligence through scalable analytics, interactive dashboards, and data-driven decision support."

1.1 Project Overview

The **Healthcare Analytics Platform** is a comprehensive Business Intelligence (BI) solution designed to demonstrate the end-to-end development of an analytical reporting environment for the healthcare industry. The project was conceived as a professional portfolio initiative that simulates analytical requirements commonly encountered in healthcare organizations, integrating PostgreSQL-based data management, SQL development, KPI design, interactive Tableau dashboards, and technical documentation into a cohesive analytical solution.

The primary objective of the platform is to transform healthcare data into meaningful analytical information that supports operational monitoring, financial analysis, patient and clinical analysis, comparative evaluation, and decision-making. Rather than focusing exclusively on data visualization, the project demonstrates a broader Business Intelligence workflow that includes data preparation, analytical logic development, KPI implementation, dashboard development, user interaction design, validation, and technical documentation.

The analytical environment is built using PostgreSQL as the relational database management system and Tableau Desktop as the analytical, visualization, and dashboard development environment. Within PostgreSQL, the `healthcare_db` database provides the structured healthcare data environment, while the centralized analytical View `vw_healthcare_master` provides the standardized data structure used by the final portfolio dashboards. Tableau consumes this View through `DS_vw_healthcare_master` (`healthcare_db`) and extends the centralized data foundation through Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, KPI calculations, ranking, segmentation, and other analytical mechanisms.

The platform consists of five executive analytical dashboards, each addressing a distinct healthcare analytical perspective: **Comparative Analytics, Revenue Distribution, Patient Demographics, Clinical &**

Medical Conditions, and Healthcare Operations. Together, these dashboards provide a coordinated analytical environment for exploring financial, demographic, clinical, comparative, and operational information through consistent visual and interaction principles.

Throughout the development process, particular emphasis was placed on creating reusable analytical components rather than isolated visualizations. Standardized KPI definitions, reusable Tableau calculations, modular worksheets, consistent design principles, and controlled dashboard interactions help maintain analytical consistency across the solution while allowing individual components to be developed, validated, and maintained according to their architectural responsibilities.

Although developed as a portfolio project using an educational healthcare dataset, the Healthcare Analytics Platform applies professional Business Intelligence concepts and methodologies. The resulting solution demonstrates competencies in PostgreSQL and SQL, centralized analytical data preparation, Tableau calculations, KPI development, dashboard architecture, visualization design, interactive analytics, and technical documentation. More importantly, it demonstrates the ability to transform structured healthcare data into analytical information designed to support exploration, interpretation, and decision-making.

1.2 Business Context

Modern healthcare organizations operate in complex environments where significant volumes of clinical, operational, demographic, and financial data can be generated across healthcare activities. Hospitals and healthcare organizations must coordinate patient information, admissions, discharges, medical conditions, insurance activity, billing information, medications, test results, and other operational data while maintaining effective visibility into organizational performance.

Traditional reporting approaches may rely on static spreadsheets, fragmented reports, and manual data-consolidation processes. Although these methods can provide useful operational information, they may limit the ability of users to explore analytical relationships, compare performance across different dimensions, and investigate changing patterns efficiently. Executives, managers, analysts, and other decision-makers can benefit from analytical environments that provide consistent performance indicators and interactive access to relevant information.

Business Intelligence (BI) can address these analytical requirements by organizing structured data into a consistent environment in which information can be analyzed, compared, and explored interactively. Through standardized Key Performance Indicators (KPIs), trend analysis, comparative analysis, and executive dashboards, healthcare information can be examined from operational, financial, demographic, clinical, and other analytical perspectives. This approach can improve analytical visibility and support more structured, data-informed decision-making.

The healthcare dataset used in this project provides an educational analytical scenario containing patient demographics, admission and discharge information, medical conditions, insurance providers, billing amounts, medications, test results, and other hospitalization-related attributes. Although the dataset is used for educational and portfolio purposes, it provides sufficient analytical diversity to support the design and implementation of a multi-domain Business Intelligence solution.

Based on this business context, the **Healthcare Analytics Platform** was designed to demonstrate how PostgreSQL and Tableau can be combined to transform structured healthcare data into an integrated analytical environment. Centralized data preparation, standardized analytical calculations, KPIs, interactive visualizations, executive dashboards, and controlled dashboard interactions provide the foundation for exploring operational, financial, demographic, clinical, and comparative healthcare information through a structured and maintainable analytical architecture.

Business Insight

Healthcare data can provide greater analytical value when operational, financial, demographic, and clinical information is organized into a consistent analytical environment. By combining standardized KPIs, interactive visualizations, and coordinated analytical perspectives, the Healthcare Analytics Platform demonstrates how Business Intelligence can help transform structured healthcare data into information that is easier to explore, interpret, and use in decision-making.

1.3 Project Objectives

The **Healthcare Analytics Platform** was designed to demonstrate how Business Intelligence methodologies can be applied to transform structured healthcare data into meaningful analytical information. The project combines PostgreSQL-based data management and preparation, SQL development, Tableau analytical logic, interactive visualization, and technical documentation within a coordinated analytical solution.

Beyond demonstrating technical proficiency with PostgreSQL, SQL, and Tableau Desktop, the platform emphasizes analytical consistency, standardized KPI definitions, modular dashboard development, controlled user interaction, and business-oriented presentation of healthcare information.

Primary Objective

Develop an end-to-end Business Intelligence solution that transforms structured healthcare data into an interactive analytical environment capable of supporting operational, financial, demographic, clinical, and comparative analysis through standardized KPIs, analytical visualizations, and executive dashboards.

Specific Objectives

- Establish a structured PostgreSQL data environment through `healthcare_db` to support analytical development.
- Develop `vw_healthcare_master` as the centralized PostgreSQL analytical View used by the final portfolio dashboards.
- Integrate `vw_healthcare_master` with Tableau Desktop through `DS_vw_healthcare_master` (`healthcare_db`) to provide a consistent analytical data foundation.
- Develop reusable analytical logic in Tableau using Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, and related analytical mechanisms.
- Build standardized Key Performance Indicators (KPIs) across the financial, demographic, clinical, comparative, and operational analytical domains.
- Create five executive Tableau dashboards: Comparative Analytics, Revenue Distribution, Patient Demographics, Clinical & Medical Conditions, and Healthcare Operations.
- Apply consistent visualization and presentation principles to improve readability, analytical interpretation, and user experience.
- Implement interactive filters, Filter Actions, and Highlight Actions to support controlled analytical exploration.
- Produce professional technical documentation describing the solution architecture, analytical methodology, implementation decisions, visualization and interaction design, project scope, and supporting portfolio assets.
- Demonstrate an end-to-end Business Intelligence workflow suitable for professional portfolio presentation and technical interview discussion.

Business Insight

Clear project objectives help ensure that data preparation, analytical calculations, KPIs, visualizations, interactions, and dashboards contribute to a consistent analytical purpose. In a Business Intelligence solution, technical components create greater value when they are aligned with defined analytical questions and organized to support understandable, reusable, and decision-oriented information.

1.4 Solution Overview

The **Healthcare Analytics Platform** was developed as a modular Business Intelligence solution that combines PostgreSQL-based data management and preparation, Tableau analytical logic, interactive visualization, dashboard interaction, and technical documentation within a coordinated analytical environment.

The solution follows a layered architectural approach in which responsibilities are logically separated according to their analytical purpose. Healthcare data is stored and organized within the PostgreSQL database `healthcare_db`. Within this environment, the centralized analytical View `vw_healthcare_master` provides the standardized data structure required by the final portfolio dashboards and establishes the primary interface between PostgreSQL and Tableau Desktop.

Tableau consumes the centralized PostgreSQL View through `DS_vw_healthcare_master (healthcare_db)`. Once the data is available within Tableau, the analytical foundation is extended through Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage-based calculations, time-based analysis, and standardized KPI calculations. This division of responsibilities allows PostgreSQL to provide the centralized analytical data structure while Tableau implements analytical logic that depends on calculation granularity, visualization context, comparison, segmentation, ranking, and user-controlled analytical behavior.

Tableau worksheets function as modular visualization components that transform analytical measures and classifications into KPIs, trends, comparisons, distributions, rankings, and other analytical views. These worksheets are subsequently assembled into executive dashboards according to specific analytical objectives and consistent presentation principles.

The final solution is organized around five executive analytical dashboards:

- **Comparative Analytics Dashboard**
- **Revenue Distribution Dashboard**
- **Patient Demographics Dashboard**
- **Clinical & Medical Conditions Dashboard**
- **Healthcare Operations Dashboard**

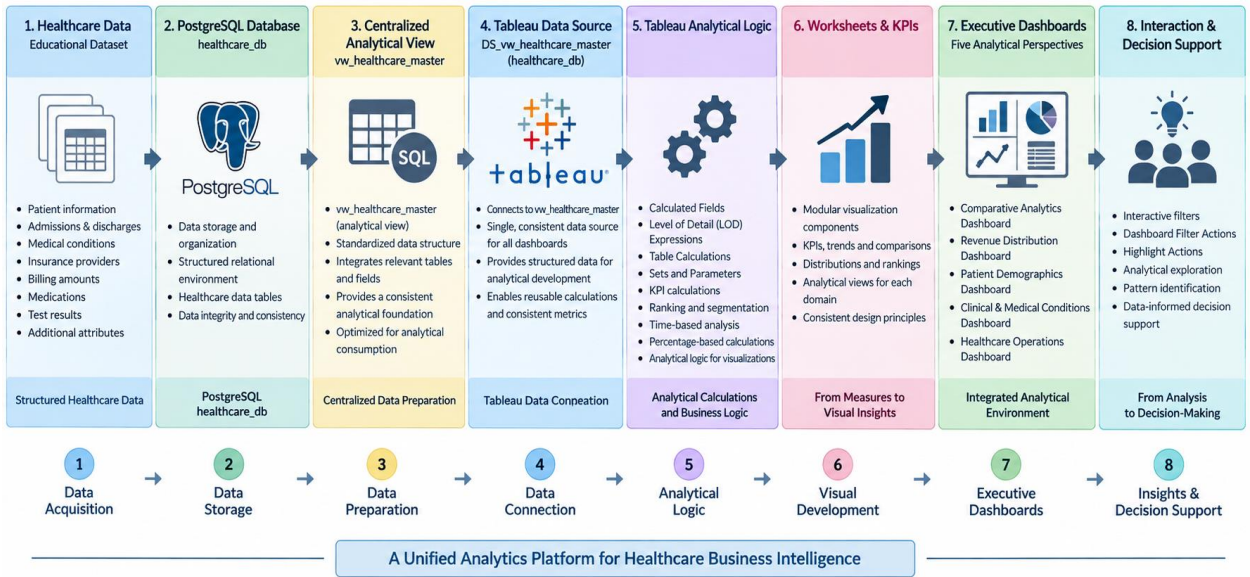
Together, these dashboards provide complementary perspectives across comparative, financial, demographic, clinical, medical-condition, and operational analysis while sharing the same centralized PostgreSQL analytical foundation through `vw_healthcare_master` and `DS_vw_healthcare_master (healthcare_db)`.

The solution also incorporates an interaction architecture implemented through interactive filters, Dashboard Filter Actions, and Highlight Actions. These mechanisms allow users to refine analytical

context, investigate selected information, and explore relationships across related visualizations while preserving controlled and predictable dashboard behavior.

The overall solution can therefore be summarized as:

Figure 1.1 – Overall Solution Architecture
From Healthcare Data to Analytical Insights



Business Insight

The value of a Business Intelligence solution lies not only in its visualizations, but also in the architecture that supports the analytical workflow. A well-designed analytical platform establishes clear responsibilities for data management, data preparation, analytical logic, visualization, interaction, and presentation while maintaining integration between these components. This separation of responsibilities improves analytical consistency, modularity, reusability, and maintainability while providing a structured foundation for future analytical expansion.

1.5 Technology Stack

The **Healthcare Analytics Platform** was developed using a Business Intelligence technology stack centered on PostgreSQL, SQL, Tableau Desktop, Microsoft Word, and PDF. These technologies support the principal responsibilities of the analytical workflow, including data management, centralized data preparation, analytical calculations, visualization, interaction, dashboard presentation, and technical documentation.

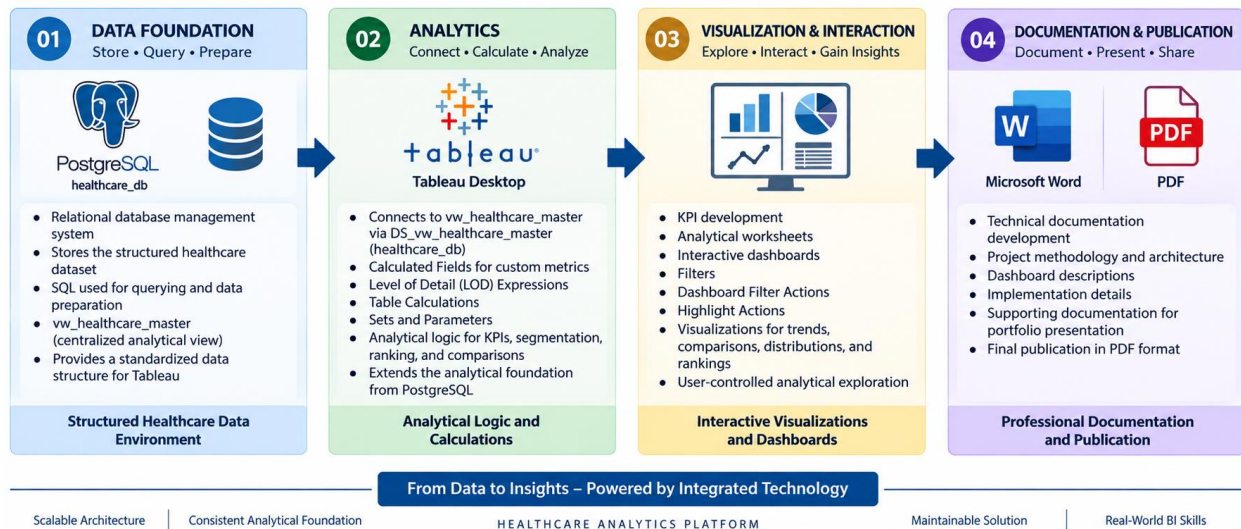
The solution follows a layered architectural approach in which each technology performs responsibilities appropriate to its role. PostgreSQL provides the relational data environment through `healthcare_db`, while SQL supports data querying and centralized analytical preparation through `vw_healthcare_master`. Tableau Desktop consumes this centralized View through `DS_vw_healthcare_master` (`healthcare_db`) and extends the analytical foundation through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, KPIs, worksheets, dashboards, filters, and Dashboard Actions.

Microsoft Word provides the authoring environment for the project's technical documentation, while PDF is planned as the final publication format for professional presentation and distribution. Together, these technologies support a structured analytical workflow from healthcare data management and preparation through interactive analysis, visualization, dashboard presentation, and technical documentation.

Component	Technology	Purpose
Database Management	PostgreSQL – <code>healthcare_db</code>	Store and organize the structured healthcare dataset within a relational database environment.
Query & Data Preparation	SQL	Query, organize, and prepare healthcare data within PostgreSQL.
Centralized Analytical View	<code>vw_healthcare_master</code>	Provide the standardized PostgreSQL analytical structure used by the five final executive dashboards.
Tableau Data Source	<code>DS_vw_healthcare_master</code> (<code>healthcare_db</code>)	Provide Tableau with a consistent connection to the centralized PostgreSQL analytical View.
Analytics & Visualization	Tableau Desktop	Develop analytical calculations, worksheets, KPIs, visualizations, and executive dashboards.
Analytical Calculations	Calculated Fields & LOD Expressions	Implement reusable derived measures, classifications, and calculations requiring controlled analytical granularity.
Contextual Analysis	Table Calculations	Support ranking, percentages, Running Totals, Moving Averages, and other visualization-context calculations.
Dynamic Analysis	Sets & Parameters	Support segmentation, analytical selection, and user-controlled analytical behavior.

User Interaction	Filters & Dashboard Actions	Support controlled analytical exploration through interactive filters, Filter Actions, and Highlight Actions.
Documentation	Microsoft Word	Develop and maintain the professional technical documentation for the project.
Publication	PDF	Provide the planned publication and distribution format while preserving document structure and presentation.

Figure 1.2 – Technology Stack Overview
Integrated Technologies for a Unified Healthcare Analytics Platform



Technology Integration

The Healthcare Analytics Platform integrates PostgreSQL, SQL, Tableau Desktop, and technical documentation into a coordinated analytical workflow. PostgreSQL provides the structured data environment, SQL supports querying and centralized analytical preparation through vw_healthcare_master, and Tableau Desktop consumes this standardized structure through DS_vw_healthcare_master (healthcare_db) to perform analytical calculations, visualization, dashboard development, and interactive exploration.

Microsoft Word and PDF support the documentation and publication responsibilities of the project. Together, these technologies form logically distinct but integrated components of the Business Intelligence workflow, with responsibilities assigned according to the analytical purpose of each technology.

Business Insight

Technology selection becomes more effective when each component has a clearly defined architectural responsibility. By using PostgreSQL and SQL for structured data management and centralized analytical preparation, Tableau Desktop for analytical logic, visualization, and interaction, and Microsoft Word and PDF for technical documentation and publication, the platform maintains a clear separation of responsibilities while supporting an integrated and maintainable Business Intelligence workflow.

1.6 Key Features

The **Healthcare Analytics Platform** incorporates a comprehensive set of Business Intelligence capabilities designed to support structured healthcare analysis, performance monitoring, comparative exploration, and decision-oriented analytical reporting. The platform combines centralized PostgreSQL data preparation, reusable Tableau analytical logic, standardized KPIs, modular visualizations, interactive dashboards, and controlled analytical interactions within a consistent architectural framework.

Rather than developing isolated visualizations, the project emphasizes reusable analytical components, standardized definitions, modular worksheet development, and consistent interaction and presentation practices. This approach improves analytical consistency and maintainability while providing a coordinated experience across the different healthcare analytical perspectives represented by the platform.

The primary features of the platform include:

- **Integrated Business Intelligence Architecture** — A layered analytical environment connecting PostgreSQL healthcare_db, the centralized analytical View vw_healthcare_master, the Tableau data source DS_vw_healthcare_master (healthcare_db), Tableau analytical logic, worksheets, KPIs, executive dashboards, and interactive analytical exploration.
- **Centralized Analytical Data Preparation** — vw_healthcare_master provides the standardized PostgreSQL analytical structure consumed by all five final executive dashboards, reducing source fragmentation and establishing a consistent interface between PostgreSQL and Tableau.
- **Standardized KPI Framework** — Reusable KPI definitions support consistent measurement across financial, demographic, clinical, comparative, and operational analytical perspectives.
- **Five Executive Analytical Dashboards** — Comparative Analytics, Revenue Distribution, Patient Demographics, Clinical & Medical Conditions, and Healthcare Operations provide complementary analytical perspectives while following consistent visualization and interaction principles.
- **Advanced Tableau Analytical Logic** — Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentages, Running Totals, Moving Averages, and other analytical mechanisms extend the centralized PostgreSQL data foundation.
- **Dynamic Segmentation and Comparative Analysis** — Sets, calculated classifications, and comparative measures support analytical scenarios such as Selected versus Others analysis and other controlled comparisons.
- **Interactive Filters and Filter Actions** — User-controlled filters and Dashboard Filter Actions allow selected analytical context to be applied across related visualizations for focused investigation.
- **Highlight Actions** — Highlight interactions emphasize related information while preserving the broader analytical context, supporting relationship-oriented exploration without filtering unrelated marks from the view.

- **Multi-Dimensional Healthcare Analysis** — The platform supports analysis across patient demographics, medical conditions, admissions, discharges, medications, test results, insurance providers, billing activity, and length-of-stay information.
- **Trend, Ranking, and Contribution Analysis** — Time-based analysis, ranking, percentage contribution, Running Totals, and Moving Averages support investigation of patterns, relative performance, and analytical change.
- **Consistent Dashboard Design Standards** — Standardized layouts, typography, KPI cards, labels, filters, tooltips, visual hierarchy, and interaction patterns support a cohesive analytical experience.
- **Modular and Maintainable Design** — Centralized data preparation, reusable analytical logic, modular worksheets, and controlled dashboard interactions allow analytical components to be developed, validated, and maintained according to their respective responsibilities.
- **Professional Technical Documentation** — The project documents business context, architecture, PostgreSQL data preparation, Tableau analytical techniques, KPIs, visualizations, interactions, design decisions, project scope, and planned portfolio presentation.

Analytical Coverage

The combination of these capabilities enables the platform to support several complementary analytical perspectives. Financial analysis provides visibility into billing and revenue distribution; operational analysis evaluates admissions, discharges, length of stay, and utilization patterns; demographic analysis explores patient characteristics; and clinical analysis examines medical conditions, medications, and test results. Together, these analytical domains provide a broader understanding of healthcare performance than any individual visualization could provide independently.

Business Insight

The combination of these capabilities enables the platform to support several complementary analytical perspectives. Financial analysis examines billing and revenue distribution; operational analysis evaluates admissions, discharges, length of stay, throughput, and related operational patterns; demographic analysis explores patient characteristics; clinical and medical-condition analysis examines conditions, medications, test results, and related healthcare attributes; and comparative analysis supports controlled evaluation of selected analytical groups and categories. Together, these analytical perspectives provide a multi-domain view of the educational healthcare dataset while using a common analytical foundation and consistent analytical definitions across the final dashboard environment.

1.7 Business Value

The **Healthcare Analytics Platform** demonstrates how a structured Business Intelligence environment can transform healthcare data into accessible, consistent, and decision-oriented analytical information. By organizing financial, operational, demographic, clinical, medical-condition, and comparative perspectives within a coordinated analytical framework, the platform demonstrates how decision-makers can move from high-level performance indicators to more focused analytical investigation.

One of the principal analytical benefits demonstrated by the solution is improved **operational visibility**. The Healthcare Operations Dashboard supports analysis of admissions, discharges, average length of stay, monthly throughput, admission-type patterns, billing trends, insurance-provider revenue, and test-result activity. Interactive visualizations allow these operational indicators to be explored from multiple perspectives within a coordinated dashboard environment.

The platform also demonstrates the value of structured **financial analysis**. Revenue distribution, billing activity, medical conditions, insurance providers, ranking, and contribution analysis provide different perspectives for examining financial patterns within the dataset. Standardized KPIs and comparative visualizations help users identify concentration, relative contribution, and trends that may require additional analytical investigation.

From a **patient demographic and clinical perspective**, the platform supports exploration of age, gender, blood type, insurance provider, medical conditions, medications, test results, and other healthcare-related attributes. These analytical perspectives demonstrate how demographic and clinical information can be organized into understandable visual components while maintaining consistent analytical definitions across the dashboard environment.

Comparative analytics provides an additional level of analytical value by enabling controlled comparisons between selected analytical groups and the remaining population. Tableau Sets, calculated classifications, and comparative measures support Selected versus Others analysis, allowing users to investigate differences in revenue, patient counts, and related analytical measures within a consistent analytical framework.

Another important benefit demonstrated by the platform is **analytical consistency**. The centralized PostgreSQL View `vw_healthcare_master`, the Tableau data source `DS_vw_healthcare_master` (`healthcare_db`), standardized KPI definitions, reusable Tableau calculations, and consistent visualization and interaction practices reduce unnecessary analytical fragmentation and provide a common foundation for the five executive dashboards.

The platform also demonstrates the value of **interactive analytical exploration**. Interactive filters, Dashboard Filter Actions, and Highlight Actions allow users to adjust analytical context, investigate selected subsets, and explore relationships between related visualizations. Filter Actions support focused investigation by narrowing target views, while Highlight Actions preserve the broader context and emphasize related information.

Finally, the project provides professional value as a **portfolio demonstration of end-to-end Business Intelligence development**. It documents the progression from PostgreSQL-based data management and centralized analytical preparation through Tableau calculations, KPI development, visualization, dashboard interaction, and technical documentation. This combination demonstrates not only the construction of analytical outputs, but also the architectural and methodological decisions used to organize the solution.

Although the Healthcare Analytics Platform was developed using an educational healthcare dataset and represents a portfolio analytical implementation rather than a production enterprise deployment, it demonstrates how Business Intelligence principles can be applied to organize healthcare information into a structured analytical environment. The project therefore provides a practical example of how data preparation, analytical logic, visualization, interaction, and documentation can work together to support analytical interpretation and decision-oriented exploration.

Business Insight

Business value in analytics is created when structured data, consistent analytical definitions, meaningful KPIs, appropriate visualizations, and controlled interactions work together to answer relevant analytical questions. The Healthcare Analytics Platform demonstrates this principle by organizing multiple healthcare perspectives within a common analytical foundation while allowing users to progress from summary-level information to focused analytical exploration.

CHAPTER 1

Chapter Summary

The **Healthcare Analytics Platform** represents an end-to-end Business Intelligence solution designed to transform structured healthcare data into consistent, interactive, and decision-oriented analytical information. This Executive Summary established the business context, project objectives, solution architecture, technology stack, analytical capabilities, interaction model, and potential business value that define the foundation of the platform.

The project combines PostgreSQL-based data management and centralized analytical preparation, Tableau analytical logic and visualization, standardized KPIs, modular worksheets, five executive dashboards, interactive analytical exploration, and professional technical documentation within a coordinated Business Intelligence framework.

The principal conclusions established in this chapter are summarized below.

Key Takeaways

✓ End-to-End Business Intelligence Solution

The project demonstrates a structured Business Intelligence workflow that progresses from healthcare data management in PostgreSQL and centralized analytical preparation through `vw_healthcare_master` to Tableau analytical calculations, KPIs, worksheets, executive dashboards, interactive exploration, and professional technical documentation.

✓ Healthcare-Focused Analytical Environment

The platform provides complementary analytical perspectives across financial performance, patient demographics, clinical and medical conditions, healthcare operations, and comparative analysis. These perspectives incorporate information related to admissions, discharges, billing activity, insurance providers, medical conditions, medications, test results, patient characteristics, and length-of-stay analysis.

✓ Layered and Modular Architecture

The platform follows a layered and modular architecture in which data management, centralized data preparation, analytical logic, visualization, interaction, presentation, and documentation operate as logically separated but integrated responsibilities.

PostgreSQL `healthcare_db` provides the structured data environment, `vw_healthcare_master` provides the centralized analytical data structure, and `DS_vw_healthcare_master (healthcare_db)` provides Tableau with a consistent connection to that analytical foundation. Tableau then extends the centralized data structure through reusable analytical calculations, KPIs, worksheets, visualizations, dashboards, and interactive analytical behavior.

✓ **Standardized KPI Framework**

Standardized KPI definitions and reusable analytical calculations provide consistent measures across the financial, demographic, clinical, comparative, and operational analytical perspectives represented by the platform. Separating KPI calculation logic from visual KPI presentation also supports greater analytical consistency and component reusability.

✓ **Five Executive Analytical Dashboards**

The final platform is organized around five executive analytical dashboards: **Comparative Analytics Dashboard, Revenue Distribution Dashboard, Patient Demographics Dashboard, Clinical & Medical Conditions Dashboard, and Healthcare Operations Dashboard**. Each dashboard addresses a distinct analytical perspective while sharing the same centralized PostgreSQL analytical foundation and consistent Tableau design principles.

✓ **Advanced Tableau Capabilities**

The solution incorporates Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking, percentage calculations, Running Totals, Moving Averages, segmentation, comparative analysis, standardized KPI calculations, and other Tableau analytical mechanisms required to transform the centralized PostgreSQL data foundation into reusable analytical information.

✓ **Interactive Analytical Exploration**

Interactive filters, Dashboard Filter Actions, and Highlight Actions allow users to refine analytical context, investigate selected information, and explore relationships across related visualizations. Filter Actions support focused investigation by narrowing target views, while Highlight Actions emphasize related information without removing the broader analytical context.

✓ **Business-Oriented Analytics**

The platform organizes financial, demographic, clinical, medical-condition, comparative, and operational information around understandable analytical questions and business-oriented terminology. This approach demonstrates how Business Intelligence can transform structured healthcare data into information designed to support interpretation, exploration, and data-informed decision-making.

✓ **Maintainability and Reusability**

Centralized data preparation through `vw_healthcare_master`, reusable Tableau analytical logic, standardized KPI definitions, modular worksheets, consistent visualization principles, controlled dashboard interactions, and structured technical documentation provide a maintainable foundation that can accommodate future analytical requirements without unnecessarily fragmenting the existing architecture.

✓ **Professional Portfolio Implementation**

Although developed using an educational healthcare dataset, the project applies professional Business Intelligence concepts and documents the architecture, PostgreSQL data environment, centralized analytical preparation, Tableau analytical techniques, KPI framework, visualization and interaction design, implementation decisions, project scope, and methodology used throughout the solution.

The current implementation represents a **portfolio analytical solution rather than a production enterprise deployment**, maintaining a clear distinction between the capabilities actually implemented and potential future production-oriented extensions.

The following chapters provide a detailed technical examination of the solution architecture, PostgreSQL data environment, centralized analytical View, Tableau analytical components, KPI framework, worksheets, visualizations, dashboard interactions, executive dashboards, implementation methodology, and design decisions that support the Healthcare Analytics Platform.

Together, these components demonstrate the progression from structured healthcare data to interactive analytical information within a documented and maintainable Business Intelligence solution.

Healthcare Analytics Platform

Technical Documentation

CHAPTER 2

Solution Architecture

" Designing a scalable Business Intelligence architecture that transforms healthcare data into consistent, reusable, and decision-ready analytical information."

2.1 Architecture Overview

The **Healthcare Analytics Platform** was designed using a layered Business Intelligence architecture in which data management, data preparation, analytical logic, visualization, interaction, presentation, and documentation operate as logically separated but integrated responsibilities. This architectural approach establishes a structured flow from the underlying healthcare dataset to interactive executive dashboards and decision-oriented analytical information while maintaining clear responsibility boundaries between PostgreSQL, Tableau, and the documentation environment.

At the foundation of the architecture, PostgreSQL provides the relational data environment through `healthcare_db`. The database stores and organizes the healthcare dataset used throughout the analytical platform, including patient demographics, medical conditions, admissions, discharge information, insurance providers, billing amounts, medications, test results, and other healthcare-related attributes.

Within PostgreSQL, the final portfolio architecture uses the centralized analytical View `vw_healthcare_master` as the standardized data-preparation structure for the five executive dashboards. This View organizes the healthcare attributes required across financial, demographic, clinical, medical-condition, comparative, insurance, and operational analysis and establishes a consistent interface between the PostgreSQL data environment and Tableau Desktop.

Tableau consumes `vw_healthcare_master` through `DS_vw_healthcare_master (healthcare_db)`, providing a common analytical data foundation for the final dashboard environment. Tableau then extends this centralized structure through Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage calculations, time-based calculations, standardized KPIs, and other analytical mechanisms.

This division of responsibilities allows PostgreSQL to provide the centralized data environment and standardized analytical structure while Tableau implements analytical logic that depends on calculation granularity, visualization context, comparison, segmentation, ranking, and user-controlled analytical behavior.

The Visualization Layer is composed of modular Tableau worksheets, charts, and KPI components that transform analytical measures and classifications into understandable visual representations. These components can be developed, validated, formatted, and maintained independently before being assembled into executive dashboards according to specific analytical objectives.

At the Presentation Layer, the platform integrates these analytical components into five executive dashboards: **Comparative Analytics Dashboard**, **Revenue Distribution Dashboard**, **Patient Demographics Dashboard**, **Clinical & Medical Conditions Dashboard**, and **Healthcare Operations Dashboard**. Together, these dashboards provide complementary comparative, financial, demographic, clinical, medical-condition, and operational perspectives while sharing the same centralized PostgreSQL analytical foundation.

The architecture also includes an Interaction Layer implemented through interactive filters, Dashboard Filter Actions, and Highlight Actions. These mechanisms support controlled analytical exploration by allowing users to refine analytical context, investigate selected information, and examine relationships across related visualizations.

Finally, the Documentation Layer records the solution architecture, analytical methodology, implementation decisions, project scope, design standards, and supporting technical information required to explain and maintain the portfolio solution.

Collectively, these architectural responsibilities establish a modular Business Intelligence environment in which data management, data preparation, analytical logic, visualization, interaction, presentation, and documentation remain logically separated while operating as an integrated analytical solution. This structure supports analytical consistency, modularity, reusability, maintainability, and future analytical extension.

Architectural Flow

The end-to-end architectural flow of the final Healthcare Analytics Platform can be summarized as follows:

1. Healthcare Data → PostgreSQL Database

The healthcare dataset is stored and organized within the PostgreSQL database `healthcare_db`, which provides the centralized relational data environment for the analytical solution.

2. PostgreSQL Database → `vw_healthcare_master`

The `vw_healthcare_master` View provides the centralized analytical structure used by the final portfolio implementation. It consolidates the healthcare attributes required for financial, demographic, clinical, comparative, and operational analysis.

3. vw_healthcare_master → Tableau Data Source

Tableau Desktop connects to the PostgreSQL View through DS_vw_healthcare_master (healthcare_db), establishing a consistent analytical data source for the five final executive dashboards.

4. Tableau Data Source → Analytical Logic

The standardized data source is extended within Tableau through Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking logic, segmentation techniques, and other analytical constructs.

5. Analytical Logic → KPIs & Worksheets

Reusable analytical measures are transformed into KPI cards, charts, trend analyses, distributions, comparisons, rankings, and other visualization components.

6. KPIs & Worksheets → Interactive Executive Dashboards

Validated visual components are assembled into the five final portfolio dashboards:

- Comparative Analytics Dashboard
- Revenue Distribution Dashboard
- Patient Demographics Dashboard
- Clinical & Medical Conditions Dashboard
- Healthcare Operations Dashboard

7. Executive Dashboards → Interactive Analytical Exploration

Filters, Filter Actions, Highlight Actions, and other dashboard interactions allow users to investigate relationships and analyze selected healthcare dimensions dynamically.

8. Interactive Analytical Exploration → Decision Support

The resulting analytical environment transforms healthcare data into financial, demographic, clinical, comparative, and operational insights that support data-driven analysis and decision-making.

Business Insight

A well-structured Business Intelligence architecture depends on clearly defined responsibilities across its analytical components. Separating data management, data preparation, analytical logic, visualization, interaction, presentation, and documentation while maintaining integration between these responsibilities reduces unnecessary dependencies and creates an analytical environment that is easier to understand, validate, maintain, and extend as analytical requirements evolve.

2.2 Architectural Layers

The **Healthcare Analytics Platform** is organized into seven logical architectural layers, each responsible for a specific part of the Business Intelligence workflow. These layers are the **Data Layer**, **Data Preparation Layer**, **Analytics Layer**, **Visualization Layer**, **Interaction Layer**, **Presentation Layer**, and **Documentation Layer**.

This layered structure follows a separation-of-responsibilities approach in which each component is assigned a clearly defined analytical purpose while remaining integrated with the broader solution. PostgreSQL provides the structured data environment and centralized analytical preparation, Tableau Desktop provides analytical calculations, visualization, interaction, and dashboard presentation, and the documentation environment records the architecture, methodology, implementation decisions, project scope, and supporting technical information.

The architectural layers represent **logical responsibilities rather than independently deployed physical systems**. Several responsibilities are therefore implemented within the same technology. For example, Tableau Desktop supports the Analytics, Visualization, Interaction, and Presentation Layers, while PostgreSQL supports the Data and Data Preparation Layers. Separating these responsibilities conceptually improves the organization, validation, maintainability, and documentation of the solution while preserving a controlled flow of information between components.

The seven logical architectural layers are described below.

2.2.1 Data Layer

The **Data Layer** forms the foundation of the Healthcare Analytics Platform and is implemented through the PostgreSQL database `healthcare_db`. Its primary responsibility is to store, organize, and provide structured relational access to the healthcare dataset used throughout the analytical solution.

The database contains healthcare information related to patient demographics, medical conditions, admission and discharge activity, doctors, hospitals, insurance providers, billing amounts, medications, test results, room information, admission types, and other attributes required by the analytical environment.

Within the architecture, the Data Layer focuses on the structured storage and availability of the underlying healthcare information. It provides the relational foundation from which SQL queries and the centralized analytical View `vw_healthcare_master` can access the fields required for downstream analysis.

Maintaining the healthcare dataset within PostgreSQL establishes a centralized data environment rather than embedding the underlying data structure independently within Tableau worksheets or dashboards. This creates a clear architectural boundary between the storage and organization of source information and the analytical preparation performed in the subsequent Data Preparation Layer.

Primary responsibilities:

- Storage and organization of the healthcare dataset within healthcare_db.
- Structured relational access to healthcare information.
- Preservation of the underlying healthcare data used by the analytical solution.
- Support for SQL-based querying of the healthcare dataset.
- Provision of the centralized data foundation required by vw_healthcare_master and downstream Tableau analytical components.

2.2.2 Data Preparation Layer

The **Data Preparation Layer** is implemented through the centralized PostgreSQL analytical View vw_healthcare_master. This layer sits between the underlying healthcare data stored within healthcare_db and the Tableau analytical environment used by the five final executive dashboards.

Its primary purpose is to provide a standardized analytical structure containing the healthcare attributes required for downstream analysis. Rather than creating separate PostgreSQL Views for each final dashboard or analytical domain, the final portfolio architecture uses vw_healthcare_master as the common data-preparation interface between PostgreSQL and Tableau.

This centralized approach establishes a consistent analytical foundation for financial, demographic, clinical, medical-condition, comparative, insurance, and operational analysis while reducing unnecessary data-source fragmentation across the final dashboard environment.

The Data Preparation Layer intentionally focuses on organizing and exposing the structured healthcare information required by Tableau. Analytical logic that depends on visualization context, calculation granularity, ranking, segmentation, percentages, time-based analysis, or user-controlled behavior is implemented within the Tableau Analytics Layer rather than unnecessarily embedded in separate PostgreSQL Views.

Tableau consumes vw_healthcare_master through DS_vw_healthcare_master (healthcare_db), creating a clear architectural transition from centralized PostgreSQL data preparation to Tableau-based analytical logic, KPI development, visualization, and dashboard interaction.

Primary responsibilities:

- Provide the centralized analytical structure through vw_healthcare_master.
- Standardize the healthcare attributes required by the five final executive dashboards.
- Establish a consistent interface between PostgreSQL and Tableau Desktop.
- Reduce unnecessary fragmentation across final Tableau data sources.
- Support downstream Tableau analytical logic without duplicating dashboard-specific analytical calculations in PostgreSQL.

2.2.3 Analytics Layer

The **Analytics Layer** extends the centralized analytical structure provided by vw_healthcare_master within Tableau Desktop. Tableau accesses this structure through DS_vw_healthcare_master (healthcare_db) and applies the analytical logic required to transform prepared healthcare data into business-oriented measures, classifications, comparisons, and performance indicators.

This layer uses Tableau Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage calculations, Running Totals, Moving Averages, and other analytical mechanisms to support requirements that extend beyond the standardized structure provided directly by PostgreSQL.

The Analytics Layer is particularly important because it separates analytical definitions from their visual representation. KPIs, derived measures, classifications, comparative calculations, and other analytical components can therefore be developed and validated before being incorporated into worksheets and executive dashboards.

Analytical logic is implemented within Tableau when it depends on calculation granularity, visualization context, ranking, segmentation, comparison, time-based behavior, or user-controlled analytical interaction. This allows the centralized PostgreSQL View to remain focused on standardized data preparation while Tableau manages analytical behavior appropriate to the visualization and dashboard environment.

Primary responsibilities:

- Development of derived analytical measures and classifications.
- KPI calculation and standardization.
- Implementation of Level of Detail (LOD) Expressions.
- Implementation of Table Calculations, including ranking, percentages, Running Totals, and Moving Averages.
- Development of Sets and Parameters for segmentation and dynamic analytical behavior.
- Support for comparative analysis and Selected versus Others analytical scenarios.
- Preparation of reusable analytical logic for downstream worksheets and dashboards.

2.2.4 Visualization Layer

The **Visualization Layer** transforms analytical measures, classifications, and calculations into visual components that users can interpret and explore efficiently. Within the Healthcare Analytics Platform, this layer is implemented primarily through Tableau worksheets, including KPI cards, trend visualizations, distributions, comparative charts, rankings, and other analytical views.

Each visualization is designed around a defined analytical purpose rather than functioning as an isolated graphic. Individual worksheets can be developed, validated, formatted, and refined independently before being incorporated into an executive dashboard, supporting a modular visualization-development approach.

The Visualization Layer also applies consistent presentation standards across analytical components. Typography, titles, labels, number formatting, KPI presentation, spacing, tooltips, and other visual conventions are coordinated to improve readability and maintain a cohesive analytical experience across the platform.

Importantly, this layer focuses on the **visual representation of analytical information**. The underlying analytical calculations are primarily defined within the Analytics Layer, while cross-visualization behavior and user interaction are handled separately within the Interaction Layer.

Primary responsibilities:

- Transform analytical measures and classifications into visual information.
- Develop modular Tableau worksheets and analytical charts.
- Present standardized KPI components.
- Visualize trends, distributions, comparisons, rankings, and other analytical relationships.
- Apply consistent formatting, titles, labels, and visual hierarchy.
- Design informative tooltips and supporting visual context.
- Prepare validated visualization components for dashboard assembly.

2.2.5 Interaction Layer

The **Interaction Layer** provides the mechanisms through which users explore and investigate information within the executive dashboards. Rather than limiting the platform to static analytical reporting, Tableau interaction capabilities allow users to refine analytical context, investigate selected information, and examine relationships across related visualizations.

The Healthcare Analytics Platform uses interactive filters, Dashboard Filter Actions, and Highlight Actions to coordinate analytical behavior across dashboard components. These interaction mechanisms are configured according to the analytical purpose of each dashboard and the relationships that users are expected to investigate.

Dashboard Filter Actions support focused analytical investigation by using selections in one visualization to filter related target worksheets. This allows users to narrow the analytical context and examine a selected category, period, demographic group, medical condition, insurance provider, or other relevant analytical dimension across coordinated visualizations.

Highlight Actions provide a different analytical behavior. Instead of removing unrelated information from the view, they emphasize marks associated with the selected analytical value while preserving the surrounding context. This supports relationship-oriented exploration and contextual comparison without unnecessarily restricting the visible dataset.

The distinction between filtering and highlighting allows the platform to support both focused investigation and contextual analytical exploration while maintaining controlled and predictable dashboard behavior.

Primary responsibilities:

- Provide interactive filtering within executive dashboards.
- Coordinate analytical behavior through Dashboard Filter Actions.
- Support relationship-oriented exploration through Highlight Actions.
- Allow users to refine analytical context through selected dimensions and categories.
- Coordinate interactions between related dashboard worksheets.
- Preserve broader analytical context when highlighting is more appropriate than filtering.
- Support controlled and predictable analytical exploration.

2.2.6 Presentation Layer

The **Presentation Layer** represents the final user-facing analytical layer of the Healthcare Analytics Platform. It integrates validated KPIs, worksheets, filters, dashboard interactions, titles, supporting text, and layout structures into coordinated executive dashboard experiences.

The platform organizes its principal analytical capabilities into five executive dashboards:

- **Comparative Analytics Dashboard**
- **Revenue Distribution Dashboard**
- **Patient Demographics Dashboard**
- **Clinical & Medical Conditions Dashboard**
- **Healthcare Operations Dashboard**

Each dashboard addresses a distinct analytical perspective while following common visualization, formatting, and interaction principles. This provides a consistent analytical experience as users move between comparative, financial, demographic, clinical, medical-condition, and operational perspectives.

The Presentation Layer is responsible for how previously developed analytical components are organized and communicated to the user. Analytical calculations remain within the Analytics Layer, individual visual representations are developed within the Visualization Layer, and interactive behavior is configured within the Interaction Layer. The Presentation Layer brings these components together into structured dashboards designed for analytical interpretation and exploration.

Primary responsibilities:

- Assemble validated analytical components into executive dashboards.
- Organize KPIs, charts, filters, interactions, titles, and supporting information within coordinated layouts.
- Establish dashboard-level visual hierarchy and information organization.
- Maintain consistent presentation principles across the five executive dashboards.

- Support clear interpretation of comparative, financial, demographic, clinical, medical-condition, and operational information.
- Provide the final user-facing analytical experience of the Tableau solution.

2.2.7 Documentation Layer

The **Documentation Layer** records and communicates the architecture, analytical methodology, implementation decisions, project scope, design standards, and supporting technical information associated with the Healthcare Analytics Platform.

This layer provides a structured explanation of how the PostgreSQL and Tableau components are organized and how data preparation, analytical logic, visualization, interaction, and dashboard presentation responsibilities are distributed across the solution. It also documents the principal analytical techniques, dashboard components, architectural decisions, and development practices used throughout the project.

Microsoft Word is used as the primary documentation environment for preparing the professional technical documentation, while PDF represents the intended publication format for the completed portfolio document. This documentation provides supporting context for understanding, reviewing, presenting, and maintaining the analytical solution.

The Documentation Layer does not perform data processing or analytical calculations. Instead, it provides the technical and methodological record required to explain how the analytical platform was designed and implemented.

Primary responsibilities:

- Document the overall Business Intelligence architecture.
- Record analytical methodology and implementation decisions.
- Explain the responsibilities of PostgreSQL and Tableau within the solution.
- Document dashboard structures, analytical techniques, and interaction mechanisms.
- Maintain supporting technical information and project scope.
- Prepare professional technical documentation for portfolio presentation and publication.

Layer	Primary Technology / Component	Core Responsibility	Primary Output
Data Layer	PostgreSQL – healthcare_db	Store and organize the healthcare dataset	Structured healthcare data
Data Preparation Layer	PostgreSQL View – vw_healthcare_master	Provide a centralized and standardized analytical structure for the final portfolio dashboards	Master analytical dataset
Analytics Layer	Tableau – DS_vw_healthcare_master (healthcare_db) + Tableau	Extend the master dataset with analytical and business logic	KPIs and derived analytical metrics

	Calculations		
Visualization Layer	Tableau Worksheets	Convert analytical measures into visual components	Charts, KPIs and analytical views
Interaction Layer	Tableau Dashboard Actions & Filters	Enable interactive analytical exploration	Filters and cross-view interactions
Presentation Layer	Tableau Dashboards	Integrate analytical components for end users	Five executive analytical dashboards
Documentation Layer	Microsoft Word / PDF	Document the solution architecture, analytical methodology, implementation decisions, project scope, and supporting technical information	Professional technical documentation and portfolio publication

Table 2.1 – Architectural Responsibility Matrix**Technical Note**

The architectural layers represent logical responsibilities rather than independently deployed physical systems. Several layers are implemented within the same technology. For example, PostgreSQL supports the Data and Data Preparation Layers, while Tableau Desktop supports the Analytics, Visualization, Interaction, and Presentation Layers. The Documentation Layer is maintained through the project's technical documentation environment.

The layered model is therefore used to separate responsibilities conceptually rather than to imply seven independent technology platforms. This separation improves architectural clarity, organization, validation, maintainability, and documentation while preserving integration across the complete analytical solution.

Business Insight

A layered Business Intelligence architecture provides value by establishing clear responsibilities across data management, data preparation, analytical logic, visualization, interaction, presentation, and documentation. When these responsibilities are logically separated while remaining integrated within the same analytical solution, changes can be evaluated and implemented with less unnecessary impact on unrelated components.

This approach supports greater consistency, maintainability, traceability, and future analytical extension while making the overall solution easier to understand, validate, communicate, and evolve as analytical requirements change.

2.3 Data Layer – PostgreSQL

The **Data Layer** represents the relational foundation of the Healthcare Analytics Platform and is implemented using PostgreSQL. Within the solution architecture, PostgreSQL provides the centralized database environment used to store, organize, and query the healthcare data required by the analytical solution.

The project uses the PostgreSQL database `healthcare_db` as its primary data environment. Healthcare information stored within this database provides the structured relational foundation from which the centralized analytical View `vw_healthcare_master` is created and subsequently consumed by Tableau Desktop through `DS_vw_healthcare_master` (`healthcare_db`).

The healthcare dataset contains information across multiple analytical dimensions, including patient demographics, medical conditions, admission and discharge activity, physicians, hospitals, insurance providers, billing amounts, medications, test results, room information, and admission types. These attributes provide the multidimensional foundation required to support comparative, financial, demographic, clinical, medical-condition, insurance, and operational analysis.

Within the layered architecture, PostgreSQL separates relational data management from the advanced analytical, visualization, interaction, and presentation responsibilities implemented within Tableau Desktop. The Data Layer therefore focuses on maintaining the structured healthcare information required by the solution, while the subsequent Data Preparation Layer uses `vw_healthcare_master` to provide the standardized analytical structure consumed by Tableau.

2.3.1 Core Healthcare Dataset

The core healthcare dataset stored within `healthcare_db` contains the fields required to support the analytical requirements of the Healthcare Analytics Platform. These attributes provide the underlying healthcare information that is subsequently organized and exposed to Tableau through the centralized analytical View `vw_healthcare_master`.

The dataset includes patient, demographic, clinical, financial, admission, discharge, insurance, medication, test-result, and hospitalization-related attributes used across the analytical environment.

Table 2.2 summarizes the principal fields and their analytical purpose.

Field	Analytical Purpose
Patient Name	Patient-level identification used in patient-count calculations and analytical segmentation
Age	Patient demographic and age-group analysis
Gender	Gender distribution and demographic analysis
Blood Type	Patient demographic and clinical categorization
Medical Condition	Analysis of patient populations and healthcare conditions
Date of Admission	Time-series, admission trend, and monthly analysis
Doctor	Physician-related healthcare information

Hospital	Healthcare facility information
Insurance Provider	Insurance segmentation and revenue analysis
Billing Amount	Financial, revenue, and billing calculations
Room Number	Hospitalization-related information
Discharge Date	Discharge activity and hospitalization analysis
Medication	Medication distribution and clinical analysis
Test Results	Laboratory result analysis and clinical categorization

Table 2.2 – Core Healthcare Dataset Fields

2.3.2 Analytical Date Attributes

Several analytical attributes used throughout the Healthcare Analytics Platform are derived from the underlying admission and discharge dates to support time-based analysis. These attributes provide consistent temporal dimensions that allow Tableau to organize and analyze healthcare activity at monthly and chronological levels.

Examples of analytical date attributes include:

- **Admission Month**
- **Admission Month Name**
- **Discharge Month**
- **Billing-related monthly periods**
- **Test-result analytical periods**

These temporal attributes support time-based visualizations within the Healthcare Operations Dashboard, including **Monthly Admissions Trend**, **Monthly Discharge Throughput**, **Monthly Billing Trend**, **Monthly Admission Type Mix**, **Insurance Provider Revenue Trend**, and **Monthly Test Results Trend**.

By providing consistent temporal references for admission, discharge, billing, insurance, and test-result analysis, these attributes support chronological comparison and trend analysis across the operational analytical environment.

2.3.3 Master Analytical View

The centralized analytical structure developed within the PostgreSQL environment for the final Healthcare Analytics Platform is:

vw_healthcare_master

This PostgreSQL View serves as the **centralized analytical source for all five executive dashboards** included in the final portfolio implementation. It provides a standardized representation of the healthcare data stored within healthcare_db and establishes the common PostgreSQL analytical foundation consumed by Tableau Desktop through DS_vw_healthcare_master (healthcare_db).

The View exposes healthcare attributes required across comparative, financial, demographic, clinical, medical-condition, insurance, and operational analysis, allowing the final Tableau environment to operate from a consistent PostgreSQL analytical structure.

patient_name
age
gender
blood_type
medical_condition
date_of_admission
admission_month
admission_month_name
doctor
hospital
insurance_provider
billing_amount
room_number
admission_type
discharge_date
medication
test_results

The purpose of vw_healthcare_master is to provide Tableau with a common and reusable analytical structure containing the healthcare attributes required by the final portfolio dashboards. Centralizing these attributes within a single PostgreSQL View reduces unnecessary source fragmentation and avoids requiring separate dashboard-specific Views for the final implementation.

Within Tableau Desktop, this View is consumed through DS_vw_healthcare_master (healthcare_db), where the standardized PostgreSQL data foundation is extended through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage calculations, time-based calculations, KPIs, and other analytical mechanisms required by the dashboard environment.

2.3.4 Supporting Analytical Views

During the earlier development and learning stages of the Healthcare Analytics Platform, additional PostgreSQL Views were created to support SQL practice, analytical experimentation, and initial Tableau prototyping.

Two examples developed during this phase were:

- **vw_revenue_by_condition** — created to support the exploration of aggregated billing and revenue information by medical condition.
- **vw_revenue_by_gender** — created to support the exploration of aggregated billing and revenue information by gender.

These specialized Views were useful for understanding PostgreSQL aggregation techniques and the workflow for consuming database-level analytical structures within Tableau. However, they represent **development and learning artifacts rather than data sources used by the five final executive dashboards**.

As the project evolved, the final portfolio architecture was consolidated around the centralized PostgreSQL View `vw_healthcare_master`, consumed within Tableau through `DS_vw_healthcare_master (healthcare_db)`. Advanced calculations, KPIs, ranking, segmentation, comparative logic, percentages, time-based calculations, and interactive analytical behavior were then implemented primarily within Tableau.

This architectural evolution reduced unnecessary source fragmentation and established a common analytical foundation for the final dashboard environment while preserving the earlier specialized Views as evidence of the SQL and Tableau learning process.

2.3.5 PostgreSQL-to-Tableau Data Flow

The PostgreSQL environment acts as the centralized relational data provider for Tableau Desktop. In the final Healthcare Analytics Platform architecture, the flow between the database and analytical environments is standardized around the master analytical View `vw_healthcare_master`.

The final PostgreSQL-to-Tableau data flow can be summarized as:

Healthcare Dataset → PostgreSQL Database `healthcare_db` → Master Analytical View `vw_healthcare_master` → Tableau Data Source `DS_vw_healthcare_master (healthcare_db)` → Tableau Analytical Logic → KPIs & Worksheets → Five Executive Dashboards

This architecture establishes a clear boundary between database-oriented data management and preparation and Tableau-oriented analytical development. PostgreSQL stores and organizes the healthcare dataset and provides the standardized analytical structure through `vw_healthcare_master`, while Tableau consumes this structure through `DS_vw_healthcare_master (healthcare_db)`.

Once the centralized analytical source is available within Tableau, it is extended through Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage calculations, time-based calculations, KPIs, and other analytical mechanisms required by the dashboard environment.

The resulting analytical components are transformed into worksheets and subsequently assembled into the five executive dashboards. This centralized flow reduces unnecessary source fragmentation and

ensures that the final dashboard environment operates from a common PostgreSQL analytical foundation.

2.3.6 Role of PostgreSQL in the Architecture

Within the overall Business Intelligence architecture, PostgreSQL performs four principal functions:

- **Centralized Data Storage** — Maintains the healthcare dataset within `healthcare_db`, providing the structured relational foundation for the analytical solution.
- **SQL Query Processing** — Provides the database engine used to query, organize, and prepare healthcare information for analytical consumption.
- **Master Analytical View Management** — Hosts and maintains `vw_healthcare_master`, the centralized PostgreSQL analytical View consumed by Tableau through `DS_vw_healthcare_master` (`healthcare_db`) for the five final executive dashboards.
- **Data Foundation for Business Intelligence** — Provides the consistent relational and analytical data structure required to support downstream Tableau calculations, KPIs, worksheets, visualizations, interactions, and dashboards.

This division of responsibilities allows PostgreSQL and Tableau to remain logically separated while operating as integrated components of the same analytical platform. PostgreSQL focuses on relational data management and centralized analytical preparation, while Tableau extends the prepared data foundation with the analytical, visualization, interaction, and presentation capabilities required by the final dashboard environment.

Technical Note

PostgreSQL is used primarily as the relational data-management and centralized data-preparation environment of the Healthcare Analytics Platform. Through `healthcare_db` and `vw_healthcare_master`, PostgreSQL provides the structured analytical foundation consumed by Tableau Desktop. Advanced analytical calculations, visualization-specific logic, and interactive behavior remain primarily within Tableau, allowing each technology to perform the responsibilities most appropriate to its role within the architecture.

Business Insight

A centralized relational data environment provides a consistent foundation for Business Intelligence analysis. By maintaining the healthcare dataset within `healthcare_db` and providing the standardized analytical structure through `vw_healthcare_master`, the platform reduces unnecessary source fragmentation and establishes a common PostgreSQL foundation for Tableau. This separation allows Tableau to focus on advanced analytical calculations, visualization, interaction, and dashboard presentation while preserving a clear and maintainable data architecture.

2.4 Data Preparation Layer – Master Analytical View

The **Data Preparation Layer** provides the analytical bridge between the PostgreSQL data environment and Tableau Desktop. For the final Healthcare Analytics Platform portfolio implementation, this layer is centered on a single standardized PostgreSQL View: **vw_healthcare_master**.

The purpose of **vw_healthcare_master** is to provide Tableau with a centralized analytical structure containing the healthcare attributes required by the five final executive dashboards. Rather than relying on multiple specialized SQL Views for different reporting requirements, the final architecture uses this master View as the common PostgreSQL source for financial, demographic, clinical, comparative, and operational analysis.

Within Tableau Desktop, **vw_healthcare_master** is exposed through the data source **DS_vw_healthcare_master (healthcare_db)**. This establishes a consistent analytical foundation from which Tableau Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, KPIs, worksheets, and dashboard interactions can be developed.

This centralized approach simplifies the final portfolio architecture by reducing source fragmentation and maintaining a clear separation of responsibilities. PostgreSQL provides the standardized analytical dataset, while Tableau extends that dataset with the advanced analytical and interactive logic required by the final dashboards.

SQL View / View Type	Primary Purpose	Analytical Domain	Consumed By
vw_healthcare_master	Final Analytical Source	Provides the standardized multi-domain healthcare dataset used for analysis in Tableau	Yes – Primary source for all five final dashboards
vw_revenue_by_condition	Development / Learning View	Supports exploration of aggregated revenue by medical condition	No – Used during earlier Tableau development and learning
vw_revenue_by_gender	Development / Learning View	Supports exploration of aggregated revenue by gender	No – Used during earlier Tableau development and learning

Table 2.3 – SQL View Responsibility Matrix

2.4.1 Master Analytical View – vw_healthcare_master

The PostgreSQL View `vw_healthcare_master` represents the **centralized analytical data-preparation structure** used by the final Healthcare Analytics Platform. It provides a standardized relational interface between the healthcare data stored within `healthcare_db` and the analytical environment implemented in Tableau Desktop.

The View consolidates the healthcare attributes required across the five final executive dashboards into a common analytical structure. Tableau consumes this View through the data source `DS_vw_healthcare_master (healthcare_db)`, ensuring that the final dashboard environment operates from a consistent PostgreSQL foundation rather than from multiple dashboard-specific SQL Views.

The principal purpose of `vw_healthcare_master` is to standardize and expose the healthcare data required for downstream analysis while maintaining a clear separation between database-oriented data preparation and Tableau-oriented analytical logic.

Advanced analytical requirements—including Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage calculations, Running Totals, Moving Averages, KPIs, and interactive analytical behavior—are implemented primarily within Tableau rather than being distributed across multiple specialized PostgreSQL Views.

This centralized-source strategy improves architectural consistency, reduces unnecessary source fragmentation, and provides a reusable analytical foundation for the five final dashboards.

2.4.2 Development and Learning Views

During the earlier development, learning, and experimentation stages of the Healthcare Analytics Platform, additional PostgreSQL Views were created to practice SQL aggregation techniques and explore alternative approaches for preparing analytical data for Tableau.

Two specialized Views developed during this phase were:

- **vw_revenue_by_condition** — used to explore aggregated billing and revenue analysis by medical condition.
- **vw_revenue_by_gender** — used to explore aggregated billing and revenue analysis by gender.

These Views supported SQL practice, analytical experimentation, and early Tableau prototyping. They provided useful experience in creating reusable database-level analytical structures and connecting PostgreSQL-prepared results to Tableau.

However, these specialized Views are **not data sources for the five final executive dashboards**. As the project matured, the final portfolio architecture was consolidated around the centralized View `vw_healthcare_master`, consumed through `DS_vw_healthcare_master (healthcare_db)`.

The earlier Views are therefore retained as **development and learning artifacts** that document the evolution of the solution rather than as components of the final dashboard data architecture.

2.4.3 Master Analytical View Design Principles

The design of the master analytical View `vw_healthcare_master` follows a set of principles intended to provide a consistent, reusable, and maintainable analytical foundation for the final Healthcare Analytics Platform. These principles define how the centralized PostgreSQL analytical source supports the five final executive dashboards while maintaining a clear separation between database-oriented data preparation and Tableau-oriented analytical logic.

Reusability — `vw_healthcare_master` provides a common PostgreSQL analytical structure that supports multiple analytical domains and dashboard components without requiring separate specialized Views for each final dashboard.

Clear Analytical Purpose — The master View provides a clearly defined analytical foundation containing the healthcare attributes required for comparative, financial, demographic, clinical, medical-condition, insurance, and operational analysis.

Simplified Tableau Consumption — Tableau consumes the standardized PostgreSQL analytical structure through `DS_vw_healthcare_master` (`healthcare_db`), allowing the five final dashboards to operate from a common data source and reducing unnecessary source-level complexity.

Consistent Field Definitions — Common healthcare attributes are exposed through the same master analytical source, helping maintain consistent field usage across Tableau calculations, KPIs, worksheets, visualizations, and dashboards.

Separation of Responsibilities — PostgreSQL provides the standardized analytical data structure through `vw_healthcare_master`, while Tableau implements the advanced analytical and presentation logic required by the final dashboard environment, including Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage calculations, time-based calculations, KPIs, visualization logic, and interactive analytical behavior.

2.4.4 Transformation Responsibilities

One of the key architectural decisions within the Healthcare Analytics Platform is determining where data-preparation and analytical responsibilities should be implemented. The final portfolio architecture separates responsibilities between PostgreSQL and Tableau according to the purpose, reusability, and analytical context of each operation.

PostgreSQL provides the centralized and standardized analytical data foundation through `vw_healthcare_master`, while Tableau consumes this structure through `DS_vw_healthcare_master` (`healthcare_db`) and extends it with the calculations, segmentation, comparisons, KPIs, and interactive analytical logic required by the five executive dashboards.

The Data Preparation Layer is therefore responsible primarily for establishing a consistent PostgreSQL analytical foundation for Tableau.

SQL / PostgreSQL responsibilities

- Selecting and organizing source fields required for analytical consumption.
- Providing the centralized analytical structure through vw_healthcare_master.
- Standardizing commonly required healthcare attributes exposed to Tableau.
- Maintaining consistent field availability across analytical domains.
- Supporting database-level transformations or aggregations when appropriate.
- Providing a stable PostgreSQL source for DS_vw_healthcare_master (healthcare_db).

The **Tableau Analytics Layer**, by contrast, extends the standardized PostgreSQL source with advanced analytical and visualization-dependent logic such as:

Tableau responsibilities

- Calculated Fields
- Level of Detail (LOD) Expressions
- Table Calculations
- Ranking
- Running Totals
- Moving Averages
- Dynamic Calculations
- Sets and Parameters
- Segmentation and comparative calculations
- Visualization-specific percentages and comparisons
- KPI calculations and presentation logic

This distinction demonstrates that the architecture does not attempt to force all data-preparation and analytical logic into a single technology. PostgreSQL provides the centralized and standardized analytical data foundation, while Tableau provides the flexible analytical, calculation, visualization, and interaction capabilities required by the final portfolio dashboards. This separation establishes a clear division of responsibilities and accurately reflects the implementation used in the final Healthcare Analytics Platform.

2.4.5 Master Analytical View and Tableau Data Source

The integration between PostgreSQL and Tableau Desktop is established through the centralized analytical View vw_healthcare_master, which serves as the **centralized PostgreSQL analytical source for all five executive dashboards** included in the final portfolio.

Within Tableau Desktop, this PostgreSQL View is consumed through the data source DS_vw_healthcare_master (healthcare_db). This data source provides Tableau with access to the standardized healthcare attributes required for comparative, financial, demographic, clinical, medical-condition, insurance, and operational analysis.

The relationship between the database and analytical environments can therefore be represented as:

PostgreSQL Database – healthcare_db

↓

Master Analytical View – vw_healthcare_master

↓

Tableau Data Source – DS_vw_healthcare_master (healthcare_db)

↓

Tableau Analytical Logic

↓

KPIs & Worksheets

↓

Five Executive Dashboards

Once the centralized data source is available within Tableau, it is extended through Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage calculations, Running Totals, Moving Averages, KPIs, and other analytical mechanisms required by the dashboard environment.

This centralized Tableau Data Source provides an important consistency benefit. KPIs, worksheets, calculations, visualizations, and dashboard components operate from the same underlying PostgreSQL analytical foundation, reducing unnecessary source fragmentation and helping maintain consistent field usage throughout the final reporting environment.

The five final executive dashboards consuming this analytical foundation are:

- **Comparative Analytics Dashboard**
- **Revenue Distribution Dashboard**
- **Patient Demographics Dashboard**
- **Clinical & Medical Conditions Dashboard**
- **Healthcare Operations Dashboard**

Although additional specialized PostgreSQL Views were developed during earlier learning, experimentation, and prototyping stages, as documented in **Section 2.4.2 – Development and Learning Views**, these Views are not used as data sources for the five final executive dashboards. The final portfolio implementation is standardized around vw_healthcare_master and DS_vw_healthcare_master (healthcare_db).

2.4.6 Benefits of the Centralized Analytical Source

The use of vw_healthcare_master as the centralized PostgreSQL analytical source provides several architectural and analytical benefits to the final Healthcare Analytics Platform. By establishing a common data foundation for all five executive dashboards, the architecture reduces unnecessary source fragmentation while maintaining consistency between PostgreSQL and Tableau.

The primary benefits include:

Centralized Analytical Foundation — All five final executive dashboards share `vw_healthcare_master` as their centralized PostgreSQL analytical source, providing a common foundation for comparative, financial, demographic, clinical, medical-condition, insurance, and operational analysis.

Consistent Tableau Integration — The master View is consumed through `DS_vw_healthcare_master` (`healthcare_db`), providing Tableau with a standardized connection to the healthcare analytical dataset.

Reduced Source Fragmentation — The final portfolio architecture does not require a separate specialized PostgreSQL View for each dashboard or analytical requirement. This reduces unnecessary duplication and simplifies the data flow between PostgreSQL and Tableau.

Reusable Data Foundation — The same underlying analytical source supports multiple KPIs, worksheets, calculations, visualizations, and dashboards, allowing analytical components to be developed from a shared healthcare data foundation.

Clear Separation of Responsibilities — PostgreSQL provides the centralized and standardized analytical data structure, while Tableau extends that structure through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, segmentation, ranking, percentages, time-based calculations, KPIs, visualization logic, and interactive analytical behavior.

Improved Maintainability — A centralized analytical source makes the final architecture easier to understand, validate, and maintain because the five portfolio dashboards share a common PostgreSQL foundation rather than depending on multiple dashboard-specific database Views.

Extensibility for Future Analysis — The centralized architecture provides a foundation that can be extended with additional Tableau calculations, worksheets, or dashboards as new analytical requirements emerge, while preserving the existing PostgreSQL-to-Tableau integration pattern.

The centralized-source strategy represents the mature architecture of the final portfolio implementation. Specialized Views created during earlier learning, experimentation, and prototyping stages remain valuable development artifacts, but the five executive dashboards are standardized around `vw_healthcare_master` and `DS_vw_healthcare_master` (`healthcare_db`).

Technical Note

The centralized PostgreSQL View `vw_healthcare_master` does not replace Tableau calculations. It provides the standardized database-oriented data-preparation foundation consumed through `DS_vw_healthcare_master` (`healthcare_db`), while Tableau remains primarily responsible for advanced analytical and visualization-specific logic such as LOD Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage calculations, Running Totals, Moving Averages, KPIs, dynamic calculations, and dashboard interactions.

Maintaining this distinction helps prevent unnecessary duplication of responsibilities across the architecture and preserves a clear separation between centralized PostgreSQL data preparation and Tableau analytical logic.

Business Insight

A centralized analytical View provides a stable interface between structured healthcare data and Tableau. By standardizing the final portfolio architecture around `vw_healthcare_master` and `DS_vw_healthcare_master` (`healthcare_db`), the Healthcare Analytics Platform provides the five executive dashboards with a consistent analytical foundation while preserving the flexibility required for Tableau-based calculations, visualization, segmentation, comparison, and interactive analysis. This approach reduces unnecessary source fragmentation, strengthens consistency across the reporting environment, and maintains a clear architectural boundary between PostgreSQL data preparation and Tableau analytical development.

2.5 Analytics Layer – Tableau

The **Analytics Layer** represents the stage of the Healthcare Analytics Platform where the standardized healthcare data provided by PostgreSQL is transformed into business-oriented measures, analytical calculations, classifications, comparisons, and reusable metrics. This layer is implemented primarily within Tableau Desktop and operates between the centralized PostgreSQL analytical source and the Visualization Layer.

For the final portfolio implementation, Tableau consumes the PostgreSQL View `vw_healthcare_master` through `DS_vw_healthcare_master` (`healthcare_db`). Tableau extends this standardized analytical foundation through Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage calculations, Running Totals, Moving Averages, KPIs, and other derived analytical mechanisms required by the five executive dashboards.

These analytical constructs transform standardized healthcare attributes into measures and classifications that support comparative, financial, demographic, clinical, medical-condition, insurance, and operational analysis. They provide the analytical logic required to answer specific business questions while preserving a common underlying PostgreSQL data foundation.

A central principle of this layer is the separation between **analytical logic and visual presentation**. Whenever appropriate, analytical calculations are defined and validated independently before being incorporated into worksheets and dashboards. This approach improves consistency by allowing the same analytical definitions to support multiple visual components without unnecessarily recreating business logic for each visualization.

The Analytics Layer therefore serves as the analytical bridge between the standardized PostgreSQL data foundation and business interpretation. PostgreSQL and `vw_healthcare_master` provide the centralized analytical structure consumed by Tableau, while Tableau determines how that information is calculated, compared, segmented, ranked, summarized, and transformed into decision-oriented analytical measures used throughout the final dashboard environment.

2.5.1 Calculated Fields

Calculated Fields are one of the principal analytical mechanisms used throughout the final Healthcare Analytics Platform. They extend the standardized healthcare attributes supplied by the PostgreSQL View `vw_healthcare_master`, consumed in Tableau through `DS_vw_healthcare_master` (`healthcare_db`), with derived measures, dimensions, classifications, and analytical logic required to support specific business and analytical requirements.

Calculated Fields are used when analytical logic is more appropriately managed within Tableau, particularly when calculations support visualization behavior, interactive analysis, segmentation, comparisons, KPI development, percentage calculations, dynamic analysis, or dashboard-specific analytical requirements.

Examples developed and applied throughout the project include calculations supporting:

- **Selected versus Others analysis**
- **Patient segmentation**
- **KPI calculations**
- **Medical-condition analysis**
- **Gender-based analysis**
- **Percentage contributions**
- **Dynamic analytical classifications**
- **Dashboard-specific analytical metrics**

Calculated Fields provide a reusable analytical mechanism between the standardized PostgreSQL source and the visualization components developed in Tableau. Once defined and validated, these calculations can be reused across KPIs, worksheets, and dashboards, helping maintain consistent analytical definitions while avoiding unnecessary duplication of analytical logic.

Within the final portfolio architecture, this approach allows PostgreSQL to provide the centralized and standardized analytical data foundation while Tableau implements the derived analytical logic required by the five executive dashboards.

2.5.2 Level of Detail (LOD) Expressions

Level of Detail (LOD) Expressions provide analytical control over the granularity at which calculations are evaluated within Tableau. They are particularly valuable when a metric must be calculated at a defined level of detail independently of, or in combination with, the dimensions currently displayed within a visualization.

Within the Healthcare Analytics Platform, LOD Expressions support analytical requirements in which patient counts, revenue values, totals, percentages, or other measures must be evaluated at a controlled level of granularity rather than relying exclusively on the dimensions present in a worksheet.

LOD Expressions extend the standardized analytical foundation provided through DS_vw_healthcare_master (healthcare_db) by allowing Tableau to define calculation behavior according to specific analytical requirements. This provides greater flexibility while preserving the centralized PostgreSQL architecture and avoiding the need to create additional database Views for changes in analytical granularity.

A typical conceptual pattern is:

```
{ FIXED [Analytical Dimension] : AGGREGATION([Measure]) }
```

Technical Insight

LOD Expressions allow the analytical granularity of a calculation to be controlled independently from the visual structure of a worksheet. This capability supports reusable and consistent analytical measures across KPIs, worksheets, and dashboard components while maintaining a common underlying data foundation.

Through this approach, LOD Expressions support analytical requirements that cannot always be addressed through standard aggregation alone and help maintain consistent metric definitions across the final dashboard environment.

2.5.3 Table Calculations

Table Calculations extend the analytical capabilities of Tableau by performing calculations over the aggregated results represented within a visualization. Unlike row-level calculations, Table Calculations operate in relation to the structure, partitioning, addressing, and ordering of the Tableau view, making them particularly useful for analytical requirements that depend on relationships between displayed values.

Within the Healthcare Analytics Platform, Table Calculations extend the standardized data foundation provided through `DS_vw_healthcare_master` (`healthcare_db`) by enabling visualization-dependent analytical logic directly within Tableau.

Capabilities practiced and applied throughout the project include:

- **Ranking**
- **Percent of Total**
- **Running Totals**
- **Moving Averages**
- **Comparative Calculations**

These techniques support analytical requirements such as evaluating relative position, measuring contribution to a total, analyzing cumulative performance, smoothing time-based trends, and comparing values across categories or periods.

Because Table Calculations are evaluated in relation to the structure of the Tableau view, their configuration—including **Compute Using, addressing, partitioning, sort order, and visualization context**—must be carefully controlled to ensure that the resulting metrics accurately represent the intended analytical behavior.

Within the final portfolio architecture, Table Calculations complement Calculated Fields and LOD Expressions by providing context-dependent analytical capabilities while continuing to operate from the common analytical foundation supplied through `DS_vw_healthcare_master` (`healthcare_db`).

2.5.4 Sets and Analytical Segmentation

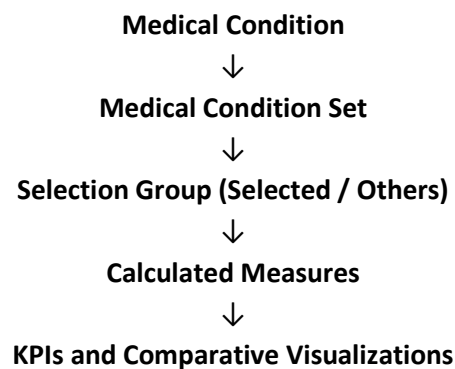
Tableau Sets provide a flexible mechanism for separating members of a dimension into analytical groups based on user selection or predefined analytical criteria. Within the Healthcare Analytics Platform, Sets were particularly important in the development of dynamic comparative and segmentation analysis.

A key implementation developed during the project is the **Medical Condition Set**, which allows selected medical conditions to be analytically separated from the remaining population. Set membership is used by Calculated Fields to support comparative measures and visualizations such as:

- **Selected Revenue**
- **Others Revenue**
- **Selected Patients**
- **Others Patients**
- **Selected vs Others Revenue Comparison**
- **Selected vs Others Patient Comparison**

A calculated classification such as **Selection Group** translates Set membership into business-oriented categories, dynamically classifying the analytical population as **Selected** or **Others**. This classification can then be reused across KPIs, worksheets, and comparative visualizations.

The resulting analytical pattern can be represented conceptually as:



This approach enables dynamic analytical segmentation directly within Tableau while continuing to operate from the common analytical foundation provided through DS_vw_healthcare_master (healthcare_db). It avoids the need to create separate PostgreSQL datasets or database Views for individual comparison scenarios and allows the same segmentation logic to be reused across multiple analytical components.

Within the final portfolio, this analytical pattern supports the comparative analysis developed around selected medical conditions and their relationship to the remaining population, demonstrating how Tableau Sets and Calculated Fields can work together to create reusable, interactive analytical segmentation.

2.5.5 Parameters and Dynamic Analysis

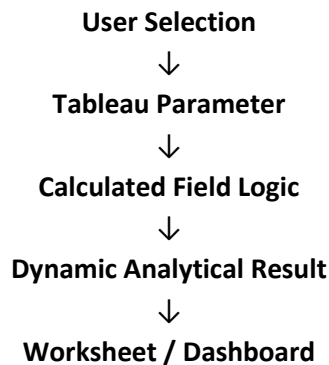
Tableau Parameters provide user-controlled values that can influence calculations, analytical behavior, and visualization logic. Unlike conventional filters, which primarily determine which records or dimension members are included in a view, Parameters can operate independently from the underlying data and provide values that can be referenced by Calculated Fields and other analytical components.

Within the Healthcare Analytics Platform, Parameters were practiced and applied to support dynamic analytical capabilities such as:

- **Dynamic Measure Selection**
- **Dynamic Dimension Selection**
- **Combined Dynamic Measure and Dimension Analysis**
- **Top N Analysis**
- **Analytical Switching**
- **User-Controlled Analytical Selections**

For example, a Parameter can allow the user to select which measure or dimension should drive an analytical view. A corresponding Calculated Field interprets the selected Parameter value and dynamically returns the appropriate measure, dimension, or analytical result.

The relationship can be represented conceptually as:



Parameters therefore complement Calculated Fields and Sets by providing an additional layer of analytical flexibility. While Sets support membership-based segmentation and comparative analysis, Parameters provide user-controlled values that can modify how calculations and visualizations behave.

This approach enables flexible analytical experiences while continuing to operate from the centralized data foundation provided through DS_vw_healthcare_master (healthcare_db). It allows multiple analytical configurations to be supported within Tableau without requiring separate PostgreSQL Views or datasets for each user-controlled analytical scenario.

2.5.6 Standardized KPI Framework

One of the most important responsibilities of the Tableau Analytics Layer is the development of a standardized KPI framework. KPIs transform the standardized healthcare data provided through DS_vw_healthcare_master (healthcare_db) into business-oriented measures that summarize key aspects of comparative, financial, demographic, clinical, and operational performance.

The Healthcare Analytics Platform includes KPIs across multiple analytical domains. Examples developed and applied throughout the project include:

- **Total Patients**
- **Total Admissions**
- **Total Revenue**
- **Average Billing**
- **Average Age**
- **Male Patients**
- **Female Patients**
- **Average Monthly Admissions**
- **Average Length of Stay**
- **Average Monthly Discharges**
- **Top Medical Condition**
- **Selected versus Others metrics**

These KPIs are supported by analytical logic implemented through the appropriate Tableau mechanisms, including Calculated Fields, aggregations, LOD Expressions, Table Calculations, Sets, and other derived analytical logic, depending on the requirements of each metric.

The objective of the KPI framework is not simply to display prominent numerical values on dashboards. Each KPI represents a defined analytical measure whose calculation, aggregation level, filtering behavior, analytical context, and business meaning must remain consistent with the underlying requirements of the analysis.

This distinction is fundamental to the architecture:

The KPI card is a visualization. The KPI calculation belongs to the Analytics Layer.

The Analytics Layer determines how each metric is calculated and interpreted, while the Visualization and Presentation Layers determine how that metric is displayed to the end user. Separating these responsibilities helps maintain analytical consistency while allowing KPI presentation to be adapted to the visual requirements of each executive dashboard.

2.5.7 Ranking and Contribution Analysis

Ranking and Contribution Analysis provide analytical context beyond absolute values by evaluating both the relative position of a category and its proportional contribution to an overall measure.

Within the Healthcare Analytics Platform, these techniques are applied to analyze medical conditions according to their financial performance and relative contribution to total revenue. This allows the analysis to move beyond reporting absolute revenue values and instead evaluate how individual medical conditions compare with one another and how strongly each contributes to the overall financial distribution.

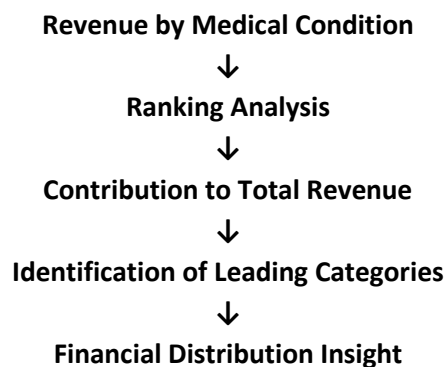
Analytical components developed during the project include:

- **Revenue Distribution Ranking**
- **Revenue Contribution Percent**
- **Top Medical Condition**

Ranking Analysis establishes the relative position of medical conditions according to a defined financial measure, allowing leading and lower-ranked categories to be identified within the revenue distribution.

Contribution Analysis evaluates the proportional importance of each medical condition relative to total revenue, providing additional context for understanding how financial activity is distributed across the analyzed categories.

When combined, these techniques support the following analytical progression:



Within the final portfolio architecture, ranking and contribution calculations are implemented within the Tableau Analytics Layer using the appropriate analytical mechanisms, including Calculated Fields and Table Calculations where required. The resulting measures can then support KPIs, worksheets, and dashboard visualizations while continuing to operate from the centralized analytical foundation provided through DS_vw_healthcare_master (healthcare_db).

By combining absolute revenue values with ranking and contribution analysis, the platform provides a clearer understanding of which medical conditions generate the greatest financial activity and how individual categories contribute to the overall revenue distribution.

2.5.8 Analytical Logic Reusability

Reusability is a major design objective of the Tableau Analytics Layer. When multiple analytical components require the same business definition or calculation logic, the preferred approach is to create and maintain a reusable analytical component rather than independently recreating equivalent logic across different worksheets or dashboards.

Within the Healthcare Analytics Platform, reusable analytical logic is implemented through Tableau mechanisms such as Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, and standardized KPI definitions, depending on the analytical requirement.

This approach provides several advantages:

- **Improved analytical consistency**
- **Reduced duplication of calculation logic**
- **Easier validation and testing**
- **Simplified maintenance**
- **Greater confidence that dashboards interpret metrics consistently**
- **More efficient development of new worksheets and dashboards**
- **Easier extension of the analytical environment**

Reusability is particularly important because all five executive dashboards share the centralized data foundation provided through DS_vw_healthcare_master (healthcare_db) while applying different analytical perspectives to the same healthcare data. Reusable calculations allow common analytical definitions to remain consistent even when they are presented through different KPIs, worksheets, visualizations, or dashboards.

This design principle contributes directly to the modular architecture of the Healthcare Analytics Platform by separating reusable analytical definitions from individual visual presentations. As a result, analytical logic can be maintained, validated, and extended without unnecessarily recreating equivalent calculations for each visualization.

Analytical Component	Primary Purpose	Example Application
Calculated Fields	Create reusable derived measures, dimensions, and analytical classifications.	KPI calculations, segmentation, and comparative logic.
LOD Expressions	Control calculation granularity independently of the visualization level of detail.	Fixed analytical measures, totals, and controlled aggregations.
Table Calculations	Analyze relationships across aggregated results within Tableau views.	Ranking, Percent of Total, Running Totals, and Moving Averages.
Sets	Create dynamic analytical groups and segmentation.	Medical Condition Set; Selected vs Others analysis.
Parameters	Provide user-controlled values that influence analytical logic.	Dynamic Measure Selection, Dynamic Dimension Selection, and analytical switching.
KPI Calculations	Standardize business-oriented performance measures.	Revenue, admissions, patients, billing, age, and length of stay.
Ranking Logic	Determine the relative analytical position of categories.	Revenue Distribution Ranking and Top Medical Condition.
Contribution Analysis	Measure the proportional	Revenue Contribution Percent by

	contribution of categories to an overall total.	Medical Condition.
Time-Based Calculations	Evaluate trends, cumulative behavior, and changes over time.	Monthly trends, Running Totals, and Moving Averages.
Reusable Analytical Logic	Maintain consistent analytical definitions across multiple components.	Shared calculations across KPIs, worksheets, and dashboards.

Table 2.4 – Tableau Analytical Components

Technical Note

Not all data-preparation and analytical logic is implemented at the same architectural level. PostgreSQL provides the centralized and standardized analytical data foundation through vw_healthcare_master, which is consumed in Tableau through DS_vw_healthcare_master (healthcare_db).

Analytical logic that depends on calculation granularity, visualization context, dynamic segmentation, comparative behavior, ranking, time-based analysis, or user-controlled analytical behavior is implemented within Tableau using the appropriate mechanisms, including Calculated Fields, LOD Expressions, Table Calculations, Sets, and Parameters.

This division allows PostgreSQL and Tableau to perform the responsibilities most appropriate to their respective roles while maintaining a clear separation between centralized data preparation and Tableau analytical logic.

Business Insight

A well-designed Analytics Layer transforms standardized healthcare data into consistent, reusable, and business-oriented analytical definitions. Calculated Fields, LOD Expressions, Table Calculations, standardized KPIs, Sets, Parameters, dynamic segmentation, ranking, contribution analysis, and time-based calculations allow the five executive dashboards to interpret healthcare information through a common analytical framework rather than relying on independently defined metrics.

By combining a centralized PostgreSQL analytical foundation with reusable Tableau analytical logic, the Healthcare Analytics Platform improves consistency, reduces unnecessary duplication, and supports multiple analytical perspectives while preserving a common underlying data structure.

2.6 Visualization & Presentation Layer

The Visualization & Presentation Layer transforms the analytical measures, classifications, and business-oriented definitions produced by the Tableau Analytics Layer into visual components and integrated dashboard experiences designed for business interpretation, analytical exploration, and decision support.

Within the Healthcare Analytics Platform, this layer is implemented primarily through Tableau worksheets, KPI cards, charts, dashboard layouts, titles, labels, legends, tooltips, filters, and supporting visual elements. Its purpose is not simply to display data, but to organize analytical information in a way that enables users to identify trends, compare performance, evaluate distributions, recognize patterns, and investigate relationships across healthcare domains.

The architecture maintains a conceptual distinction between visualization and presentation. The **Visualization Layer** focuses on individual analytical components such as worksheets, charts, and KPI cards, while the **Presentation Layer** integrates those components into coordinated executive dashboards with a consistent information hierarchy and analytical experience.

This separation supports a modular development process. Individual worksheets can be created, validated, formatted, and tested independently before being incorporated into a larger dashboard environment. Dashboard layouts can then focus on information hierarchy, usability, visual consistency, interaction, and analytical storytelling without requiring the underlying analytical calculations to be recreated.

In the final portfolio implementation, these principles are applied across five executive dashboards:

- **Comparative Analytics Dashboard**
- **Revenue Distribution Dashboard**
- **Patient Demographics Dashboard**
- **Clinical & Medical Conditions Dashboard**
- **Healthcare Operations Dashboard**

Together, these dashboards present different analytical perspectives while remaining connected to the common analytical architecture established through the PostgreSQL and Tableau layers.

2.6.1 Worksheet-Based Visualization Architecture

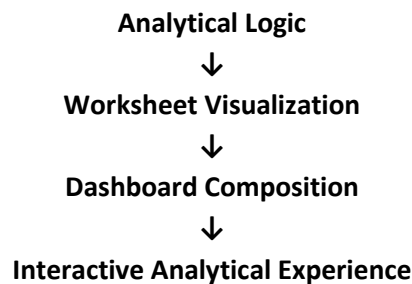
Tableau worksheets represent the fundamental visual building blocks of the Healthcare Analytics Platform. Each worksheet is developed to address a defined analytical requirement and visually represent one or more measures, dimensions, classifications, KPIs, or analytical results produced within the Tableau Analytics Layer.

Rather than treating worksheets as isolated graphics, the platform uses them as reusable visualization components that can be independently developed, validated, formatted, and subsequently integrated into executive dashboards.

Worksheet responsibilities include:

- **Presenting defined analytical questions**
- **Visualizing KPIs and derived analytical measures**
- **Displaying trends, distributions, rankings, comparisons, and time-based analysis**
- **Supporting comparative and segmentation analysis**
- **Providing source and target components for dashboard interactions**
- **Presenting contextual information through tooltips**
- **Supporting consistent titles, labels, legends, and formatting**
- **Acting as reusable visual components within dashboard layouts**

This worksheet-based architecture establishes a clear progression between analytical logic, visual representation, dashboard composition, and user interaction:



This modular approach allows analytical calculations to remain logically separated from their visual representation. A calculation can therefore support one or more worksheet components, while individual worksheets can be integrated into dashboards without requiring the underlying analytical logic to be recreated.

This separation improves consistency and reusability across the five executive dashboards while allowing each dashboard to present a distinct analytical perspective.

2.6.2 KPI Presentation

KPI cards provide high-level visual summaries of important analytical measures and are positioned prominently within the executive dashboards to communicate key information before users explore more detailed visualizations.

Within the Healthcare Analytics Platform, KPI cards are used across multiple analytical domains to present metrics such as:

- **Total Patients**
- **Total Admissions**

- **Total Revenue**
- **Average Billing**
- **Average Age**
- **Male Patients**
- **Female Patients**
- **Average Monthly Admissions**
- **Average Length of Stay**
- **Average Monthly Discharges**
- **Top Medical Condition**
- **Selected versus Others metrics**

An important architectural distinction is maintained between **KPI calculation** and **KPI presentation**.

The analytical definition and calculation of a KPI belong to the **Tableau Analytics Layer**, while the KPI card used to visually communicate that metric belongs to the **Visualization & Presentation Layer**.

This separation allows the analytical definition of a KPI to remain independent from presentation decisions involving font size, number formatting, layout, alignment, labels, visual emphasis, and dashboard placement. As a result, an analytical measure can maintain a consistent business definition while its visual presentation is adapted to the requirements of a specific dashboard.

KPI cards therefore function as a high-level summary component of the dashboard experience, providing users with immediate visibility into key analytical results before they investigate the supporting trends, distributions, comparisons, and detailed visualizations.

2.6.3 Analytical Visualization Types

The Healthcare Analytics Platform uses different visualization types according to the analytical question and business relationship being evaluated. Rather than applying a single chart style throughout the solution, visual forms are selected according to the type of comparison, trend, distribution, composition, ranking, or performance measure that needs to be communicated.

The final portfolio dashboards incorporate visualization approaches supporting the following analytical purposes:

Trend Analysis

Used to examine changes over time in measures such as admissions, discharges, billing activity, insurance-provider revenue, length of stay, and test-result patterns.

Comparative Analysis

Used to evaluate differences between analytical groups, including Selected versus Others populations and other categorical comparisons.

Distribution Analysis

Used to understand how measures such as revenue and patient volume are distributed across medical conditions, demographic groups, insurance providers, and other healthcare dimensions.

Ranking Analysis

Used to establish the relative position of categories and identify leading contributors, including revenue ranking by medical condition and the identification of the Top Medical Condition.

Composition Analysis

Used to evaluate proportional relationships and category mix, including gender distribution and admission-type composition.

KPI Analysis

Used to communicate high-level analytical measures that provide an immediate summary of financial, demographic, clinical, comparative, or operational information before users investigate supporting visualizations.

The selection of visualization types is therefore driven by analytical purpose rather than decorative preference. Each visual component is intended to communicate a specific analytical relationship while supporting clarity, comparison, consistency, and business interpretation across the dashboard environment.

2.6.4 Dashboard Composition

The Presentation Layer combines validated worksheets, KPI cards, and supporting visual components into integrated executive dashboards. Dashboard composition establishes the information hierarchy through which users move from high-level analytical summaries to more detailed trends, distributions, comparisons, demographic patterns, clinical perspectives, and operational analysis.

The final Healthcare Analytics Platform includes five executive dashboards:

Comparative Analytics Dashboard

Provides comparative analysis between selected analytical populations and the remaining dataset, supporting Selected versus Others analysis across revenue, patient populations, and related comparative measures.

Revenue Distribution Dashboard

Examines financial distribution across medical conditions through revenue KPIs, ranking analysis, contribution percentages, and distribution-oriented visualizations.

Patient Demographics Dashboard

Provides insight into patient characteristics through demographic KPIs and visual analysis of age, gender, blood type, age-group composition, and insurance-provider distribution.

Clinical & Medical Conditions Dashboard

Focuses on medical conditions and related clinical characteristics, supporting the evaluation of patient populations, condition-level patterns, and relevant clinical indicators.

Healthcare Operations Dashboard

Provides operational visibility into admissions, discharges, billing activity, length of stay, admission types, insurance-provider revenue, test results, and time-based healthcare trends.

Together, these dashboards organize the analytical environment around distinct business perspectives while maintaining a consistent visual and presentation framework. Although each dashboard addresses a different analytical domain, they share common architectural principles involving KPI placement, worksheet composition, visual hierarchy, formatting consistency, and interactive analytical exploration.

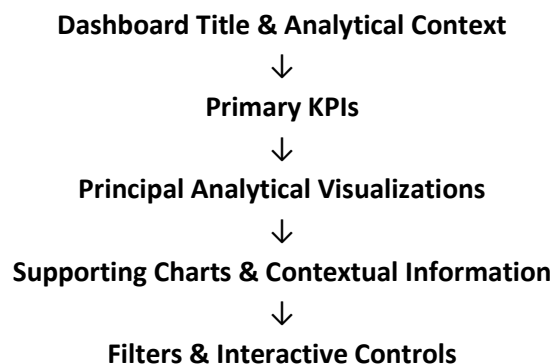
All five dashboards remain connected to the common analytical architecture of the Healthcare Analytics Platform while presenting different business perspectives through their respective worksheets, KPIs, visualizations, and interactions.

2.6.5 Visual Hierarchy

A major design objective of the Presentation Layer is the establishment of a clear and consistent visual hierarchy. Visual hierarchy determines how analytical information is prioritized and guides users through each dashboard from high-level business context toward increasingly detailed analytical exploration.

Users should be able to distinguish clearly between dashboard context, high-level KPIs, principal analytical visualizations, supporting information, and interactive controls. Elements with greater analytical importance receive greater visual emphasis through their placement, size, formatting, and relationship to surrounding components, while supporting elements provide additional context without competing with the principal analytical message.

The dashboard hierarchy generally follows this progression:



This structure allows users to move naturally from summary information toward more detailed analytical exploration. High-level KPIs provide immediate context, principal visualizations communicate

the most important trends, comparisons, distributions, or operational patterns, and supporting charts provide additional detail when deeper investigation is required.

A consistent visual hierarchy across the five executive dashboards also improves usability by establishing a predictable analytical experience. Users can more easily identify where to locate summary metrics, principal visualizations, supporting analysis, and interactive controls without having to interpret a completely different visual structure for each dashboard.

This consistent hierarchy supports both analytical clarity and presentation consistency while still allowing each dashboard to emphasize the information most relevant to its specific analytical purpose.

2.6.6 Layout and Container Strategy

The executive dashboards use Tableau **Horizontal and Vertical Containers** to organize analytical components, maintain consistent alignment, and establish a structured presentation environment.

Containers provide a controlled layout framework for organizing:

- **KPI cards**
- **Analytical charts**
- **Filter and control areas**
- **Dashboard titles and subtitles**
- **Supporting text and contextual information**
- **Spacing and layout elements**

During the dashboard refinement process, visual separation between components was achieved primarily through spacing, padding, alignment, and controlled container placement rather than relying heavily on visible separator lines or unnecessary graphical elements.

This approach creates a cleaner analytical interface while preserving clear visual boundaries between dashboard sections. Consistent spacing and alignment help users distinguish KPI areas, principal visualizations, supporting charts, and interactive controls without introducing unnecessary visual complexity.

The container-based layout strategy also supports modularity and maintainability. Individual worksheets, KPI cards, and supporting components can be adjusted within the dashboard structure while preserving the overall organization and visual hierarchy of the presentation.

Together, Horizontal and Vertical Containers, spacing, padding, and alignment provide the structural foundation for organizing the visual components of the five executive dashboards.

2.6.7 Filter Panel Design

Interactive filters are organized within dedicated dashboard areas so that users can quickly identify the analytical controls available for refining the displayed information without confusing those controls with the analytical visualizations themselves.

Examples of filtering dimensions used across the Healthcare Analytics Platform include:

- **Admission Date**
- **Medical Condition**
- **Insurance Provider**

Filter placement is designed to preserve the primary analytical area of each dashboard while maintaining accessibility for users who need to refine the displayed information. Filters are positioned as supporting controls rather than dominant visual elements, allowing KPIs and principal analytical visualizations to retain greater visual emphasis.

The filter panel therefore serves both an **interaction function** and a **presentation function**. From an interaction perspective, it provides users with direct control over the analytical context displayed in the dashboard. From a presentation perspective, it organizes those controls within a clearly identifiable area that remains visually integrated with the overall dashboard layout.

This design supports a clear separation between analytical content and analytical controls, improving dashboard usability while preserving the visual hierarchy established across the presentation environment.

Filter panels complement the broader dashboard interaction architecture, where Tableau Filter Actions and Highlight Actions can provide additional analytical exploration directly through worksheet interaction.

2.6.8 Formatting and Visual Consistency

The Healthcare Analytics Platform applies consistent formatting and visual design principles across its five executive dashboards to create a unified, professional, and predictable analytical experience.

These principles include:

- **Consistent typography**
- **Standardized KPI presentation**
- **Controlled and purposeful use of color**
- **Consistent chart titles and subtitles**
- **Clear and meaningful axis labels**
- **Appropriate number, currency, and percentage formatting**
- **Consistent spacing, padding, and alignment**
- **Structured filter and control placement**
- **Clear visual hierarchy**

- **Reduced unnecessary visual clutter**

Visual consistency is particularly important in a multi-dashboard analytical platform because users should not need to relearn the visual language when moving from one analytical domain to another. Similar components follow comparable presentation conventions so that users can more easily recognize KPI areas, principal analytical visualizations, supporting information, and interactive controls.

Consistent formatting also improves readability and analytical interpretation. Titles communicate the purpose of each visualization, axis labels use business-oriented terminology, numerical formats reflect the meaning of the underlying measure, and spacing and alignment help distinguish analytical sections without introducing unnecessary visual complexity.

These formatting standards allow the five executive dashboards to maintain their individual analytical purposes while still functioning as coordinated components of a unified Healthcare Analytics Platform.

The objective is therefore not to make every dashboard visually identical, but to maintain a common presentation language while allowing each dashboard to emphasize the analytical information most relevant to its specific business perspective.

2.6.9 Axis Labels and Analytical Terminology

Technical field names and automatically generated labels do not always provide the clearest terminology for business-facing analytical environments. For this reason, axis labels, chart titles, legends, KPI labels, filter descriptions, and other user-facing terminology were reviewed during dashboard refinement to ensure that analytical information is communicated clearly and consistently.

When appropriate, technical or automatically generated descriptions were replaced with more concise, business-oriented terminology without modifying the underlying field, calculation, or analytical definition.

For example, an automatically generated axis label such as:

Month of Date Of Admission

can be presented to the user as:

Admission Month

This change improves readability while preserving the underlying analytical meaning and calculation.

The same principle applies across dashboard titles, axis labels, legends, KPI labels, filter descriptions, tooltips, and other user-facing analytical elements. Terminology should communicate the business meaning of the information rather than unnecessarily expose the technical structure used to generate it.

This design principle reinforces an important Business Intelligence practice:

Technical field structures should not unnecessarily dictate the terminology presented to business users.

By separating technical field definitions from business-facing terminology, the dashboards provide a clearer and more intuitive analytical experience while preserving the integrity of the underlying data and analytical logic.

2.6.10 Dashboard Usability

The Presentation Layer is designed not only for visual quality but also for analytical usability. Dashboard components are organized so that users can identify important information, understand the purpose of each visualization, access analytical controls, and progressively explore supporting detail without needing to understand the underlying Tableau implementation.

Dashboard usability considerations include:

- **Clear identification of KPIs**
- **Logical chart placement**
- **Predictable filter and control placement**
- **Consistent interaction patterns**
- **Readable labels and business-oriented terminology**
- **Appropriate spacing and alignment**
- **Reduced unnecessary visual clutter**
- **Meaningful chart titles**
- **Preservation of analytical context**
- **Consistent behavior across related visualizations**

These principles help ensure that the dashboards communicate information effectively and provide an intuitive analytical experience. Users can focus on interpreting healthcare information, comparing analytical results, and investigating relationships rather than understanding the technical structure used to construct the dashboards.

Usability therefore functions as a bridge between analytical accuracy and effective communication. A technically correct calculation provides limited business value if users cannot easily identify, interpret, and explore the resulting information.

Within the Healthcare Analytics Platform, dashboard usability is supported through the coordinated application of visual hierarchy, consistent formatting, business-oriented terminology, structured layouts, accessible analytical controls, and interactive analytical behavior.

Component	Architectural Role	Primary Purpose
KPI Cards	Visualization	Present high-level analytical measures and summaries.
Worksheets	Visualization	Represent individual analytical questions and reusable visual components.
Charts	Visualization	Communicate trends, comparisons, distributions, rankings, and composition.
Dashboard Containers	Presentation	Organize visual components and maintain structured alignment.
Dashboard Layout	Presentation	Establish information hierarchy and analytical flow.
Filter Panel	Presentation / Interaction	Provide accessible analytical controls while preserving the primary visualization area.
Titles & Labels	Presentation	Communicate clear, business-facing analytical terminology.
Tooltips	Visualization / Interaction	Provide contextual analytical detail without overcrowding the dashboard.
Formatting Standards	Presentation	Maintain visual consistency across dashboards and analytical domains.

Table 2.5 – Visualization & Presentation Components

Technical Note

The Visualization and Presentation Layers represent logically distinct architectural responsibilities but are implemented within the same Tableau environment. Worksheets, charts, and KPI cards function as reusable visualization components, while dashboards provide the presentation structure in which those components are organized, formatted, aligned, and integrated into a coordinated analytical experience.

Maintaining this conceptual separation supports modular development, improves maintainability, and allows visual components to be developed, validated, and refined independently before being incorporated into the final executive dashboards. The underlying analytical definitions remain part of the Tableau Analytics Layer, while the Visualization & Presentation Layer focuses on communicating those analytical results effectively to users.

Business Insight

Effective Business Intelligence presentation requires more than accurate calculations. Analytical information must also be organized according to business priority, communicated through consistent and understandable terminology, and structured so that users can move naturally from high-level KPIs to increasingly detailed analytical investigation.

A well-designed presentation environment reduces the effort required to interpret information, improves analytical usability, and increases the practical business value of the underlying analytical model by helping users identify trends, compare results, evaluate distributions, and explore relevant analytical relationships more efficiently.

Within the Healthcare Analytics Platform, consistent visual hierarchy, business-oriented terminology, structured dashboard layouts, and accessible interactive controls work together to transform analytical results into a clearer and more usable decision-support experience.

2.7 Interaction Architecture

The Interaction Architecture defines how users interact with the analytical components of the Healthcare Analytics Platform and how Tableau worksheets coordinate with one another within the dashboard environment.

While the Visualization & Presentation Layer determines how analytical information is visually organized and communicated, the Interaction Layer determines how that information responds to user selections and analytical controls. Through interactive filters, Dashboard Filter Actions, Highlight Actions, and worksheet selections, the platform enables users to move from high-level analytical summaries to focused investigation without modifying the underlying data structure or analytical definitions.

The interaction strategy is designed around two complementary analytical behaviors: **filtering** and **highlighting**. Filtering restricts the information displayed in target visualizations according to the selected analytical context, while highlighting preserves the broader visualization context and visually emphasizes marks associated with the selected information.

This distinction supports two different forms of analytical exploration. **Filtering** enables focused investigation by narrowing the displayed analytical scope, whereas **highlighting** enables contextual investigation by emphasizing relationships while retaining surrounding information for comparison.

As a result, dashboards function as coordinated analytical systems rather than collections of independent charts. User selections in one visual component can influence related components, creating an integrated analytical experience across the dashboard environment.

This interaction architecture complements the analytical and visualization layers by allowing users to explore different perspectives of the same underlying analytical foundation without requiring separate datasets or dashboard-specific PostgreSQL Views.

2.7.1 Interactive Filters

Interactive filters provide users with direct control over the analytical context displayed within a dashboard. They allow users to refine the information shown by selecting specific dimensions, categories, or time periods without modifying the underlying worksheets, analytical calculations, or data structure.

Examples of filtering dimensions incorporated into the Healthcare Analytics Platform include:

- **Admission Date**
- **Medical Condition**
- **Insurance Provider**

These controls are positioned within dedicated filter areas to maintain a clear separation between analytical controls and visualization components. Their placement supports accessibility while preserving the primary dashboard area for KPIs, charts, and supporting analytical visualizations.

The filter panel is therefore designed to satisfy two objectives simultaneously:

Analytical Control — allowing users to define or refine the analytical context displayed within the dashboard.

Presentation Consistency — maintaining an organized and predictable location for interactive controls across the dashboard environment.

Where appropriate, filter selections can be applied across multiple related worksheets so that analytical components respond consistently to the selected criteria. This coordinated filtering behavior helps preserve a common analytical context across the dashboard while allowing users to focus their investigation on specific dimensions, categories, or time periods.

Interactive filters therefore provide an explicit user-controlled mechanism for refining dashboard context, complementing Dashboard Filter Actions and Highlight Actions that are initiated through direct interaction with worksheet visualizations.

2.7.2 Dashboard Filter Actions

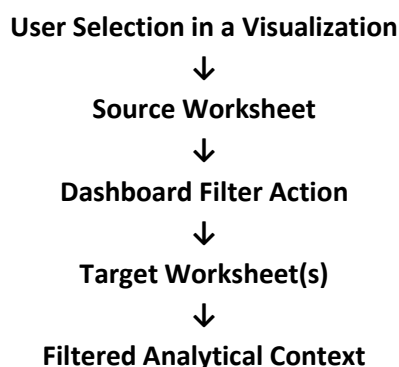
Dashboard Filter Actions allow a user selection made within one Tableau worksheet to dynamically filter the information displayed in one or more related worksheets within the same dashboard environment.

Unlike interactive filter controls, which allow users to refine analytical context through dedicated filter selections, a Dashboard Filter Action is initiated directly through interaction with a visualization. When a user selects a relevant mark within a source worksheet, Tableau applies the configured Filter Action to the designated target worksheet or worksheets.

Dashboard Filter Actions support cross-view analytical exploration by allowing users to investigate relationships between visual components without navigating away from the dashboard. A selected category, time period, provider, or other analytical element can therefore become the context through which related visualizations are examined.

Within the Healthcare Analytics Platform, Dashboard Filter Actions are used to coordinate related analytical components so that a user selection in one visualization can update other relevant views. This behavior transforms individual worksheets into connected components of a broader interactive analytical environment.

The general interaction pattern can be represented as:



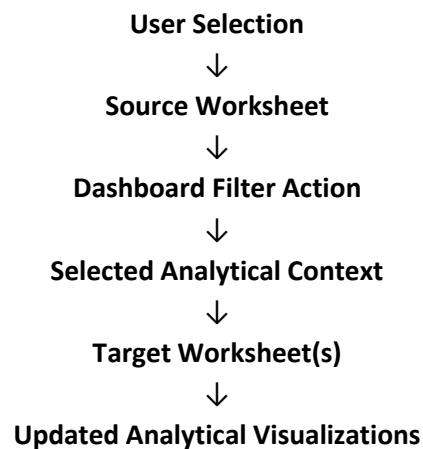
The specific relationship between source and target worksheets, together with the configuration of the Filter Action, determines how each interaction affects the dashboard. This architecture allows interactive analytical behavior to be implemented within Tableau while preserving the centralized data foundation provided by DS_vw_healthcare_master (healthcare_db).

2.7.3 Source and Target Architecture

Every Dashboard Filter Action establishes an interaction relationship between one or more source worksheets and one or more target worksheets within the dashboard environment.

The **source worksheet** initiates the interaction when the user selects a relevant analytical mark. The configured Dashboard Filter Action then transfers the selected analytical context to the designated **target worksheet or worksheets**, which update their displayed information according to the filtering criteria defined by the Action.

Conceptually, the interaction follows this architecture:



This source-and-target architecture allows individual worksheets to participate in a coordinated analytical workflow. A visualization can therefore function not only as a mechanism for presenting information, but also as an interactive entry point for investigating related analytical components.

The relationship between source and target worksheets is defined according to the analytical purpose of each dashboard. A source worksheet provides the selected context, while the designated target worksheets respond to that context, allowing multiple visual components to remain analytically coordinated during user exploration.

This interaction occurs within Tableau and operates on the common analytical foundation provided through DS_vw_healthcare_master (healthcare_db), allowing coordinated dashboard behavior without requiring separate data sources for individual source-and-target relationships.

Conceptually:

2.7.3 Source and Target Architecture

Every Dashboard Filter Action establishes a relationship between one or more source worksheets and one or more target worksheets.

Conceptually:

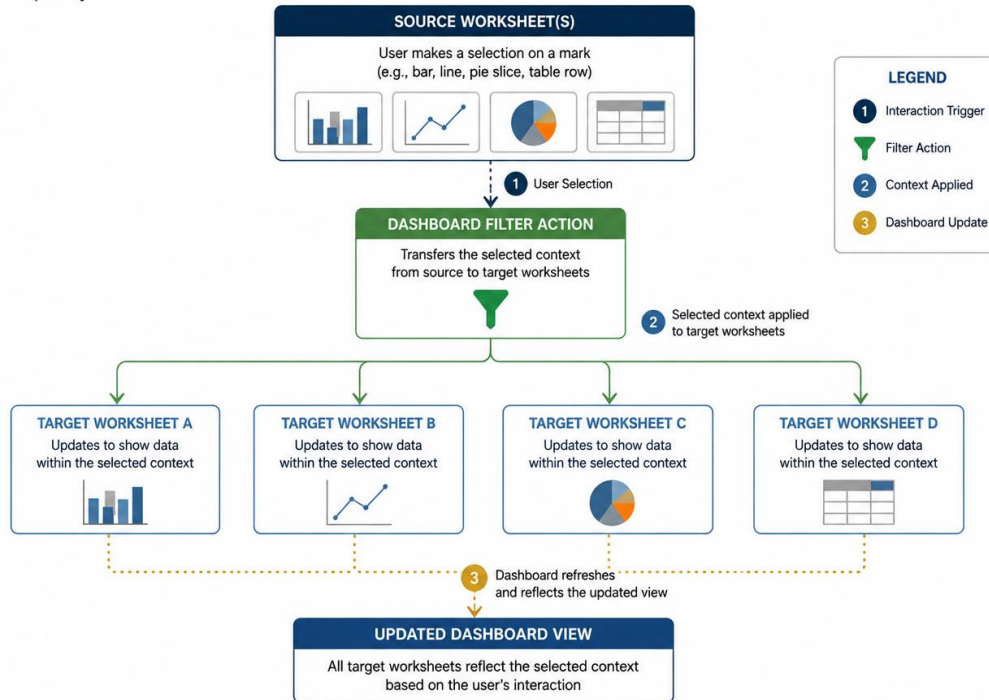


Figure 2.3 – Source and Target Architecture (Conceptual)

This source-and-target model is fundamental to understanding Tableau dashboard interaction architecture because it defines where an interaction originates, what analytical context is applied, and which visual components respond to the user selection.

A well-designed Dashboard Action should therefore answer three fundamental questions:

What initiates the interaction?

The designated source worksheet, through a user interaction with a relevant analytical mark.

Which analytical context is applied?

The dimension value or analytical context associated with the selected mark and defined by the Action configuration.

Which components should respond?

The designated target worksheet or worksheets configured to receive the Action.

Clearly defining these three elements helps ensure that dashboard interactions are intentional, analytically meaningful, and consistent with the purpose of the dashboard.

2.7.4 Filter Action Behavior

Filter Actions can be configured to execute according to different user interaction behaviors. The selected execution method determines when the Action is triggered and should be aligned with the intended analytical experience of the dashboard.

Tableau supports several common execution behaviors:

Select — The Action executes when the user selects a mark within the source worksheet.

Hover — The Action executes when the user positions the pointer over a mark, allowing the interaction to occur without requiring an explicit selection.

Menu — The Action becomes available through a contextual menu associated with the selected mark.

Within the Healthcare Analytics Platform, Select is used as the primary execution behavior for Dashboard Filter Actions because it provides intentional analytical control. Users explicitly choose the analytical mark that initiates the filtering interaction, reducing unintended changes to the dashboard while navigating across visualizations.

The behavior that occurs when a selection is cleared is also an important part of the Action configuration. Depending on the analytical requirement, the Action configuration determines how target worksheets respond when the initiating selection is no longer active.

These configuration decisions influence both technical behavior and analytical usability. A well-configured Filter Action should respond predictably to user interaction, preserve the intended analytical context, and remain consistent with the purpose of the dashboard.

2.7.5 Highlight Actions

Highlight Actions provide a different interaction model from Dashboard Filter Actions. Rather than restricting the target visualization to a selected analytical context, a Highlight Action preserves the broader visual context while emphasizing marks associated with the selected value.

Conceptually, the distinction can be represented as follows:

Filter Action

Selected / Related Data → **Remains Visible**

Unrelated Data → **Filtered from the Target View**

Highlight Action

Selected / Related Data → **Visually Emphasized**

Other Data → **Remains Visible but De-emphasized**

This behavior is particularly useful when the analytical objective is to identify relationships, associations, or patterns across visualizations without removing the surrounding information required for comparison.

Within the Healthcare Analytics Platform, this interaction model is demonstrated in the **Healthcare Operations Dashboard** through the Insurance Provider analysis. The Highlight Action allows a user to interact with an insurance provider in the revenue visualization and visually emphasize related marks across connected analytical views while preserving the broader dashboard context.

The implemented Highlight Action, HA_Insurance_Provider_Highlight, uses CH_Insurance_Provider_Revenue_Trend as its source worksheet and applies **Insurance Provider** as the selected analytical field. The Action executes on **Hover**, allowing related information to be visually emphasized as the user moves across insurance-provider marks without narrowing the overall analytical scope.

This behavior is intentional and represents the fundamental distinction between filtering and highlighting: **filtering narrows the analytical scope displayed in the target views, whereas highlighting preserves that scope and changes the visual emphasis applied to related information.**

2.7.6 Filter Actions vs. Highlight Actions

Filter Actions and Highlight Actions support different analytical objectives within an interactive Tableau dashboard. Although both mechanisms allow user interactions in one visualization to influence related views, they differ fundamentally in how the analytical context is presented to the user.

Filter Actions narrow the displayed analytical scope by restricting target visualizations to information associated with the selected context. This behavior is appropriate when the analytical objective is to focus investigation on a specific subset of the data.

Highlight Actions, by contrast, preserve the broader analytical context while visually emphasizing information associated with the selected value. This behavior is appropriate when users need to investigate relationships and patterns without removing surrounding information required for comparison.

The selection between filtering and highlighting should therefore be based on the analytical purpose of the interaction rather than treating the two mechanisms as interchangeable.

Table 2.6 summarizes the principal differences between these two interaction mechanisms.

Characteristic	Filter Action	Highlight Action
Primary Purpose	Focus analysis on a selected analytical context	Emphasize related analytical information
Unrelated Marks	Filtered from the target view	Remain visible but de-emphasized
Analytical Context	Narrowed to the selected context	Broader context preserved
Typical Use	Focused investigation	Relationship exploration
User Experience	Changes the displayed analytical scope	Changes visual emphasis

Best Used When	User needs to investigate a specific analytical subset	User needs to understand relationships within the broader context
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Table 2.6 – Filter Actions vs. Highlight Actions

The selection between a Filter Action and a Highlight Action should therefore be based on the intended analytical objective. **Filter Actions** are appropriate when users need to narrow the analytical scope for focused investigation, while **Highlight Actions** are appropriate when users need to explore relationships without losing the surrounding analytical context.

2.7.7 Interaction Naming Standards

As the number of dashboard interactions increases, a consistent naming convention becomes important for maintenance, configuration review, and troubleshooting. Descriptive Action names make it easier to identify the purpose of an interaction and distinguish between different Action types within the Tableau environment.

The Healthcare Analytics Platform uses descriptive Action names that combine an Action-type prefix, the analytical component or context, and the interaction behavior or purpose.

For example:

FA_Monthly_Billing_Filter

can be interpreted as:

FA → Filter Action

Monthly Billing → Analytical component or context

Filter → Interaction behavior

The general naming convention can therefore be represented as:

[Action Type]_[Analytical Component]_[Behavior]

Examples implemented within the Healthcare Analytics Platform include:

- FA_Monthly_Discharges_Filter
- FA_Monthly_Billing_Filter
- FA_Monthly_Admission_Type_Mix_Filter
- FA_Insurance_Provider_Filter
- FA_Monthly_Test_Results_Filter

The same naming principle is applied to Highlight Actions using the HA prefix. For example:

HA_Insurance_Provider_Highlight

where:

HA → Highlight Action

Insurance Provider → Analytical component or context

Highlight → Interaction behavior

This naming convention improves readability within the Tableau Action configuration environment and makes the interaction architecture easier to review, maintain, and explain as the analytical platform evolves.

2.7.8 Interaction Testing and Validation

Interactive functionality should be validated independently from visual formatting to ensure that dashboard interactions produce the intended analytical behavior and preserve the correct analytical context.

During dashboard testing, each configured Action should be evaluated to confirm:

- The correct source worksheet initiates the Action.
- The intended target worksheet or worksheets respond to the interaction.
- The appropriate analytical dimension or context is applied through the Action.
- Selecting, hovering over, or otherwise interacting with a relevant mark produces the expected analytical result according to the configured Action behavior.
- Clearing the interaction returns the dashboard to the intended analytical state.
- Filter Actions do not unintentionally affect unrelated analytical components.
- Highlight Actions emphasize related marks while preserving the broader analytical context.
- Multiple Actions operating within the same dashboard do not create unintended interaction conflicts.
- Dashboard interactions remain consistent, understandable, and predictable for the user.

This validation becomes particularly important when multiple Actions operate within the same dashboard. Incorrect source-and-target relationships, inappropriate Action configurations, or overlapping interactions can produce results that appear visually plausible while representing an unintended analytical context.

Interaction testing therefore serves both a **technical** and an **analytical** purpose: it confirms that the configured Actions function as intended while also verifying that the resulting dashboard behavior supports the analytical objectives of the visualization and the broader dashboard experience.

Validation should therefore consider not only whether an interaction executes successfully, but also whether it preserves analytical meaning, produces the expected relationship between source and target worksheets, and remains understandable from the user's perspective.

2.7.9 Interaction Design Principles

The Interaction Architecture follows a set of design principles intended to ensure that dashboard interactions remain analytically meaningful, predictable, controlled, consistent, and maintainable.

Purposeful Interaction

Every Action should support a defined analytical objective or user investigation. Interactive functionality should be implemented because it contributes meaningful analytical value, rather than simply because the functionality is available within Tableau.

Predictable Behavior

Users should be able to understand how and why dashboard information changes in response to an interaction. Actions should produce consistent and understandable results aligned with the analytical purpose of the visualization.

Controlled Scope

Actions should affect only the worksheets that logically belong to the selected analytical context. Source-and-target relationships should therefore be configured deliberately to prevent unrelated analytical components from responding unintentionally.

Context Preservation

Highlight Actions should be used when preserving the broader analytical context provides greater analytical value than restricting the analysis to a selected subset. This allows users to investigate relationships while retaining surrounding information for comparison.

Consistent Interaction Patterns

Similar interactions should follow consistent behavioral patterns across related dashboard components. Consistency reduces unnecessary variation and helps users develop a predictable understanding of how analytical interactions operate.

Maintainable Configuration

Dashboard Actions should use descriptive and consistent naming conventions so that their purpose, analytical context, and interaction type can be identified efficiently during configuration review, maintenance, and troubleshooting.

Together, these principles ensure that dashboard interaction is treated as a deliberate component of the analytical architecture rather than as an isolated visualization feature. The objective is to support analytical exploration while maintaining clarity, consistency, appropriate context, and controlled interaction behavior across the dashboard environment.

Action Name	Action Type	Primary Analytical Component	Purpose
FA_Monthly_Discharges_Filter	Filter	Monthly Discharges	Coordinate discharge-related analysis
FA_Monthly_Billing_Filter	Filter	Monthly Billing	Coordinate billing-related analysis
FA_Monthly_Admission_Type_Mix_Filter	Filter	Monthly Admission Type Mix	Analyze admission-type context
FA_Insurance_Provider_Filter	Filter	Insurance Provider Revenue	Analyze insurance-provider context
FA_Monthly_Test_Results_Filter	Filter	Monthly Test Results	Coordinate test-result analysis

Table 2.7 – Dashboard Filter Action Inventory

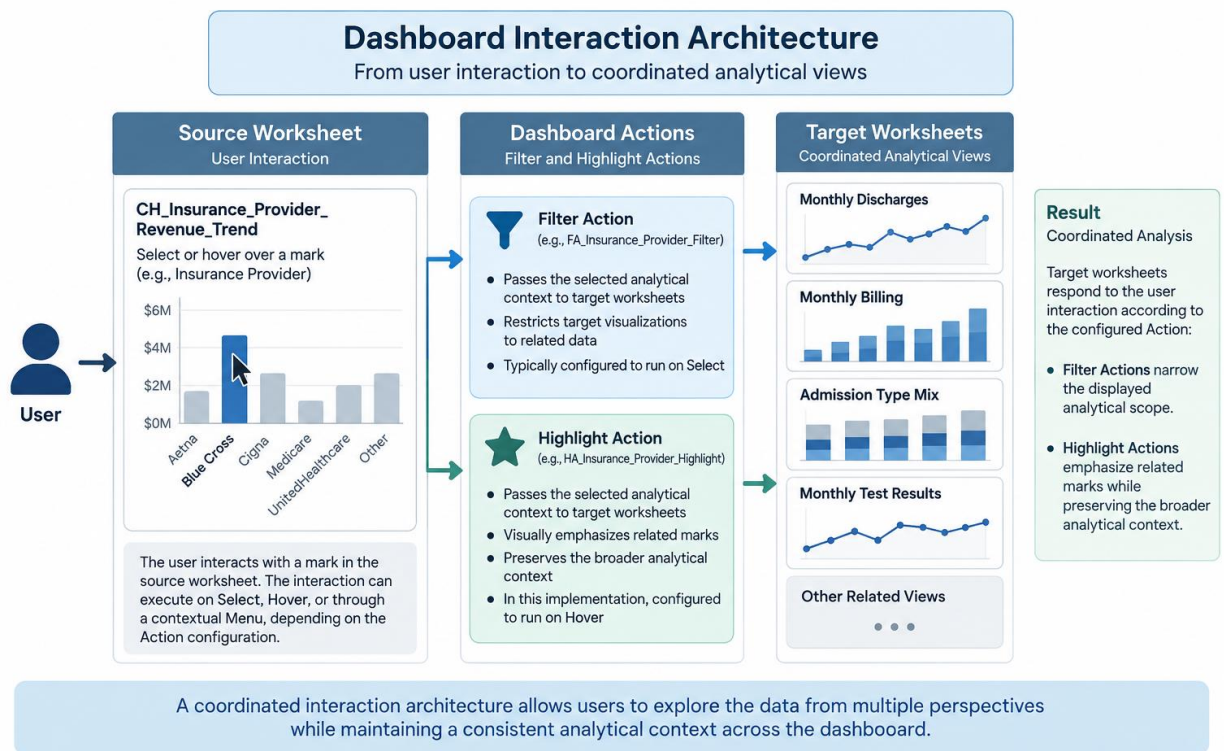


Figure 2.4 – Dashboard Interaction Architecture

Technical Note

Filter Actions and Highlight Actions may originate from similar user interactions, but they produce fundamentally different analytical behaviors. Filter Actions narrow the analytical scope displayed by target worksheets, while Highlight Actions preserve the broader analytical context and visually emphasize marks associated with the selected value.

The appropriate interaction mechanism should therefore be determined by the analytical objective of the dashboard. Filter Actions support focused investigation of a selected analytical context, whereas Highlight Actions support relationship exploration while retaining surrounding information for comparison.

Both interaction types operate within Tableau on the common analytical foundation provided by `DS_vw_healthcare_master (healthcare_db)`, preserving the centralized architecture while enabling different forms of interactive analysis.

Business Insight

Interactive dashboards create greater analytical value when user interactions produce clear, intentional, and predictable results. By coordinating interactive filters, Filter Actions, and Highlight Actions across related visualizations, the Healthcare Analytics Platform enables users to move from high-level analytical summaries to focused investigation while preserving the context required to interpret relationships correctly.

Filter Actions support focused analysis by narrowing the displayed analytical scope, while Highlight Actions support relationship exploration by emphasizing related information without removing the broader context. Together, these mechanisms allow users to investigate healthcare data from multiple analytical perspectives while maintaining a coherent dashboard experience.

This coordinated interaction model reduces the effort required to explore related information and strengthens the dashboard's ability to support analytical interpretation and informed decision-making.

2.8 Documentation & Deployment Layer

The Documentation & Deployment Layer represents the final stage of the Healthcare Analytics Platform architecture. Its purpose is to organize the technical assets, analytical components, supporting documentation, and presentation materials required to make the solution understandable, reproducible, maintainable, and suitable for professional portfolio presentation.

While the preceding architectural layers focus on data management, centralized data preparation, analytical logic, visualization, presentation, and user interaction, the Documentation & Deployment Layer consolidates these components into a coherent project package that communicates how the solution was designed, implemented, validated, and organized.

Within the Healthcare Analytics Platform, documentation provides a structured record of the project architecture, analytical methodology, PostgreSQL implementation, Tableau analytical logic, dashboard design, interaction architecture, and validation approach.

Deployment, in the context of this portfolio project, refers primarily to the structured packaging, publication, and professional presentation of the implemented analytical solution and its supporting technical assets. It does not represent a production enterprise deployment.

Because the Healthcare Analytics Platform was developed as a portfolio-based analytical project, production infrastructure such as Tableau Server, Tableau Cloud, automated CI/CD pipelines, enterprise authentication, production monitoring, automated refresh infrastructure, and cloud database deployment is outside the scope of the current implementation.

This distinction ensures that the technical documentation accurately represents the capabilities that were actually implemented while establishing a structured foundation for potential future deployment enhancements.

2.8.1 Technical Documentation

Technical documentation is treated as a fundamental component of the Healthcare Analytics Platform rather than as a separate activity performed only after development. It provides a structured record of the analytical architecture, implementation decisions, design methodology, and validation processes used throughout the project.

The Healthcare Analytics Platform is documented through a structured technical document that describes the complete analytical lifecycle, including:

- **Business context and project objectives**
- **Solution architecture**
- **PostgreSQL data environment**
- **Centralized data preparation through `vw_healthcare_master`**
- **Tableau integration through `DS_vw_healthcare_master (healthcare_db)`**

- **Tableau analytical logic and calculations**
- **KPI framework and analytical measures**
- **Worksheets and analytical visualizations**
- **Five executive dashboards**
- **Dashboard interactions and Action architecture**
- **Architectural and design decisions**
- **Visualization and presentation standards**
- **Testing and validation**
- **Documentation and portfolio organization**
- **Future improvements and potential extensions**

The objective of this documentation is to provide sufficient technical and analytical context for another reader to understand not only what was developed, but also how and why the solution was structured in its current form.

By documenting both the implemented components and the reasoning behind important architectural and analytical decisions, the technical document provides a reference for future maintenance, portfolio review, interview discussion, and potential extension of the Healthcare Analytics Platform.

2.8.2 Documentation Standards

The technical documentation follows a consistent structural and formatting standard designed to improve readability, navigation, traceability, and professional presentation throughout the document.

The documentation incorporates:

- **Chapter-based organization**
- **Hierarchical section numbering**
- **Standardized headings and subsection formatting**
- **Technical and architectural tables**
- **Architecture and analytical workflow diagrams**
- **Consistent figure and table captions**
- **Technical Note callout sections**
- **Business Insight callout sections**
- **Consistent technical and business terminology**
- **Standardized header and footer information**
- **Page numbering**
- **Document revision and status information**

Figures and tables are numbered systematically and provided with standardized captions to support navigation, cross-referencing, and the maintenance of a structured List of Figures and List of Tables as part of the final publication process.

Consistent terminology is maintained across architectural layers, technical components, Tableau objects, analytical concepts, figures, and tables. This is particularly important for named technical objects such as `vw_healthcare_master`, `DS_vw_healthcare_master (healthcare_db)`, Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, and Dashboard Actions, because consistent terminology reduces ambiguity and improves component traceability throughout the documentation.

The same structural and visual standards are maintained throughout the technical document so that the documentation itself reflects the principles of consistency, clarity, organization, and maintainability applied to the Healthcare Analytics Platform.

2.8.3 Project Deliverables

The Healthcare Analytics Platform is designed to be presented as a collection of related technical and analytical deliverables rather than as a single isolated Tableau workbook. Together, these deliverables document the data foundation, analytical implementation, visualization environment, interaction architecture, and supporting materials required to present the project as a coherent Business Intelligence solution.

The principal deliverable categories include:

Database Assets

PostgreSQL database structures supporting the analytical environment, including the `healthcare_db` database and the centralized analytical View `vw_healthcare_master`, which provides the data foundation for all five final Tableau dashboards.

SQL Assets

SQL scripts and View definitions used to create or reproduce relevant database and data-preparation structures. These assets include the final master analytical View `vw_healthcare_master`, together with selected development SQL artifacts that document earlier analytical exploration where appropriate.

Development Views such as `vw_revenue_by_condition` and `vw_revenue_by_gender` are retained as learning and development artifacts and are not presented as data sources for the five final dashboards.

Tableau Assets

The Tableau workbook containing the centralized data source `DS_vw_healthcare_master (healthcare_db)`, Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, KPI calculations, worksheets, five executive dashboards, interactive filters, Filter Actions, and Highlight Actions used throughout the analytical solution.

Technical Documentation

The complete project documentation describing the solution architecture, PostgreSQL data environment, centralized data-preparation strategy, Tableau analytical methodology, implementation decisions, visualization and interaction design, testing considerations, and project outcomes.

Visual Portfolio Assets

Selected dashboard screenshots, architecture diagrams, analytical workflow diagrams, and supporting visual materials used to communicate the project efficiently during portfolio review, professional presentation, or interviews.

Together, these assets provide a more complete representation of the Healthcare Analytics Platform than a dashboard screenshot or Tableau workbook alone. They demonstrate not only the final analytical outputs, but also the centralized data foundation, analytical logic, architectural decisions, visualization design, interaction strategy, and technical documentation supporting the complete solution.

2.8.4 Document Publication

Microsoft Word is used as the primary authoring and maintenance environment for the Healthcare Analytics Platform technical documentation. It provides the structured document capabilities required to maintain consistent headings, hierarchical section numbering, captions, tables, figures, headers and footers, page numbering, cross-references, and document navigation throughout the project.

After the technical content, figures, tables, captions, cross-references, terminology, and final formatting have been reviewed and validated, the approved document is prepared for publication as a PDF. The PDF serves as the primary distribution format for professional portfolio presentation while preserving the structure and visual consistency of the source document.

The PDF publication format provides several advantages:

- **Preserves the intended document layout and formatting**
- **Maintains consistent pagination across viewing environments**
- **Reduces the risk of accidental formatting changes during distribution**
- **Provides a convenient and widely accessible format for portfolio review**
- **Allows the technical documentation to be reviewed independently from the Tableau development environment**
- **Supports professional presentation during job applications, portfolio evaluations, and interviews**

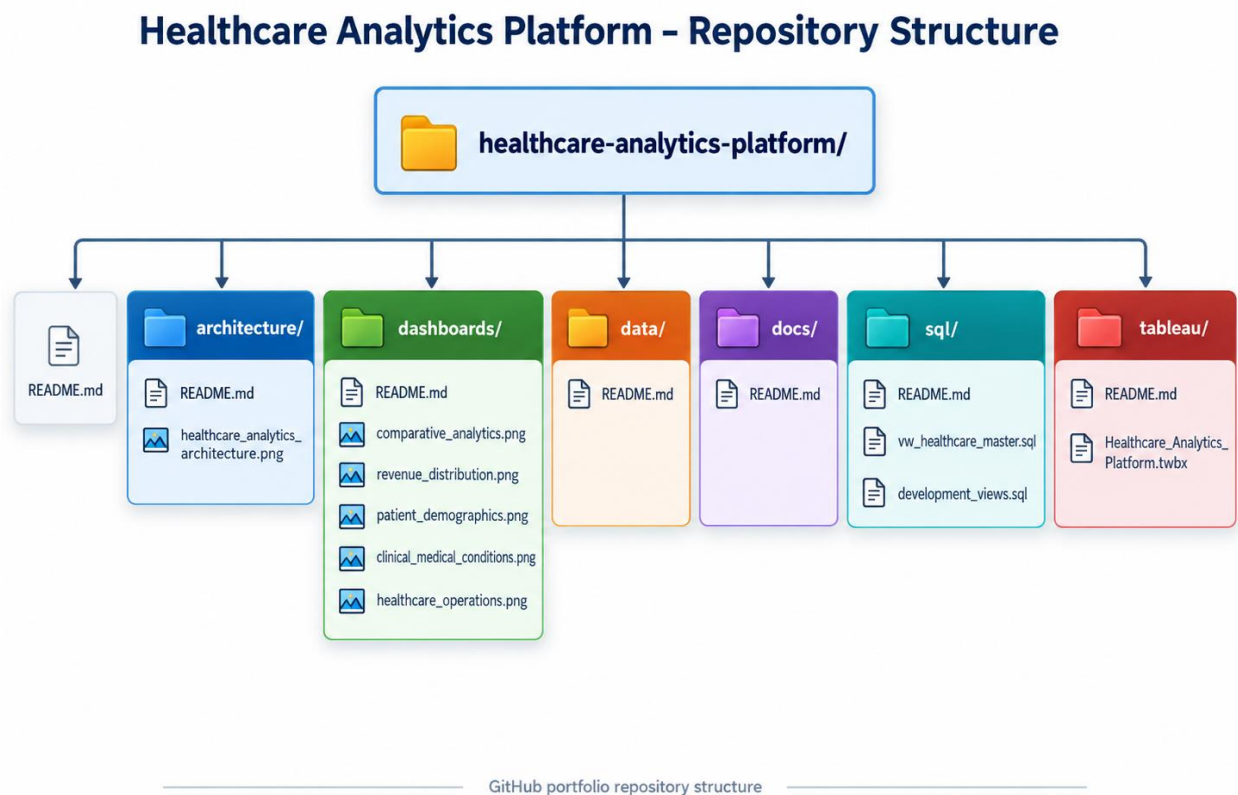
Microsoft Word therefore remains the editable source environment for maintaining and updating the technical documentation, while PDF serves as the controlled publication format used to distribute and present the finalized document.

2.8.5 Portfolio Repository Strategy

A future **GitHub portfolio repository** can serve as a centralized location for organizing and presenting the technical, analytical, documentation, and visual assets associated with the Healthcare Analytics Platform.

The repository is intended to support clear project navigation, technical transparency, portfolio presentation, and reproducibility of selected project components. Its structure should reflect the architecture of the analytical solution while maintaining a clear separation between database and SQL assets, Tableau assets, technical documentation, visual portfolio materials, and supporting dataset information.

A proposed repository structure can be organized as follows:



This repository structure represents a **completed portfolio organization model** for the Healthcare Analytics Platform and should not be interpreted as an already implemented or published GitHub repository until the corresponding repository has been created, populated with the selected project assets, and validated.

Once implemented, the repository can provide recruiters, hiring managers, and technical reviewers with a centralized location from which to understand the project architecture, navigate the principal technical deliverables, and examine selected implementation artifacts. The repository can also

complement the technical documentation by providing direct access to organized SQL assets, the Tableau workbook, portfolio documentation, architecture diagrams, dashboard screenshots, and supporting dataset information.

In this context, GitHub functions as a **portfolio publication and project organization environment** rather than as a production deployment platform. Its purpose is to improve project accessibility, technical transparency, organization, and professional presentation while preserving the distinction between the implemented analytical solution and future production deployment capabilities.

2.8.6 Portfolio Presentation Strategy

The complete technical documentation provides detailed evidence of the architecture, analytical methodology, implementation decisions, and development process of the Healthcare Analytics Platform. However, different audiences require different levels of technical detail and different methods of accessing project information.

The project can therefore be presented through multiple complementary formats, each serving a specific communication purpose:

Technical Documentation

Provides the most comprehensive description of the project, including business context, solution architecture, PostgreSQL data preparation, Tableau analytical logic, KPI methodology, visualization design, dashboard interactions, validation considerations, architectural decisions, and project outcomes.

GitHub README

Provides a concise entry point to the portfolio project, including the project overview, technology stack, architecture summary, principal analytical capabilities, selected dashboard screenshots, and navigation to supporting repository assets once the planned GitHub repository has been implemented.

Dashboard Screenshots

Provide immediate visual evidence of the completed analytical solution and allow reviewers to quickly understand the scope, design quality, and principal analytical perspectives represented across the executive dashboards.

Tableau Workbook

Provides access to the implemented Tableau analytical environment, including the centralized data source `DS_vw_healthcare_master` (`healthcare_db`), Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, KPI calculations, worksheets, five executive dashboards, interactive filters, Filter Actions, and Highlight Actions, when an appropriate workbook distribution method is used.

Interview Presentation

Allows selected architectural decisions, PostgreSQL structures, Tableau analytical mechanisms, dashboard capabilities, and interaction strategies to be explained in a focused format without requiring the interviewer to review the complete technical documentation.

This multi-format presentation strategy allows the same Healthcare Analytics Platform to communicate effectively with different audiences. Recruiters and hiring managers can obtain a rapid understanding of the project's scope, technologies, analytical capabilities, and business value, while technical reviewers and interviewers can examine progressively deeper levels of implementation detail through the technical documentation, repository assets, Tableau workbook, and supporting analytical materials.

2.8.7 Deployment Scope

The current implementation of the Healthcare Analytics Platform is classified as a portfolio analytical deployment rather than a production enterprise deployment. This classification reflects the technical and analytical capabilities implemented within the project while clearly distinguishing them from the infrastructure, automation, security, and operational capabilities that would typically be required in a production Business Intelligence environment.

The current implementation scope includes:

- **PostgreSQL-based analytical data environment using healthcare_db**
- **Centralized analytical data preparation through vw_healthcare_master**
- **Tableau Desktop analytical development using DS_vw_healthcare_master (healthcare_db)**
- **Tableau analytical logic implemented through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, and related analytical mechanisms**
- **Five interactive executive dashboards**
- **Interactive filters, Filter Actions, and Highlight Actions**
- **Technical documentation**
- **PDF publication for professional portfolio distribution**
- **Planned GitHub portfolio repository organization**

The current implementation scope does not claim implementation of:

- **Tableau Server**
- **Tableau Cloud production publishing**
- **Automated ETL or ELT orchestration**
- **CI/CD pipelines**
- **Enterprise authentication or authorization**
- **Production monitoring**
- **Automated data refresh infrastructure**
- **Cloud database deployment**

These capabilities represent potential areas for future production-oriented enhancement rather than components of the current portfolio implementation. Clearly defining these boundaries strengthens the

technical credibility of the Healthcare Analytics Platform by distinguishing implemented analytical capabilities, planned portfolio components, and potential future enterprise deployment capabilities.

2.8.8 Future Deployment Opportunities

The current architecture provides several opportunities for future expansion beyond the scope of the existing portfolio analytical implementation. These opportunities represent potential extensions toward a more production-oriented Business Intelligence environment and are not presented as capabilities already implemented within the Healthcare Analytics Platform.

Potential future enhancements include:

- Publishing dashboards through Tableau Cloud or Tableau Server
- Implementing scheduled data refresh processes
- Introducing automated ETL or ELT workflows
- Migrating or replicating the PostgreSQL environment to cloud infrastructure
- Implementing role-based access control
- Adding automated data-quality validation processes
- Establishing formal source-control practices for SQL, documentation, and related project assets
- Introducing separate development, testing, and production environments
- Expanding the platform with additional healthcare datasets and analytical domains
- Implementing production monitoring and operational support processes

These enhancements could extend the existing analytical architecture toward a more production-oriented Business Intelligence environment while preserving the fundamental separation of responsibilities established by the current layered design. The existing architecture provides a foundation from which additional deployment, automation, governance, security, and operational capabilities could be introduced as the solution evolves.

Future deployment opportunities should therefore be understood as **architectural extensions rather than components of the current implementation**. This distinction preserves the accuracy of the portfolio while demonstrating how the existing analytical solution could evolve toward a more mature production environment.

Deliverable	Technology / Format	Primary Purpose	Current Status
Analytical Database	PostgreSQL	Provide the structured healthcare data environment through healthcare_db	Implemented
Master Analytical View	PostgreSQL / SQL	Provide the centralized analytical data structure through vw_healthcare_master	Implemented
Analytical Workbook	Tableau Desktop	Provide analytical logic, worksheets, KPIs, and dashboards	Implemented
Executive Dashboards	Tableau Desktop	Provide interactive analytical presentation across five business perspectives	Implemented
Technical Documentation	Microsoft Word	Document architecture, methodology, implementation, and design decisions	Completed
Published Documentation	PDF	Provide the professional publication and distribution format	Planned
Portfolio Repository	GitHub	Organize and present selected project assets	Planned
Portfolio Visual Assets	Images / Screenshots	Present dashboards and architecture visually	In Development

Table 2.8 – Project Deliverable Matrix

The relationship between the implemented analytical components, documentation assets, planned portfolio repository, and professional presentation process can be summarized through the following documentation and portfolio deployment flow:

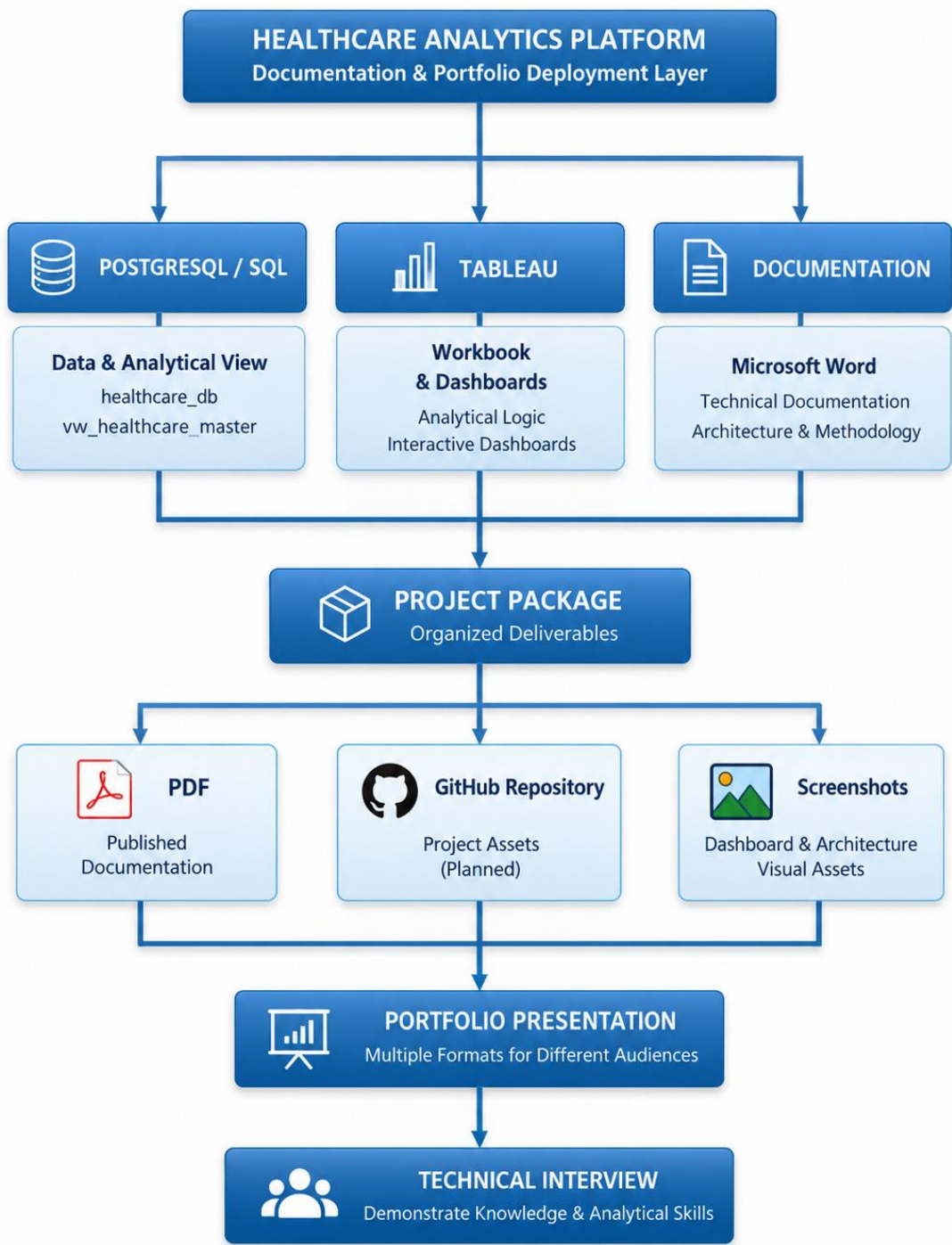


Figure 2.7 – Documentation & Portfolio Deployment Flow

Technical Note

The current Healthcare Analytics Platform does not represent a production Tableau deployment. Tableau Desktop is used for analytical development and dashboard implementation, Microsoft Word serves as the maintainable source environment for the technical documentation, and PDF serves as the primary publication and distribution format. The GitHub repository remains a planned portfolio organization and publication environment. Production capabilities such as Tableau Server, Tableau Cloud production publishing, automated data refresh infrastructure, ETL or ELT orchestration, CI/CD pipelines, enterprise authentication and authorization, production monitoring, and cloud database deployment remain potential future extensions rather than implemented project capabilities.

Business Insight

A Business Intelligence solution becomes significantly more valuable when its analytical logic, architecture, implementation decisions, and design principles can be understood beyond the development environment. Structured technical documentation and well-organized project deliverables improve maintainability, facilitate technical review, and allow analytical work to be communicated effectively to both business and technical audiences. By presenting the Healthcare Analytics Platform through complementary formats, the project can provide rapid high-level understanding for recruiters and hiring managers while preserving the technical depth required by technical reviewers and interviewers.

2.9 Architecture Design Principles

The architecture of the Healthcare Analytics Platform was developed according to a set of design principles intended to create a Business Intelligence solution that is structured, understandable, reusable, maintainable, and capable of supporting future analytical expansion.

These principles guide how responsibilities are distributed across the architectural layers of the platform, including the PostgreSQL data environment, centralized data preparation through `vw_healthcare_master`, Tableau analytical logic, worksheets and visualizations, executive dashboards, user interactions, and technical documentation.

Rather than concentrating data preparation, analytical logic, visualization, interaction, presentation, and documentation responsibilities within a single component, the platform assigns each responsibility to the architectural layer where it can be implemented and maintained most appropriately.

This layered approach establishes clear boundaries between the data foundation, data preparation, analytical logic, visualization, interaction, presentation, and documentation responsibilities of the solution. At the same time, the layers remain integrated through defined interfaces, including the connection between `vw_healthcare_master` and `DS_vw_healthcare_master` (`healthcare_db`), allowing the platform to operate as a coordinated analytical environment rather than as a collection of isolated technical components.

The principal architectural design principles are described below.

2.9.1 Separation of Concerns

Separation of Concerns is one of the fundamental architectural principles of the Healthcare Analytics Platform. It establishes clear logical ownership of responsibilities across the different layers and components of the analytical solution.

The architecture separates major responsibilities as follows:

- **PostgreSQL** manages the structured healthcare data environment through `healthcare_db`.
- **`vw_healthcare_master`** provides the centralized analytical data structure used by the final Tableau solution.
- **Tableau analytical components** implement analytical logic through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, KPI calculations, ranking, segmentation, and other context-dependent analytical mechanisms.
- **Worksheets** transform analytical measures and dimensions into reusable visual representations.
- **Dashboards** organize KPIs, worksheets, and supporting visual components into integrated analytical presentations.
- **Interactive Filters, Filter Actions, and Highlight Actions** provide controlled interactive behavior and coordinated analytical exploration across related visualizations.
- **Technical documentation** records the architecture, methodology, implementation decisions, design standards, validation considerations, and project scope.

This separation reduces unnecessary dependencies between components and helps prevent data preparation, analytical calculations, visualization logic, interaction behavior, presentation decisions, and documentation responsibilities from becoming unnecessarily intertwined.

The objective is not to create physical separation between every component, but to establish clear logical ownership of responsibilities throughout the analytical workflow. Each architectural component performs the responsibilities most appropriate to its role while remaining integrated with the other components through clearly defined relationships.

This approach improves maintainability, analytical consistency, and architectural clarity by allowing individual components to evolve without requiring unrelated responsibilities to be redesigned unnecessarily.

2.9.2 Modularity

The Healthcare Analytics Platform follows a **modular development approach** in which major data, analytical, visualization, interaction, and presentation components can be developed, validated, and refined independently before being integrated into larger analytical structures.

The centralized analytical View `vw_healthcare_master`, Tableau calculations, KPI calculations and cards, worksheets, charts, interactive filters, Filter Actions, Highlight Actions, and executive dashboards represent distinct but coordinated components within the overall solution. Each component performs a defined role while interacting with other components through the layered architecture.

For example, a KPI calculation can be developed and validated independently from the KPI card used to visually present it. Similarly, a worksheet can be tested before it is incorporated into an executive dashboard, and a Filter Action or Highlight Action can be configured and validated independently from the visual formatting of the worksheets involved in the interaction.

This modular structure simplifies development, testing, troubleshooting, and maintenance by allowing individual components to be reviewed or modified without requiring the entire analytical environment to be redesigned. It also supports future analytical expansion because new worksheets, calculations, interactions, or dashboard components can be introduced while preserving the responsibilities of existing modules.

Modularity therefore complements the principle of **Separation of Concerns**: Separation of Concerns defines where responsibilities belong within the architecture, while Modularity allows the components responsible for those functions to be developed, validated, maintained, and extended with greater independence.

2.9.3 Reusability

Reusability reduces unnecessary duplication across the Business Intelligence environment by allowing validated data structures, analytical definitions, and visualization components to support multiple analytical requirements.

Within the Healthcare Analytics Platform, `vw_healthcare_master` provides a centralized and reusable analytical data foundation for the final Tableau solution. Within Tableau, reusable Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, KPI definitions, and other analytical components allow

consistent analytical logic to support multiple worksheets and dashboard components where appropriate.

The preferred architectural approach is:

Define once → Validate → Reuse

rather than:

Recreate independently in every worksheet

This principle applies at multiple levels of the architecture. A standardized data structure can support multiple analytical domains, a validated calculation can support multiple visual components, and reusable worksheets can be incorporated into dashboards without recreating their underlying analytical logic.

Reusability improves analytical consistency, reduces unnecessary duplication, simplifies maintenance, and decreases the possibility that equivalent business metrics or analytical definitions will be implemented differently across dashboards. It also supports future analytical expansion by allowing validated components to be extended or incorporated into new analytical requirements without rebuilding the solution from the beginning.

2.9.4 Analytical Consistency

A Business Intelligence platform must provide consistent interpretations of the same analytical concepts across different reporting and visualization components. **Analytical Consistency** ensures that equivalent metrics, dimensions, classifications, and analytical relationships maintain the same meaning regardless of where they are presented within the Healthcare Analytics Platform.

The platform promotes analytical consistency through the centralized data foundation provided by `vw_healthcare_master`, standardized analytical definitions, reusable calculations, KPI definitions, naming conventions, business-facing terminology, and consistent dashboard interaction patterns.

For example, when the same metric is required by multiple worksheets or dashboard components, its analytical definition should remain consistent rather than being independently reconstructed with different logic for each visualization. Similarly, shared analytical concepts should preserve consistent aggregation methods, calculation granularity, classifications, and terminology wherever they are reused.

Consistency also applies to business-facing terminology. Technical field names or automatically generated labels can be replaced with clearer analytical terminology where appropriate without modifying the underlying data structure, calculation, or analytical definition. For example:

Month of Date Of Admission → Admission Month

This separation between technical field structure and business-facing terminology improves readability while preserving analytical meaning.

Analytical consistency also extends to dashboard interactions. Similar filtering, selection, and highlighting behaviors should follow predictable patterns so that users can interpret changes in analytical context consistently across related visualizations.

The objective is to ensure that users receive the same analytical interpretation regardless of where a metric, dimension, or analytical concept appears within the platform. Maintaining this consistency increases trust in the analytical results, reduces ambiguity, and supports more reliable comparison and decision-making across the dashboard environment.

2.9.5 Maintainability

Maintainability was considered throughout the organization and development of the Healthcare Analytics Platform. A maintainable Business Intelligence solution should allow developers to identify where a particular responsibility is implemented, understand its relationship with other components, and modify it with minimal unintended impact on unrelated parts of the analytical environment.

Several architectural practices contribute to maintainability:

- Centralized analytical data preparation through `vw_healthcare_master`
- Reusable and clearly defined Tableau calculations
- Structured worksheet and KPI naming
- Descriptive Filter Action and Highlight Action names
- Logical dashboard organization
- Consistent interaction patterns
- Standardized formatting and business-facing terminology
- Separation between data preparation, analytical logic, visualization, interaction, presentation, and documentation responsibilities
- Structured technical documentation

For example, Filter Action names such as `FA_Monthly_Billing_Filter` communicate both the interaction type and the analytical component involved, making the corresponding configuration easier to identify, review, and troubleshoot. Similarly, the centralized use of `vw_healthcare_master` and `DS_vw_healthcare_master` (`healthcare_db`) provides a clearly identifiable path between the PostgreSQL analytical data foundation and the Tableau analytical environment.

Maintainability becomes increasingly important as the number of calculations, worksheets, KPIs, dashboards, and interactions grows. Clear naming conventions, centralized data structures, reusable analytical definitions, modular components, and documented architectural responsibilities reduce the effort required to locate, understand, validate, and modify individual elements of the solution.

The objective of maintainability is not simply to make individual components easier to edit, but to preserve the clarity and stability of the overall analytical architecture as the platform evolves.

2.9.6 Scalability and Extensibility

The Healthcare Analytics Platform was designed to provide an analytical foundation that can be **scaled and extended** as business and analytical requirements evolve. Within the scope of the current portfolio implementation, scalability and extensibility refer primarily to the ability of the architecture, analytical components, and development methodology to accommodate additional requirements without requiring a complete redesign of the existing solution.

Potential extensions could include:

- Additional healthcare datasets and data attributes
- Extension of vw_healthcare_master to support new analytical requirements where appropriate
- Additional KPIs and analytical measures
- New analytical dimensions and classifications
- Additional Calculated Fields, LOD Expressions, Table Calculations, Sets, and Parameters
- Additional worksheets and analytical visualizations
- New executive dashboards
- Expanded interactive filters, Filter Actions, and Highlight Actions
- Additional time-based and comparative analysis
- New demographic, financial, clinical, or operational analytical domains
- Future production-oriented deployment capabilities

The architecture supports this extensibility through its centralized data foundation, modular analytical components, reusable calculations, standardized naming conventions, and separation of responsibilities across architectural layers. New analytical requirements can therefore be evaluated at the appropriate architectural level rather than requiring unrelated components to be redesigned.

The current project does not claim **enterprise-scale production infrastructure or enterprise production scalability**. Production capabilities such as Tableau Server or Tableau Cloud deployment, automated data pipelines, scheduled refresh infrastructure, production monitoring, enterprise security, and cloud-based database infrastructure remain potential future extensions rather than capabilities of the current implementation.

Scalability in the current Healthcare Analytics Platform therefore refers primarily to the ability of the **analytical architecture and development methodology** to accommodate additional data, calculations, visualizations, interactions, and analytical requirements while preserving the existing architectural structure and separation of responsibilities.

This distinction ensures that the scalability and extensibility claims remain appropriate to the actual scope of the portfolio implementation while demonstrating that the current architecture provides a structured foundation for future analytical growth.

2.9.7 Appropriate Placement of Business Logic

An important architectural decision within the Healthcare Analytics Platform is determining **where a particular data transformation, calculation, or analytical rule should be implemented**. The platform does not assume that all logic belongs in PostgreSQL or that all analytical logic belongs in Tableau. Instead, responsibilities are assigned according to the purpose, scope, granularity, reusability, and analytical context of each requirement.

PostgreSQL and vw_healthcare_master are appropriate for:

- Centralized and reusable data preparation
- Standardized analytical data structures
- Common field transformations where appropriate
- Database-oriented transformations and aggregations where appropriate
- Consistent field availability for downstream analysis
- Providing a stable analytical data foundation for Tableau through DS_vw_healthcare_master (healthcare_db)

Tableau is appropriate for:

- Calculated Fields and analytical measures
- LOD Expressions
- Table Calculations
- Sets and Parameters
- Dynamic segmentation and comparative analysis
- Ranking and contribution analysis
- Running Totals and Moving Averages
- Visualization-context-dependent calculations
- KPI calculations and presentation-oriented analytical logic
- Interactive analytical behavior

The placement of business and analytical logic should therefore be determined by the nature of the requirement rather than by an assumption that one technology should perform all processing. Logic that contributes to the centralized and reusable data foundation is generally more appropriate within PostgreSQL and vw_healthcare_master, while logic that depends on analytical granularity, visualization context, dynamic user selections, segmentation, ranking, or interactive behavior is generally more appropriate within Tableau.

This principle allows each technology to perform the type of processing most appropriate to its architectural role while maintaining a clear separation between data preparation and analytical logic. It also improves maintainability by making it easier to identify where a particular transformation or calculation belongs and reduces unnecessary duplication across the analytical environment.

2.9.8 User-Centered Analytical Design

Architectural and analytical design decisions ultimately support the users who consume and interpret the information presented by the Healthcare Analytics Platform. The platform therefore applies a **User-Centered Analytical Design** approach in which technical complexity is managed within the underlying architecture and should not appear unnecessarily in the final dashboard experience.

This principle is reflected through several design practices:

- Business-oriented chart titles and analytical terminology
- Clear and simplified axis labels
- Consistent KPI presentation
- Structured dashboard layouts and visual hierarchy
- Dedicated and logically organized filter areas
- Predictable Filter Action behavior
- Context-preserving Highlight Actions
- Consistent spacing, alignment, and formatting
- Contextual tooltips that provide additional analytical detail where appropriate
- Reduced unnecessary visual clutter

User-centered analytical design also requires the dashboard experience to reflect the analytical purpose of each interaction. Filter Actions should narrow the analytical context when focused investigation is required, while Highlight Actions should preserve the broader context when the objective is to explore relationships between related marks. These behaviors should remain understandable and predictable from the user's perspective.

The underlying analytical environment may contain PostgreSQL structures, Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, KPI logic, interactive filters, Filter Action configurations, and Highlight Action configurations. However, users should not need to understand these technical implementation mechanisms in order to interpret the analytical information presented by the dashboards.

The objective is to reduce unnecessary cognitive effort and allow users to concentrate on **analytical interpretation, comparison, pattern recognition, and decision support** rather than on understanding the technical implementation behind the dashboard. In this way, architectural and technical complexity remains appropriately managed within the platform while the final analytical experience remains clear, consistent, and business-oriented.

2.9.9 Controlled Interaction Design

Interactivity is treated as an **architectural capability rather than as a decorative dashboard feature**. Interactive behavior is introduced only when it supports a defined analytical objective and improves the user's ability to investigate, compare, or interpret related information.

Within the Healthcare Analytics Platform, Filter Actions are used when the analytical objective requires narrowing the context displayed by target worksheets, while Highlight Actions are used when relationships should be emphasized without removing the surrounding analytical context. Interactive filters provide an additional mechanism for allowing users to control the analytical scope of dashboard content through explicit selections.

Interactive behavior should therefore be:

- **Purposeful** — each interaction should support a defined analytical objective.
- **Predictable** — users should be able to understand the effect of an interaction.
- **Scoped appropriately** — Filter Actions and Highlight Actions should affect only the worksheets required by the intended analytical behavior.
- **Consistent** — similar interactions should behave consistently across related visualizations and dashboards.
- **Context-aware** — the selected interaction mechanism should either narrow or preserve analytical context according to the intended analytical purpose.
- **Testable** — interaction behavior should be validated independently from visual formatting and dashboard appearance.

Controlled Interaction Design also requires careful definition of source worksheets, target worksheets, triggering behavior, selected fields, and the expected analytical result of each Filter Action or Highlight Action. This allows interactions to be configured according to analytical purpose rather than applied broadly across every dashboard component.

This controlled approach prevents excessive or unnecessary interactivity from making dashboards difficult to understand, unintentionally changing analytical context, or producing unexpected analytical behavior. The objective is to ensure that every interaction contributes to analytical exploration while preserving clarity, consistency, and user confidence in the dashboard experience.

2.9.10 Documentation as an Architectural Component

Documentation is considered an architectural component of the Healthcare Analytics Platform rather than merely a final project deliverable. It provides a structured record of how the solution is organized, how responsibilities are distributed across architectural layers, which analytical mechanisms were implemented, and why important design decisions were made.

The technical documentation records:

- Architectural structure and layered solution design
- Technology and component responsibilities
- PostgreSQL data environment and centralized data-preparation strategy through `vw_healthcare_master`
- Tableau analytical techniques and calculation mechanisms
- KPI framework and analytical definitions
- Visualization and presentation principles
- Interaction architecture, interactive filters, Filter Action design, and Highlight Action design
- Architectural and analytical design decisions
- Project scope and implementation boundaries
- Deployment limitations
- Future enhancement and deployment opportunities

Documenting these elements provides traceability between the implemented analytical solution and the architectural reasoning behind its design. It also creates a reference that can support future maintenance, validation, troubleshooting, extension, portfolio review, and technical discussion.

Documentation also contributes to architectural consistency by preserving terminology, naming conventions, component responsibilities, analytical definitions, and implementation boundaries across the project. As the analytical environment evolves, this structured reference can help ensure that future modifications remain aligned with the established architecture rather than introducing undocumented or inconsistent design decisions.

This is particularly valuable for a portfolio-based Business Intelligence project because it demonstrates not only the final dashboards and analytical outputs, but also the technical methodology, architectural reasoning, analytical design principles, and implementation decisions that produced them. Documentation therefore functions both as a project deliverable and as a supporting component of the architecture itself.

Design Principle	Application within the Platform	Primary Benefit
Separation of Concerns	Responsibilities distributed across PostgreSQL, vw_healthcare_master, Tableau analytics, visualization, interaction, presentation, and documentation	Reduced coupling and clearer responsibility ownership
Modularity	Data, calculations, KPIs, worksheets, interactive filters, Filter Actions, Highlight Actions, and dashboards organized as distinct but coordinated components	Easier development, validation, troubleshooting, and maintenance
Reusability	Centralized analytical data foundation and reusable Tableau calculations, analytical definitions, and visual components	Reduced duplication and improved consistency
Analytical Consistency	Standardized metrics, calculation logic, terminology, naming conventions, and interaction patterns	Consistent analytical interpretation
Maintainability	Centralized data structure, structured naming, reusable logic, modular components, and technical documentation	Easier modification, troubleshooting, and future maintenance
Scalability & Extensibility	Architecture supports additional data, analytical requirements, calculations, visualizations, interactions, and dashboards	Structured foundation for future analytical expansion
Appropriate Placement of Business Logic	PostgreSQL and Tableau responsibilities assigned according to purpose, granularity, reusability, and analytical context	Clearer architecture and appropriate technology usage
User-Centered Analytical Design	Business-oriented terminology, visual hierarchy, consistent KPIs, organized filters, and predictable interactions	Improved usability and reduced cognitive effort
Controlled Interaction Design	Purposeful and appropriately scoped filters, Filter Actions, and Highlight Actions	Predictable and context-aware analytical exploration
Documentation as an Architectural Component	Architecture, methodology, analytical definitions, implementation decisions, scope, and limitations formally documented	Traceability, maintainability, and knowledge transfer

Table 2.9 – Architecture Design Principles Summary

The architectural design principles applied throughout the Healthcare Analytics Platform can be summarized through the following framework, which places the analytical platform at the center and organizes the principal principles that guide its structure, implementation, analytical consistency, interaction model, maintainability, and future analytical evolution.

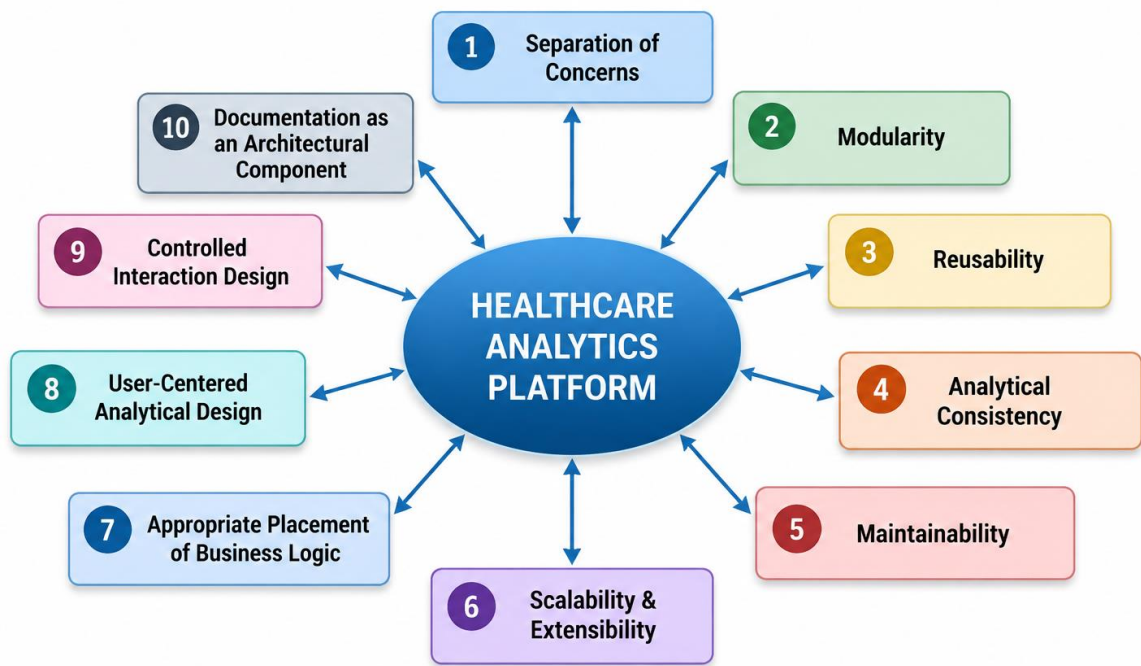


Figure 2.8 – Architecture Design Principles Framework

Technical Note

The architecture described in this chapter represents the logical organization of the Healthcare Analytics Platform rather than a production enterprise architecture composed of independently deployed services for each layer. Several responsibilities are implemented within PostgreSQL and Tableau Desktop, while the layered architectural model provides a conceptual framework for organizing, documenting, validating, and maintaining those responsibilities.

The purpose of this layered structure is to establish clear ownership of data preparation, analytical logic, visualization, interaction, presentation, and documentation responsibilities without implying physical infrastructure separation between every component.

Business Insight

Strong Business Intelligence architecture is not defined by the number of technologies used, but by how clearly responsibilities are assigned, how consistently analytical definitions are applied, and how effectively the different components work together to support analytical interpretation and decision-making.

Separation of Concerns, Modularity, Reusability, Analytical Consistency, Maintainability, Scalability & Extensibility, Appropriate Placement of Business Logic, User-Centered Analytical Design, Controlled Interaction Design, and Documentation as an Architectural Component provide a structured foundation for an analytical platform that can evolve while preserving clarity, consistency, maintainability, and reliability.

CHAPTER 2

Chapter Summary

Key Takeaways

The **Solution Architecture** of the Healthcare Analytics Platform establishes a structured Business Intelligence environment in which data management, data preparation, analytical logic, visualization, interaction, presentation, and documentation operate as logically separated but integrated responsibilities.

The architecture combines the PostgreSQL data environment `healthcare_db`, centralized analytical data preparation through `vw_healthcare_master`, Tableau integration through `DS_vw_healthcare_master` (`healthcare_db`), Tableau analytical components, worksheets, executive dashboards, interactive filters, Filter Actions, Highlight Actions, and technical documentation within a layered framework designed to promote analytical consistency, modularity, reusability, maintainability, and future extensibility.

✓ Layered Business Intelligence Architecture

The platform follows a layered architecture that separates **data management, data preparation, analytics, visualization, interaction, presentation, and documentation** into clearly defined logical responsibilities.

These responsibilities are logically separated but remain integrated through defined relationships between architectural components. This structure creates a controlled analytical flow from the PostgreSQL data environment through centralized data preparation and Tableau analytical logic to interactive executive dashboards and ultimately to decision-oriented analytical information.

The layered model represents a **logical architecture rather than a collection of independently deployed physical services**, allowing responsibilities implemented within PostgreSQL and Tableau Desktop to be organized, documented, validated, and maintained according to their architectural purpose.

✓ PostgreSQL as the Central Data Foundation

PostgreSQL provides the centralized relational data foundation of the Healthcare Analytics Platform through `healthcare_db`, where the healthcare dataset is stored, organized, and made available for SQL-based querying and analytical preparation.

Within this environment, PostgreSQL supports the structured data foundation required by the analytical solution and provides the database context in which the centralized analytical View `vw_healthcare_master` is maintained.

This centralized data environment establishes the foundation for the controlled flow of standardized healthcare information from PostgreSQL into the Tableau analytical environment.

✓ Centralized Analytical Data Preparation through `vw_healthcare_master`

The PostgreSQL View `vw_healthcare_master` provides the **centralized analytical data-preparation structure** used by the final Healthcare Analytics Platform. It organizes the healthcare attributes required

across the analytical domains represented by the five executive dashboards and establishes a consistent interface between the PostgreSQL data environment and Tableau.

In the final architecture, `vw_healthcare_master` is consumed through the Tableau data source `DS_vw_healthcare_master (healthcare_db)`, providing a common analytical foundation for financial, demographic, clinical, medical-condition, operational, comparative, segmentation, insurance, and time-based analysis.

Specialized Views such as `vw_revenue_by_condition` and `vw_revenue_by_gender` were used during earlier development, learning, and analytical experimentation, but they are **not data sources for the five final executive dashboards**. The final portfolio architecture is standardized around `vw_healthcare_master` as the centralized PostgreSQL analytical View.

✓ Tableau as the Analytical Engine

Tableau Desktop functions as the primary analytical engine of the Healthcare Analytics Platform by extending the centralized analytical data foundation provided through `DS_vw_healthcare_master (healthcare_db)` with analytical and context-dependent business logic.

The Tableau analytical environment includes:

- Calculated Fields
- Level of Detail (LOD) Expressions
- Table Calculations
- Sets
- Parameters
- Ranking and contribution analysis
- Percentage-based calculations
- Running Totals and Moving Averages
- Time-based analysis
- Dynamic segmentation and comparative analysis
- Standardized KPI calculations

These analytical mechanisms transform the standardized healthcare data provided by `vw_healthcare_master` into reusable analytical measures, classifications, comparisons, and business definitions capable of supporting KPIs, worksheets, visualizations, and executive dashboards.

This division of responsibilities allows PostgreSQL to provide the centralized analytical data foundation while Tableau performs the analytical calculations that depend on granularity, visualization context, segmentation, comparison, ranking, and user-controlled analytical behavior.

✓ Separation Between Analytical Logic and Presentation

The architecture distinguishes between **what an analytical metric or business definition means** and **how that information is visually communicated to the user**.

Analytical definitions and calculations belong primarily to the Analytics Layer, while worksheets, charts, KPI cards, formatting, labels, visual hierarchy, and dashboard layouts belong to the Visualization & Presentation Layer.

For example, a KPI calculation defines the analytical meaning and value of a metric, while the KPI card determines how that metric is visually presented within a dashboard.

This separation improves analytical consistency, modularity, and maintainability by allowing analytical definitions and presentation components to be developed, validated, and modified according to their respective responsibilities.

✓ Modular Visualization Architecture

Tableau worksheets function as **modular and reusable visualization components** that can be developed, validated, formatted, and tested independently before being incorporated into executive dashboards.

KPI cards, trend charts, comparative visualizations, ranking analysis, distributions, contribution analysis, and other analytical views are assembled into dashboards according to defined business and analytical objectives.

This modular visualization architecture allows individual visual components to evolve without requiring the complete dashboard environment to be redesigned and supports the reuse of validated analytical components where appropriate.

✓ Five Executive Analytical Dashboards

The Presentation Layer consolidates the analytical environment into five principal dashboards:

- **Comparative Analytics Dashboard**
- **Revenue Distribution Dashboard**
- **Patient Demographics Dashboard**
- **Clinical & Medical Conditions Dashboard**
- **Healthcare Operations Dashboard**

Together, these dashboards provide complementary analytical perspectives across financial performance, comparative analysis, patient demographics, clinical and medical conditions, and healthcare operations. Although each dashboard addresses a distinct analytical perspective, all five are supported by the same centralized PostgreSQL analytical View, `vw_healthcare_master`, consumed in Tableau through `DS_vw_healthcare_master (healthcare_db)`.

✓ Interactive Analytical Architecture

The platform extends beyond static reporting through interactive filters, Filter Actions, and Highlight Actions that support coordinated analytical exploration across related visualizations.

Filter Actions narrow the analytical context displayed by target worksheets when focused investigation is required, while Highlight Actions preserve the broader visualization context and emphasize related information. Interactive filters provide additional user-controlled mechanisms for adjusting the analytical scope of dashboard content.

This distinction enables both focused investigation and contextual relationship analysis while ensuring that each interaction mechanism is selected according to its intended analytical purpose.

✓ Structured Interaction Management

Dashboard interactions follow **controlled source-target relationships, defined triggering behavior, appropriate analytical scope, and descriptive naming conventions.**

Actions such as:

FA_Monthly_Discharges_Filter

FA_Monthly_Billing_Filter

FA_Monthly_Admission_Type_Mix_Filter

FA_Insurance_Provider_Filter

FA_Monthly_Test_Results_Filter

demonstrate how Filter Action behavior can be organized, identified, tested, and maintained systematically within a multi-visualization dashboard environment.

Highlight Actions follow the same controlled interaction principle. For example, HA_Insurance_Provider_Highlight provides a clearly identifiable interaction configuration for relationship-based exploration involving Insurance Provider analysis.

This structured approach improves maintainability and helps ensure that interactive behavior remains purposeful, predictable, and consistent with the analytical objective of each dashboard.

✓ Documentation as Part of the Architecture

Technical documentation is treated as an **architectural component rather than as an afterthought or merely a final project deliverable.**

The project documentation records the solution architecture, PostgreSQL data environment, centralized data-preparation strategy through vw_healthcare_master, Tableau analytical techniques, KPI framework, visualization and presentation standards, interaction architecture, architectural and analytical design decisions, project scope, deployment limitations, and future enhancement opportunities.

The final documentation package is designed to complement the Tableau workbook, SQL assets, architecture diagrams, dashboard screenshots, and other supporting technical materials used to present the Healthcare Analytics Platform.

Documentation therefore provides traceability between the implemented solution and the reasoning behind its design while supporting maintenance, validation, portfolio review, technical discussion, and potential future extension.

✓ Clearly Defined Deployment Scope

The current implementation represents a **portfolio analytical deployment rather than a production enterprise deployment**.

The project demonstrates PostgreSQL-based data management through healthcare_db, centralized analytical data preparation through vw_healthcare_master, Tableau integration through DS_vw_healthcare_master (healthcare_db), Tableau Desktop analytical development, interactive executive dashboards, interactive filters, Filter Actions, Highlight Actions, and professional technical documentation.

The current implementation does **not** claim production capabilities such as Tableau Server, Tableau Cloud production publishing, automated ETL or ELT orchestration, CI/CD pipelines, enterprise authentication and authorization, automated data refresh infrastructure, production monitoring, or cloud database deployment.

These capabilities represent potential future extensions rather than components of the current portfolio implementation. Clearly defining this boundary maintains an accurate, transparent, and technically defensible representation of the Healthcare Analytics Platform.

✓ Architecture Designed for Maintainability and Extension

The platform follows architectural principles centered on:

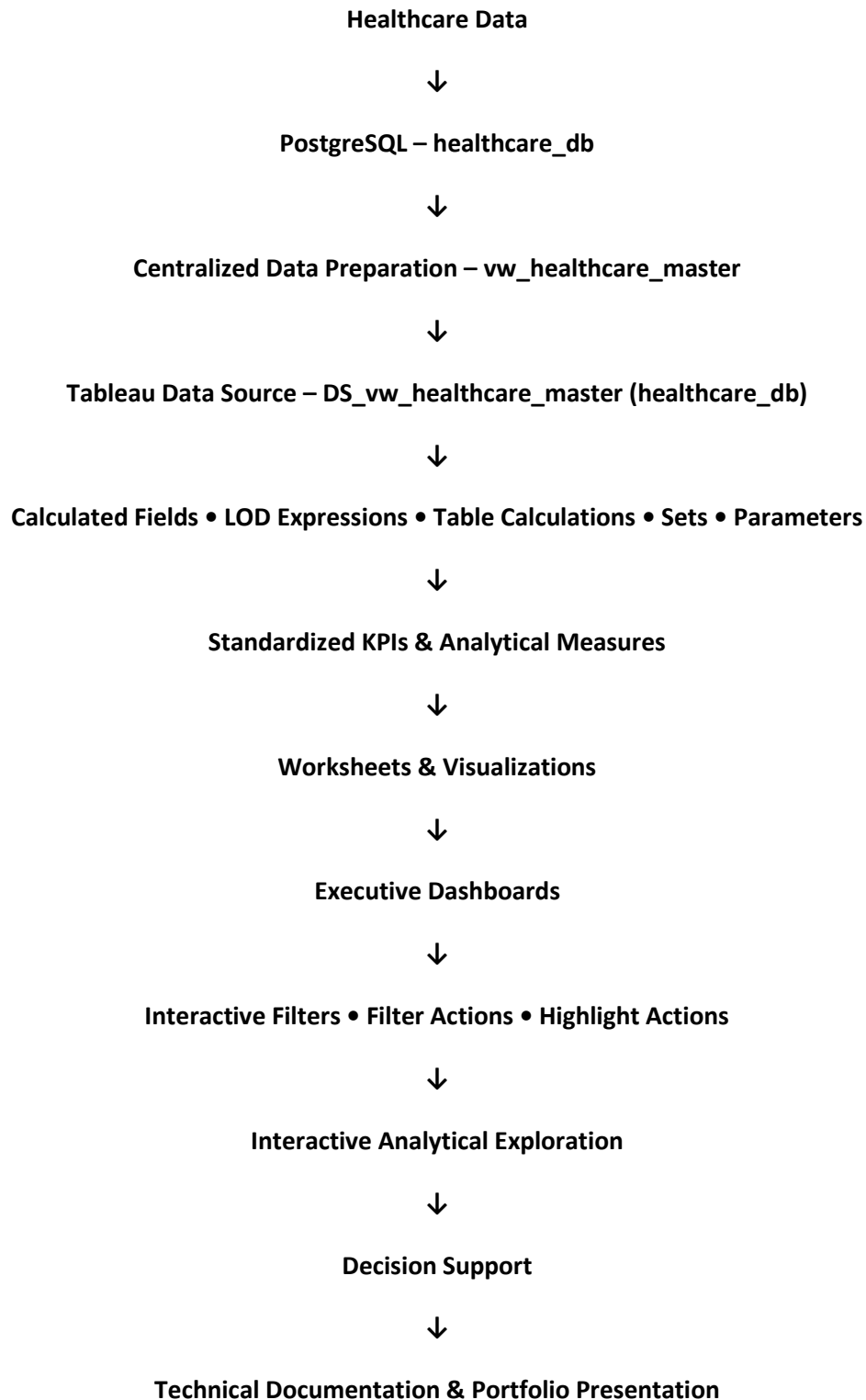
Separation of Concerns • Modularity • Reusability • Analytical Consistency • Maintainability • Scalability & Extensibility • Appropriate Placement of Business Logic • User-Centered Analytical Design • Controlled Interaction Design • Documentation as an Architectural Component

Together, these principles provide a structured foundation that can accommodate additional data, analytical requirements, calculations, visualizations, interactions, and dashboard capabilities while preserving consistency across the existing solution.

The architecture is therefore designed to support future analytical evolution without implying enterprise-scale production infrastructure. Its extensibility is based on clear responsibility boundaries, centralized data preparation, reusable analytical logic, modular visualization components, controlled interaction design, and structured technical documentation.

Architecture in One View

The complete architectural flow documented in this chapter can be summarized as:



Chapter Conclusion

The architecture developed for the **Healthcare Analytics Platform** demonstrates how multiple Business Intelligence responsibilities can be organized into a coherent, layered, and integrated analytical workflow. PostgreSQL provides the structured data foundation through `healthcare_db`, while `vw_healthcare_master` provides the centralized analytical data-preparation structure consumed in Tableau through `DS_vw_healthcare_master` (`healthcare_db`). Tableau extends this standardized data foundation through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, KPI calculations, ranking, segmentation, and other analytical mechanisms. Worksheets transform analytical measures into visual components, executive dashboards integrate those components into coordinated analytical experiences, and interactive filters, Filter Actions, and Highlight Actions enable controlled analytical exploration.

By applying **Separation of Concerns, Modularity, Reusability, Analytical Consistency, Maintainability, Scalability & Extensibility, Appropriate Placement of Business Logic, User-Centered Analytical Design, Controlled Interaction Design, and Documentation as an Architectural Component**, the platform establishes an architecture in which data preparation, analytical logic, visualization, interaction, presentation, and documentation remain logically distinct while operating as an integrated Business Intelligence solution. This structure supports analytical consistency and maintainability while providing a foundation for future analytical expansion without overstating the production maturity of the current portfolio implementation.

The architectural foundation documented in this chapter provides the basis for the detailed examination of the project's PostgreSQL data environment, centralized analytical View, Tableau analytical calculations, KPI framework, worksheets, visualizations, dashboard interactions, and executive dashboards presented in the following chapters. Together, these components demonstrate how the Healthcare Analytics Platform progresses from structured healthcare data to interactive analytical information designed to support interpretation, exploration, and decision-making.

CHAPTER 3 - PostgreSQL Data Environment & Master Analytical View

3.1 Chapter Overview

The PostgreSQL data environment provides the structured relational foundation of the Healthcare Analytics Platform. Within this environment, healthcare data is stored in the `healthcare_db` database and organized for analytical consumption through the centralized PostgreSQL View `vw_healthcare_master`.

This chapter examines the PostgreSQL implementation that supports the analytical platform, including the database environment, healthcare dataset structure, centralized analytical data preparation, the role of `vw_healthcare_master`, development and learning Views, data validation practices, and the controlled transition of standardized healthcare information from PostgreSQL into Tableau Desktop.

The final portfolio architecture is centered on `vw_healthcare_master`, which provides the standardized PostgreSQL analytical source consumed in Tableau through `DS_vw_healthcare_master` (`healthcare_db`). Specialized Views developed during earlier learning and experimentation stages are documented separately and are not presented as data sources for the five final executive dashboards.

The objective of this chapter is to demonstrate how PostgreSQL supports the Healthcare Analytics Platform as a centralized and maintainable analytical data foundation while preserving a clear separation between database-level data preparation and the advanced analytical logic implemented in Tableau.

3.2 PostgreSQL Database Environment – `healthcare_db`

The Healthcare Analytics Platform uses PostgreSQL as its relational database environment, with `healthcare_db` serving as the central database for storing and preparing the healthcare dataset used throughout the project.

The healthcare data was imported into PostgreSQL and managed through the PostgreSQL environment using pgAdmin for database administration and SQL development. Within `healthcare_db`, the source healthcare data provides the structured foundation from which the centralized analytical View `vw_healthcare_master` was developed.

PostgreSQL is responsible for maintaining the relational data environment, organizing the healthcare attributes required for analysis, supporting SQL-based validation and data preparation, and providing a stable analytical source for Tableau Desktop.

The database environment is intentionally kept separate from visualization-specific analytical logic. Advanced calculations, segmentation, ranking, comparative analysis, percentages, Running Totals, Moving Averages, and interactive analytical behavior are implemented primarily within Tableau rather than being duplicated unnecessarily at the database level.

This separation allows healthcare_db to remain a centralized and maintainable data foundation while Tableau extends the standardized PostgreSQL data through the analytical mechanisms required by the five executive dashboards.

3.3 Healthcare Dataset Structure

The healthcare dataset stored within healthcare_db contains the attributes required to support the demographic, clinical, financial, and operational analytical perspectives developed throughout the Healthcare Analytics Platform.

The dataset includes information related to patient demographics, medical conditions, admission and discharge activity, physicians, hospitals, insurance providers, billing amounts, medications, laboratory test results, room information, and admission types. These attributes provide the multidimensional foundation required by the Tableau analytical environment.

Representative fields used throughout the project include:

- **Patient and Demographic:** Patient Name, Age, Gender, Blood Type
- **Clinical:** Medical Condition, Medication, Test Results
- **Admission and Operational:** Date of Admission, Discharge Date, Admission Type, Room Number
- **Healthcare Organization:** Doctor, Hospital
- **Financial and Insurance:** Insurance Provider, Billing Amount

These fields support analytical requirements such as patient segmentation, demographic analysis, medical-condition analysis, admission and discharge trends, insurance analysis, revenue and billing analysis, medication analysis, and test-result evaluation.

The source attributes are subsequently organized through vw_healthcare_master, which exposes a standardized analytical structure for Tableau consumption. This approach allows the five executive dashboards to operate from a common set of healthcare attributes rather than requiring separate source structures for each analytical perspective.

3.4 Master Analytical View – vw_healthcare_master

The PostgreSQL View vw_healthcare_master serves as the centralized analytical source for the final Healthcare Analytics Platform. Its primary role is to organize the healthcare attributes required by Tableau into a consistent and reusable structure that can support multiple analytical domains from a common PostgreSQL source.

The View supports financial, demographic, clinical, medical-condition, operational, comparative, segmentation, insurance-related, and time-based analysis. Rather than creating a separate database View for each final dashboard, the portfolio architecture standardizes analytical consumption around vw_healthcare_master.

Within Tableau Desktop, the View is consumed through DS_vw_healthcare_master (healthcare_db). This shared data source provides the standardized healthcare attributes used by KPIs, worksheets, calculations, visualizations, and the five executive dashboards.

The centralized View also establishes a clear separation of responsibilities between PostgreSQL and Tableau. PostgreSQL organizes and standardizes the analytical data structure, while Tableau extends that structure through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentages, and interactive analytical behavior.

This design reduces unnecessary source fragmentation, promotes consistent field usage, and provides a maintainable analytical foundation that can support additional calculations, worksheets, or dashboards as future analytical requirements evolve.

3.5 Data Preparation Responsibilities

Data preparation within the Healthcare Analytics Platform is primarily focused on establishing a consistent PostgreSQL analytical foundation for Tableau rather than implementing all business and analytical logic at the database level.

The principal PostgreSQL data-preparation responsibilities include:

- Selecting and organizing source fields required for analytical consumption
- Providing the centralized analytical structure through vw_healthcare_master
- Standardizing commonly required healthcare attributes exposed to Tableau
- Maintaining consistent field availability across analytical domains
- Supporting database-level transformations or aggregations when appropriate
- Providing a stable PostgreSQL source for DS_vw_healthcare_master (healthcare_db)

Tableau extends this standardized data structure with the calculations and analytical behavior that depend on visualization context, segmentation, comparison, ranking, percentages, Sets, Parameters, LOD Expressions, Table Calculations, and interactive analysis.

This division of responsibilities avoids unnecessary duplication between PostgreSQL and Tableau while preserving a clear boundary between data preparation and advanced analytical logic.

3.6 Development and Learning Views

During the earlier development stages of the Healthcare Analytics Platform, additional PostgreSQL Views were created to support SQL learning, analytical experimentation, and Tableau integration prototyping.

Two representative development Views were:

- `vw_revenue_by_condition` — created to aggregate healthcare billing information by medical condition and support early revenue analysis across clinical categories.
- `vw_revenue_by_gender` — created to aggregate healthcare billing information by gender and support early demographic and revenue analysis in Tableau.

These Views were useful for understanding how focused database-level aggregations could be consumed and visualized within Tableau. They contributed to the development process but are not used as data sources for the five final executive dashboards.

As the project evolved, the final portfolio architecture was consolidated around `vw_healthcare_master`, which became the centralized PostgreSQL analytical source for the professional dashboard environment. The earlier specialized Views are therefore retained as development and learning artifacts rather than components of the final dashboard data architecture.

3.7 PostgreSQL-to-Tableau Integration

The integration between PostgreSQL and Tableau Desktop is established through the centralized analytical View `vw_healthcare_master`, which provides the standardized PostgreSQL source used by the five final executive dashboards.

Within Tableau Desktop, this View is represented through the data source `DS_vw_healthcare_master (healthcare_db)`, providing access to the healthcare attributes required for financial, demographic, clinical, comparative, and operational analysis.

The integration flow can be summarized as:

PostgreSQL Database – `healthcare_db`
↓
Master Analytical View – `vw_healthcare_master`
↓
Tableau Data Source – `DS_vw_healthcare_master (healthcare_db)`
↓
Tableau Analytical Components

Once the standardized data is available in Tableau, it is extended through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, ranking logic, percentages, segmentation techniques, and other derived analytical measures required by the dashboard environment.

This integration pattern provides a consistent boundary between database-level data preparation and Tableau-based analytical logic while allowing all five final dashboards to operate from the same PostgreSQL analytical foundation.

3.8 Data Validation and Analytical Readiness

Data validation within the Healthcare Analytics Platform focuses on ensuring that the PostgreSQL analytical foundation provides the fields and structure required for consistent downstream analysis in Tableau.

Analytical readiness is established by confirming that the healthcare attributes required by the final dashboards are available through `vw_healthcare_master`, consistently exposed through `DS_vw_healthcare_master` (`healthcare_db`), and suitable for use by the Tableau calculations, KPIs, worksheets, and visualizations built on top of the centralized source.

Validation is also performed progressively at the analytical-component level. Calculated measures and KPIs can be created and validated before being incorporated into worksheets or dashboards, while modular worksheets and calculations can be reviewed independently before becoming part of the broader dashboard environment.

This approach supports analytical consistency by allowing potential issues to be identified at the appropriate layer rather than only after dashboard assembly. The objective is not to represent the project as having a production-grade automated data-quality framework, but to demonstrate a structured validation process appropriate to the scope of a professional portfolio implementation.

Once the PostgreSQL source structure, Tableau calculations, and analytical components have been validated within their respective responsibilities, the data environment can be considered analytically ready for use across the five executive dashboards.

Technical Note

PostgreSQL provides the structured relational and data-preparation foundation of the Healthcare Analytics Platform through `healthcare_db` and the centralized analytical View `vw_healthcare_master`. The final portfolio architecture intentionally avoids distributing advanced analytical logic across multiple database Views. Instead, PostgreSQL standardizes and exposes the healthcare attributes required for analysis, while Tableau extends that foundation through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentages, and interactive analytical behavior.

This separation keeps the PostgreSQL layer maintainable and reusable while allowing Tableau to manage analytical logic that depends on visualization context and user interaction.

Business Insight

A centralized and consistent data foundation improves the reliability of Business Intelligence analysis by ensuring that dashboards and analytical components operate from the same standardized healthcare information.

By using `vw_healthcare_master` as the common PostgreSQL analytical source, the Healthcare Analytics Platform reduces unnecessary source fragmentation and provides Tableau with a consistent foundation for financial, demographic, clinical, comparative, and operational analysis.

This approach allows analytical teams to focus on interpretation, comparison, and decision-oriented analysis rather than repeatedly reconstructing the underlying dataset for individual reporting requirements.

Chapter Summary

This chapter documented the PostgreSQL data environment that provides the structured analytical foundation of the Healthcare Analytics Platform. The `healthcare_db` database stores the healthcare information used throughout the project, while the centralized PostgreSQL View `vw_healthcare_master` organizes the attributes required for analytical consumption in Tableau.

The principal conclusions established in this chapter are summarized below.

Key Takeaways

✓ Centralized PostgreSQL Data Environment

PostgreSQL provides the relational environment used to store, organize, query, and prepare the healthcare data supporting the analytical platform. The `healthcare_db` database functions as the central database environment from which the analytical solution is developed.

✓ Standardized Healthcare Dataset

The dataset contains demographic, clinical, operational, financial, insurance, admission, discharge, medication, test-result, physician, hospital, and related healthcare attributes that support multidimensional analysis across the final dashboard environment.

✓ Master Analytical View

`vw_healthcare_master` serves as the centralized PostgreSQL analytical source for the five final executive dashboards. It provides Tableau with a standardized and reusable healthcare data structure rather than requiring separate database Views for each final analytical requirement.

✓ Clear PostgreSQL and Tableau Responsibilities

PostgreSQL is responsible primarily for relational data management, field organization, standardized analytical preparation, and the centralized View. Tableau extends this foundation through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentages, KPIs, and interactive analytical behavior.

✓ Development Views Documented Separately

`vw_revenue_by_condition` and `vw_revenue_by_gender` were developed during earlier learning, experimentation, and prototyping stages. They contributed to SQL and Tableau integration practice but are not used as data sources for the five final executive dashboards.

✓ Consistent PostgreSQL-to-Tableau Integration

The final analytical flow is standardized through:

`healthcare_db`

↓

`vw_healthcare_master`

↓

`DS_vw_healthcare_master (healthcare_db)`



Tableau Analytical Components

✓ Validation and Analytical Readiness

Analytical components are developed and validated according to their responsibilities before being integrated into the final dashboard environment. This approach supports analytical consistency while remaining appropriate to the scope of a professional portfolio implementation rather than representing a production-grade automated data-quality framework.

Chapter Conclusion

The PostgreSQL implementation establishes a centralized and reusable analytical foundation for the Healthcare Analytics Platform. By organizing healthcare information through `healthcare_db` and standardizing analytical consumption through `vw_healthcare_master`, the project provides Tableau with a consistent source from which the five executive dashboards can be developed.

This structure reduces unnecessary source fragmentation, maintains a clear separation between database-level preparation and Tableau-based analytical logic, and provides a maintainable foundation for future analytical expansion.

CHAPTER 4 – Tableau Analytical Logic & KPI Framework

4.1 Chapter Overview

The Tableau Analytics Layer transforms the standardized healthcare data provided through `DS_vw_healthcare_master` (`healthcare_db`) into reusable business-oriented measures, analytical calculations, KPIs, segmentation logic, and comparative analytical components.

Building on the centralized PostgreSQL foundation established in the previous chapter, Tableau Desktop provides the analytical mechanisms required to interpret, compare, rank, segment, and summarize healthcare information across the financial, demographic, clinical, comparative, and operational perspectives represented by the platform.

The Healthcare Analytics Platform applies Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking, percentage calculations, segmentation techniques, and standardized KPI definitions according to the analytical requirements of the final dashboard environment. These components are developed as reusable analytical logic rather than being embedded unnecessarily within individual visualizations.

This chapter focuses on the Tableau analytical logic implemented within the project, including calculated measures, advanced analytical techniques, dynamic segmentation, comparative analysis, KPI standardization, and validation practices. Visualization layout, dashboard composition, and interaction design are addressed separately so that analytical definitions remain clearly distinguished from their visual presentation.

4.2 Tableau Analytical Environment

Tableau Desktop functions as the primary analytical environment of the Healthcare Analytics Platform. It consumes the standardized healthcare data provided through `DS_vw_healthcare_master` (`healthcare_db`) and extends that centralized source with the analytical logic required by the five final executive dashboards.

Within Tableau, healthcare attributes are transformed into reusable measures, classifications, KPIs, comparative calculations, rankings, percentages, segmentation logic, and time-based analytical measures. These components support financial, demographic, clinical, comparative, and operational analysis while maintaining a common underlying data foundation.

An important design principle is that analytical calculations are defined and validated independently from their final visual presentation whenever practical. This allows the same analytical definition to support multiple worksheets or dashboard components without requiring the logic to be recreated for each visualization.

Tableau therefore serves as the layer in which standardized healthcare data is interpreted, calculated, compared, segmented, ranked, and transformed into reusable analytical information before being presented through worksheets and executive dashboards.

4.3 Calculated Fields, LOD Expressions & Table Calculations

The Healthcare Analytics Platform uses multiple Tableau calculation mechanisms according to the analytical behavior required by each metric or visualization. Calculated Fields, Level of Detail (LOD) Expressions, and Table Calculations provide complementary capabilities that extend the standardized healthcare data supplied through DS_vw_healthcare_master (healthcare_db).

Calculated Fields are used to create derived measures, classifications, and analytical logic required for segmentation, comparisons, KPIs, medical-condition analysis, gender-based analysis, percentage contributions, dynamic classifications, and dashboard-specific metrics. They provide a reusable analytical layer between the centralized PostgreSQL source and the final visual components.

Level of Detail (LOD) Expressions are used when a calculation must be evaluated at a controlled analytical granularity rather than relying exclusively on the dimensions displayed within a worksheet. Within the project, LOD logic supports analytical requirements involving patient counts, revenue values, totals, percentages, and other measures that require consistent calculation behavior across different visualization contexts.

A conceptual LOD pattern used in Tableau can be represented as:

```
{ FIXED [Analytical Dimension] : AGGREGATION([Measure]) }
```

Table Calculations operate over aggregated values within the Tableau view and are used for analytical behavior that depends on the relationship between displayed marks. Capabilities practiced and applied within the project include ranking, Percent of Total, Running Totals, Moving Averages, comparative calculations, contribution analysis, and index-based analytical logic.

Because Table Calculations depend on the structure of the visualization, their configuration must account for factors such as Compute Using, addressing, partitioning, and sort order. This distinction is important because the analytical result can change when the structure of the Tableau view changes.

Together, these calculation mechanisms allow the Healthcare Analytics Platform to maintain reusable analytical definitions while supporting different levels of granularity, visualization-dependent calculations, comparative analysis, ranking, cumulative behavior, and proportional analysis.

4.4 Sets, Parameters & Dynamic Analysis

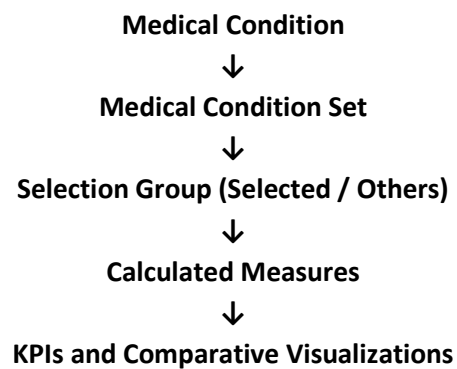
Tableau Sets and Parameters extend the analytical flexibility of the Healthcare Analytics Platform by supporting dynamic segmentation, user-controlled selections, and analytical switching without requiring additional PostgreSQL Views for each analytical scenario.

Sets are used to separate members of a dimension into analytical groups according to selected or predefined criteria. A key implementation within the project is the Medical Condition Set, which supports dynamic comparison between selected medical conditions and the remaining population. Set membership is translated through the calculated classification Selection Group, allowing records to be categorized as **Selected** or **Others**.

This logic supports reusable comparative measures and visualizations such as:

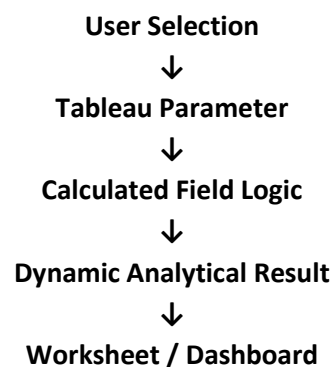
- Selected Revenue
- Others Revenue
- Selected Patients
- Others Patients
- Selected vs Others Revenue Comparison
- Selected vs Others Patient Comparison

The analytical pattern can be summarized as:



Parameters provide user-controlled values that can influence calculations and analytical behavior independently from conventional filtering. Within the project, Parameters were practiced and applied to support Dynamic Measure Selection, Dynamic Dimension Selection, combined measure-and-dimension analysis, Top N analysis, analytical switching, and other user-controlled analytical selections.

A Parameter can therefore operate together with a Calculated Field to interpret a user selection and return the appropriate analytical result:



Sets and Parameters serve complementary purposes: Sets manage analytical membership and segmentation, while Parameters provide user-controlled values that can change how calculations or visualizations behave. Both mechanisms preserve the centralized data foundation provided through DS_vw_healthcare_master (healthcare_db) while increasing analytical flexibility within Tableau.

4.5 Comparative and Segmentation Logic

Comparative analysis within the Healthcare Analytics Platform uses reusable Tableau segmentation logic to evaluate a selected analytical population against the remaining dataset.

A principal implementation is the **Selected vs Others** analytical pattern. The Medical Condition Set determines membership in the selected population, while the calculated classification Selection Group translates that membership into the business-oriented categories **Selected** and **Others**. This classification can then be reused across analytical measures, KPIs, worksheets, and comparative visualizations.

The resulting analytical structure supports measures and visualizations including **Selected Revenue**, **Others Revenue**, **Selected Patients**, **Others Patients**, **Selected vs Others Revenue Comparison**, and **Selected vs Others Patient Comparison**.

This approach allows the comparison to remain dynamic while continuing to operate from the common data source `DS_vw_healthcare_master (healthcare_db)`. Changing Set membership changes the analytical population without requiring separate PostgreSQL datasets or Views for each comparison scenario.

The same segmentation logic supports the **Comparative Analytics Dashboard**, where selected analytical populations can be evaluated against the remaining population through revenue, patient counts, and related comparative measures.

4.6 Standardized KPI Framework

The Healthcare Analytics Platform uses a standardized KPI framework to transform the healthcare data provided through `DS_vw_healthcare_master (healthcare_db)` into business-oriented summary measures across financial, demographic, clinical, comparative, and operational analytical domains.

Representative KPIs developed throughout the project include:

- Total Patients
- Total Admissions
- Total Revenue
- Average Billing
- Average Age
- Male Patients
- Female Patients
- Average Monthly Admissions
- Average Length of Stay
- Average Monthly Discharges
- Top Medical Condition
- Selected versus Others metrics

These KPIs are supported by the appropriate Tableau analytical mechanisms according to the requirements of each metric, including aggregations, Calculated Fields, LOD Expressions, Table Calculations, Sets, and other derived analytical logic.

A key design principle is that the analytical definition of a KPI is separated from its visual presentation. The KPI calculation belongs to the Tableau Analytics Layer, while the KPI card belongs to the Visualization and Presentation Layer. This separation allows the same metric definition to remain consistent even when its visual formatting, placement, or presentation changes across dashboard components.

Each KPI therefore requires a clearly defined calculation, aggregation level, filtering behavior, and business meaning. Standardizing these elements helps maintain analytical consistency across worksheets and dashboards and reduces the risk of presenting visually similar metrics with different underlying definitions.

4.7 Analytical Validation & Reusability

Reusability is an important design principle within the Tableau Analytics Layer. When multiple analytical components require the same business definition or calculation logic, the preferred approach is to create a reusable analytical component rather than independently recreating equivalent logic across different worksheets or dashboards.

Within the Healthcare Analytics Platform, reusable analytical logic is implemented through mechanisms such as Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, and standardized KPI definitions, depending on the analytical requirement.

This approach supports:

- Improved analytical consistency
- Reduced duplication of calculation logic
- Easier validation and testing
- Simplified maintenance
- More consistent metric interpretation across dashboards
- More efficient development of new worksheets and analytical components
- Easier future expansion of the analytical environment

Reusable calculations are particularly valuable because the five executive dashboards share the same centralized Tableau data source while applying different analytical perspectives to the healthcare dataset. Common analytical definitions can therefore remain consistent even when they are presented through different KPIs, worksheets, or dashboards.

Analytical components are also developed and validated according to their individual responsibilities before being integrated into the broader dashboard environment. This modular approach reduces the risk of duplicating inconsistent logic and makes troubleshooting, maintenance, and future analytical extension more manageable.

Technical Note

Tableau analytical mechanisms are selected according to the behavior required by each analytical problem rather than applying a single calculation technique universally. Calculated Fields support reusable derived logic, LOD Expressions control calculation granularity, Table Calculations operate in relation to the displayed view, Sets support analytical membership and segmentation, and Parameters provide user-controlled analytical values.

Using each mechanism according to its intended analytical role improves consistency, reusability, validation, and maintainability across the Healthcare Analytics Platform.

Business Insight

Business Intelligence creates greater value when analytical calculations are standardized, reusable, and aligned with clearly defined business questions rather than being recreated independently within individual visualizations.

Within the Healthcare Analytics Platform, Tableau transforms the standardized healthcare data into business-oriented measures that support financial, operational, demographic, clinical, and comparative analysis. Calculations, segmentation logic, ranking, percentages, and KPIs provide the analytical structure required to interpret healthcare information consistently across multiple dashboards.

Reusable analytical definitions improve consistency because the same business logic can support multiple KPIs, worksheets, and dashboards without being reconstructed for each visual component. This approach reduces duplicated logic, simplifies validation and maintenance, and helps ensure that analytical results are interpreted consistently throughout the platform.

The result is an analytical environment in which standardized healthcare data can be calculated, compared, segmented, ranked, and transformed into decision-oriented measures before being presented through executive dashboards.

Chapter Summary

This chapter documented the Tableau analytical logic that transforms the standardized healthcare data provided through DS_vw_healthcare_master (healthcare_db) into reusable business-oriented measures, calculations, KPIs, segmentation logic, and comparative analytical components.

Tableau extends the centralized PostgreSQL analytical foundation through Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, ranking, percentage calculations, segmentation techniques, and other analytical mechanisms selected according to the requirements of each analytical problem.

The principal conclusions established in this chapter are summarized below.

Key Takeaways

✓ Multiple Analytical Calculation Mechanisms

Calculated Fields support reusable derived logic, LOD Expressions control calculation granularity, and Table Calculations provide visualization-dependent capabilities such as ranking, Percent of Total, Running Totals, Moving Averages, contribution analysis, and comparative calculations.

✓ Dynamic Analysis through Sets and Parameters

Sets support analytical membership and segmentation, while Parameters provide user-controlled values that can influence calculations and analytical behavior. Together, these mechanisms enable more flexible analysis while continuing to use the centralized Tableau data source.

✓ Comparative and Segmentation Analysis

The Medical Condition Set, Selection Group, and Selected vs Others measures provide a reusable analytical pattern for comparing selected medical-condition populations with the remaining dataset across revenue, patient counts, KPIs, and comparative visualizations.

✓ Standardized KPI Framework

KPIs are treated as analytical definitions rather than only visual cards. Their calculation, aggregation level, filtering behavior, and business meaning are maintained separately from presentation decisions, supporting consistent interpretation across dashboards.

✓ Analytical Reusability and Validation

Reusable Tableau calculations reduce duplicated logic, simplify validation and maintenance, and help ensure that common analytical definitions remain consistent across KPIs, worksheets, and dashboards.

Chapter Conclusion

The Tableau Analytics Layer provides the analytical intelligence that converts standardized healthcare data into meaningful and reusable measures. By applying the appropriate combination of Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, comparative logic, and standardized KPIs, the Healthcare Analytics Platform maintains a clear separation between analytical definitions and their visual presentation.

This reusable analytical foundation supports the worksheets, visualizations, and executive dashboards that present the healthcare information to end users.

CHAPTER 5 – Executive Dashboards & Portfolio Results

5.1 Chapter Overview

The final Healthcare Analytics Platform is presented through five executive dashboards that integrate validated KPIs, analytical calculations, reusable worksheets, visualizations, and interactive analytical capabilities into a coordinated Business Intelligence environment.

Each dashboard addresses a distinct analytical perspective while operating from the same centralized data foundation and standardized Tableau analytical framework. The final dashboard portfolio includes the Comparative Analytics Dashboard, Revenue Distribution Dashboard, Patient Demographics Dashboard, Clinical & Medical Conditions Dashboard, and Healthcare Operations Dashboard.

Together, these dashboards provide complementary perspectives across comparative, financial, demographic, clinical, medical-condition, and operational analysis. They share common design principles involving KPI placement, worksheet composition, visual hierarchy, formatting consistency, and interactive analytical exploration.

This chapter provides a concise overview of the final dashboard environment, the principal analytical and interaction capabilities incorporated into the portfolio, and the overall results demonstrated by the Healthcare Analytics Platform. Detailed architectural, database, and Tableau calculation concepts have already been documented in previous chapters and are not repeated here.

5.2 Five Executive Dashboards

The final Healthcare Analytics Platform is organized around five executive dashboards, each designed to address a distinct healthcare analytical perspective while sharing the same centralized PostgreSQL and Tableau analytical foundation. Together, the dashboards provide complementary comparative, financial, demographic, clinical, and operational views of the healthcare dataset.

Comparative Analytics Dashboard

The Comparative Analytics Dashboard supports controlled comparison between selected analytical populations and the remaining dataset. Its principal analytical pattern is Selected versus Others analysis, allowing differences in revenue, patient populations, and related comparative measures to be examined within a consistent analytical framework.

Revenue Distribution Dashboard

The Revenue Distribution Dashboard focuses on financial analysis across medical conditions. It combines revenue-oriented KPIs with ranking, percentage contribution, and distribution analysis to help evaluate how financial activity is distributed across healthcare categories.

Patient Demographics Dashboard

The Patient Demographics Dashboard provides an analytical view of patient characteristics through demographic KPIs and visual analysis of age, gender, blood type, age-group composition, and insurance-provider distribution.

Clinical & Medical Conditions Dashboard

The Clinical & Medical Conditions Dashboard focuses on medical conditions and related clinical characteristics. It supports evaluation of patient populations, condition-level patterns, and relevant clinical indicators within the healthcare dataset.

Healthcare Operations Dashboard

The Healthcare Operations Dashboard provides operational visibility into admissions, discharges, billing activity, length of stay, admission types, insurance-provider revenue, test results, and time-based healthcare trends. These perspectives allow operational activity to be examined through coordinated analytical visualizations rather than isolated reports.

Although each dashboard addresses a different analytical domain, all five follow common design principles involving KPI placement, worksheet composition, visual hierarchy, formatting consistency, and interactive analytical exploration.

5.3 Key Analytical & Interaction Capabilities

The five executive dashboards extend beyond static reporting by combining standardized KPIs, comparative analysis, segmentation, ranking, contribution analysis, time-based measures, and controlled interactive exploration within a coordinated Tableau environment.

Key analytical capabilities demonstrated across the portfolio include Selected versus Others analysis, ranking and percentage contribution, Running Totals, Moving Averages, demographic and clinical segmentation, financial distribution analysis, and operational trend analysis.

Interactive exploration is supported through user-controlled filters, Filter Actions, and Highlight Actions. Filter Actions narrow the analytical context for focused investigation, while Highlight Actions preserve the broader visual context and emphasize related information. This distinction allows the interaction method to be selected according to the analytical objective.

Implemented Filter Actions include:

- FA_Monthly_Discharges_Filter
- FA_Monthly_Billing_Filter
- FA_Monthly_Admission_Type_Mix_Filter
- FA_Insurance_Provider_Filter
- FA_Monthly_Test_Results_Filter

These interactions support coordinated analysis of discharges, billing, admission-type mix, insurance-provider revenue, and test-result activity within the dashboard environment.

The portfolio also includes HA_Insurance_Provider_Highlight, which supports relationship-oriented exploration involving Insurance Provider analysis while preserving the broader analytical context.

Together, these analytical and interaction capabilities demonstrate how the platform allows users to move from high-level KPI monitoring to focused investigation and contextual exploration without requiring separate reporting environments for each analytical question.

5.4 Portfolio Results

The completed Healthcare Analytics Platform demonstrates an end-to-end Business Intelligence workflow that progresses from PostgreSQL-based healthcare data management and centralized analytical preparation through Tableau analytical calculations, KPI development, visualization, dashboard interaction, and professional technical documentation.

The final portfolio combines a structured PostgreSQL environment, the centralized analytical View `vw_healthcare_master`, the Tableau data source `DS_vw_healthcare_master (healthcare_db)`, reusable Tableau analytical logic, standardized KPIs, modular worksheets, five executive dashboards, and controlled interactive exploration within a coordinated analytical solution.

From an analytical perspective, the platform demonstrates the ability to examine healthcare information across financial, operational, demographic, clinical, medical-condition, and comparative perspectives. The dashboards support exploration of admissions, discharges, billing activity, insurance providers, medical conditions, medications, test results, patient characteristics, length of stay, ranking, contribution, and Selected versus Others analysis.

From a professional portfolio perspective, the project demonstrates not only the construction of analytical outputs but also the architectural, methodological, and implementation decisions used to organize the solution. It provides evidence of competencies in PostgreSQL, SQL-based analytical preparation, Tableau calculations, KPI design, comparative analysis, visualization, dashboard development, interactive analytics, validation, and technical documentation.

The Healthcare Analytics Platform was developed using an educational healthcare dataset and represents a portfolio analytical implementation rather than a production enterprise deployment. This distinction is intentionally maintained so that the documentation accurately reflects the capabilities that were actually implemented while still demonstrating professional Business Intelligence principles and an end-to-end analytical development methodology.

Chapter Summary & Final Conclusion

Chapter Summary

Chapter 5 presented the final portfolio results of the Healthcare Analytics Platform through five executive dashboards: Comparative Analytics, Revenue Distribution, Patient Demographics, Clinical & Medical Conditions, and Healthcare Operations.

Together, these dashboards consolidate the centralized PostgreSQL analytical foundation, reusable Tableau analytical logic, standardized KPIs, modular worksheets, and interactive capabilities into a coordinated Business Intelligence environment covering comparative, financial, demographic, clinical, medical-condition, and operational analysis.

The final dashboard environment extends beyond static reporting through interactive filters, Filter Actions, and Highlight Actions that support both focused investigation and contextual relationship analysis. These interaction mechanisms are configured according to their analytical purpose and contribute to a more controlled and interpretable user experience.

The project also demonstrates the importance of maintaining clear boundaries between portfolio implementation and production enterprise capabilities. The current solution represents a professional portfolio analytical implementation and does not claim production infrastructure such as automated ETL/ELT orchestration, enterprise authentication, CI/CD pipelines, production monitoring, or cloud deployment.

Final Conclusion

The Healthcare Analytics Platform demonstrates a complete Business Intelligence development workflow that transforms structured healthcare data into reusable analytical information and interactive executive dashboards.

PostgreSQL provides the structured data environment through `healthcare_db` and centralized analytical preparation through `vw_healthcare_master`. Tableau consumes this standardized foundation through `DS_vw_healthcare_master (healthcare_db)` and extends it with Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, standardized KPIs, reusable worksheets, visualizations, and controlled interactive analysis.

The completed project demonstrates practical competencies in SQL, PostgreSQL, analytical data preparation, Tableau analytical development, KPI design, comparative analysis, data visualization, dashboard development, interactive analytics, validation, architectural thinking, and professional technical documentation.

By maintaining separation of concerns, reusable analytical logic, centralized data preparation, modular visualization components, and controlled interaction design, the platform provides a structured and technically defensible portfolio example that can support professional presentation, technical discussion, and future analytical extension without overstating the capabilities of the current implementation.